

SECTION PLATE t=0 | Year 2001 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

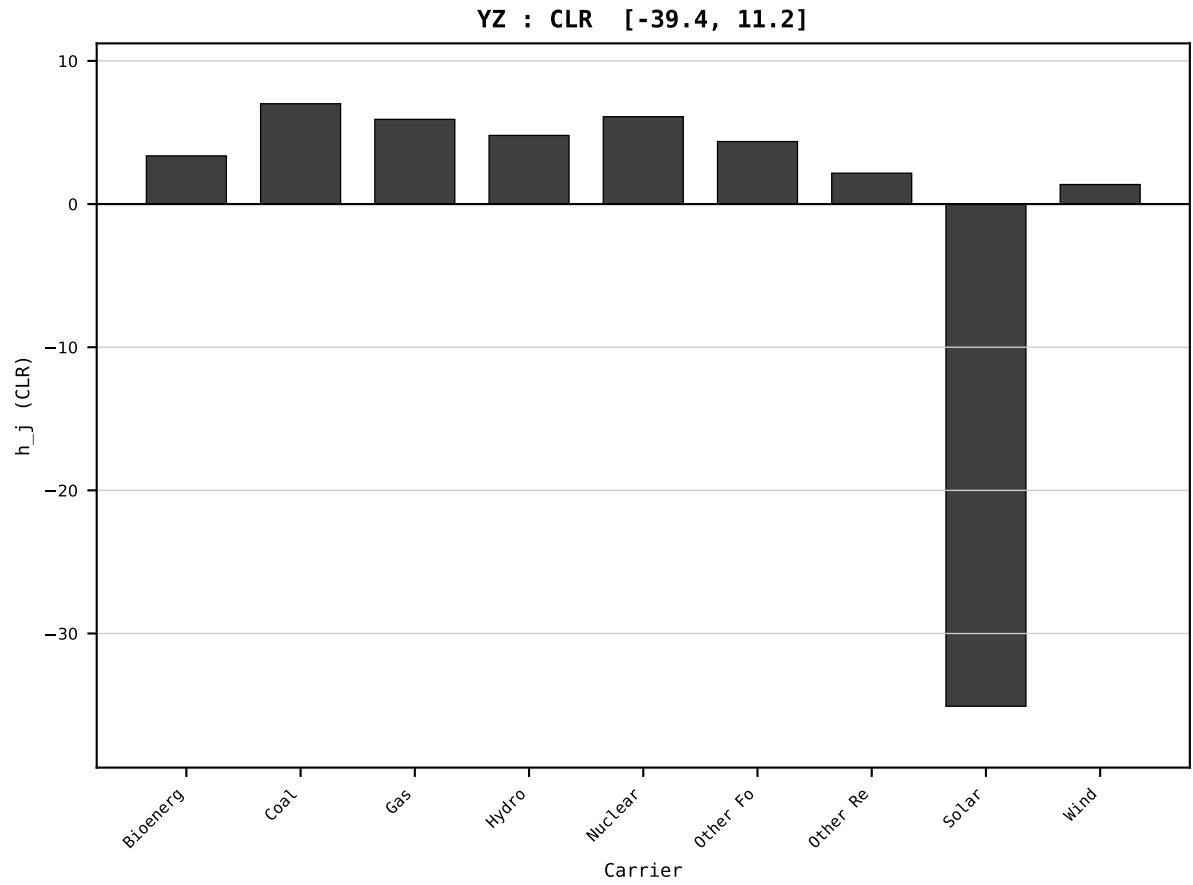
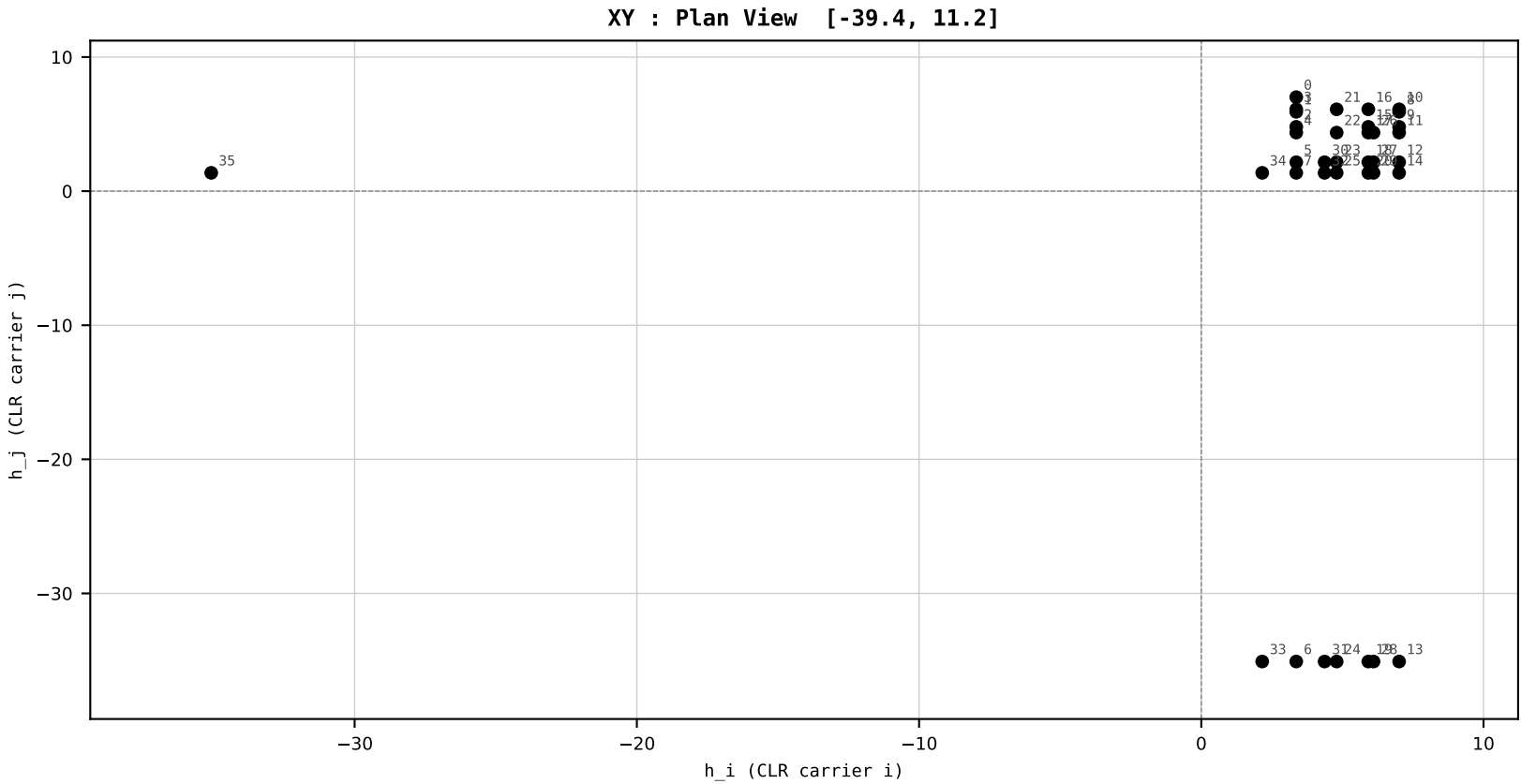
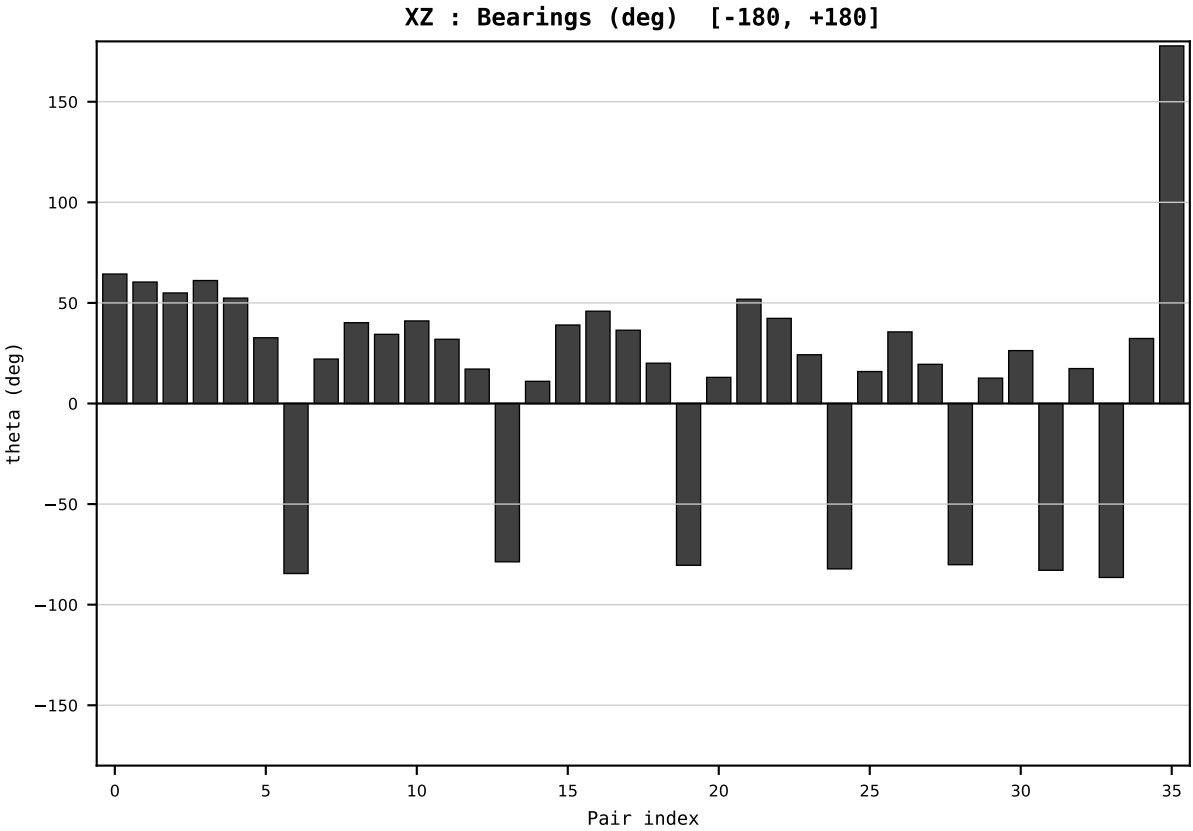
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2001
t : 0 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.388371
Ring = Hs-3
E_metric= 895.1337
kappa_HS= 190397000000000000.00
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 42.090 HLR
DR ratio= 1903969992697630976.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=1 | Year 2002 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

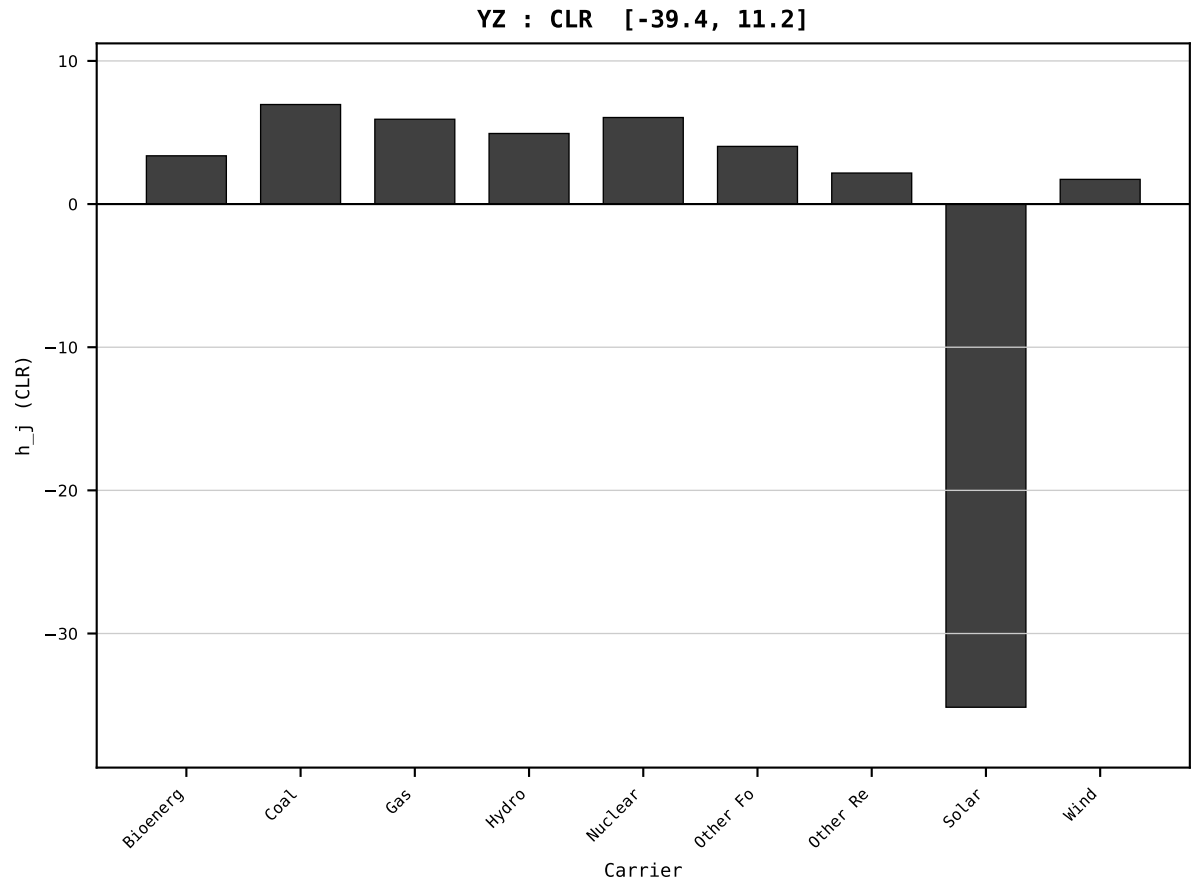
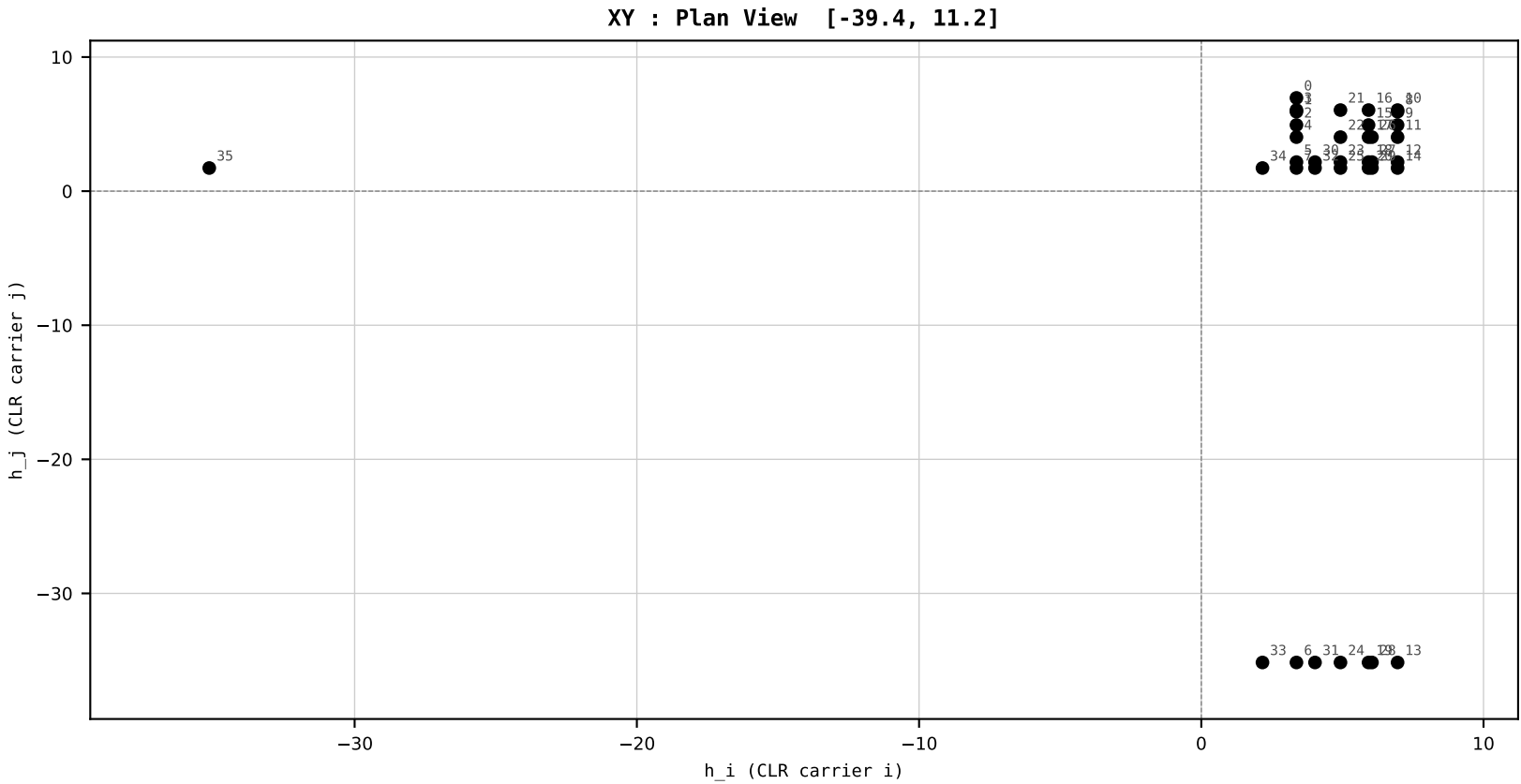
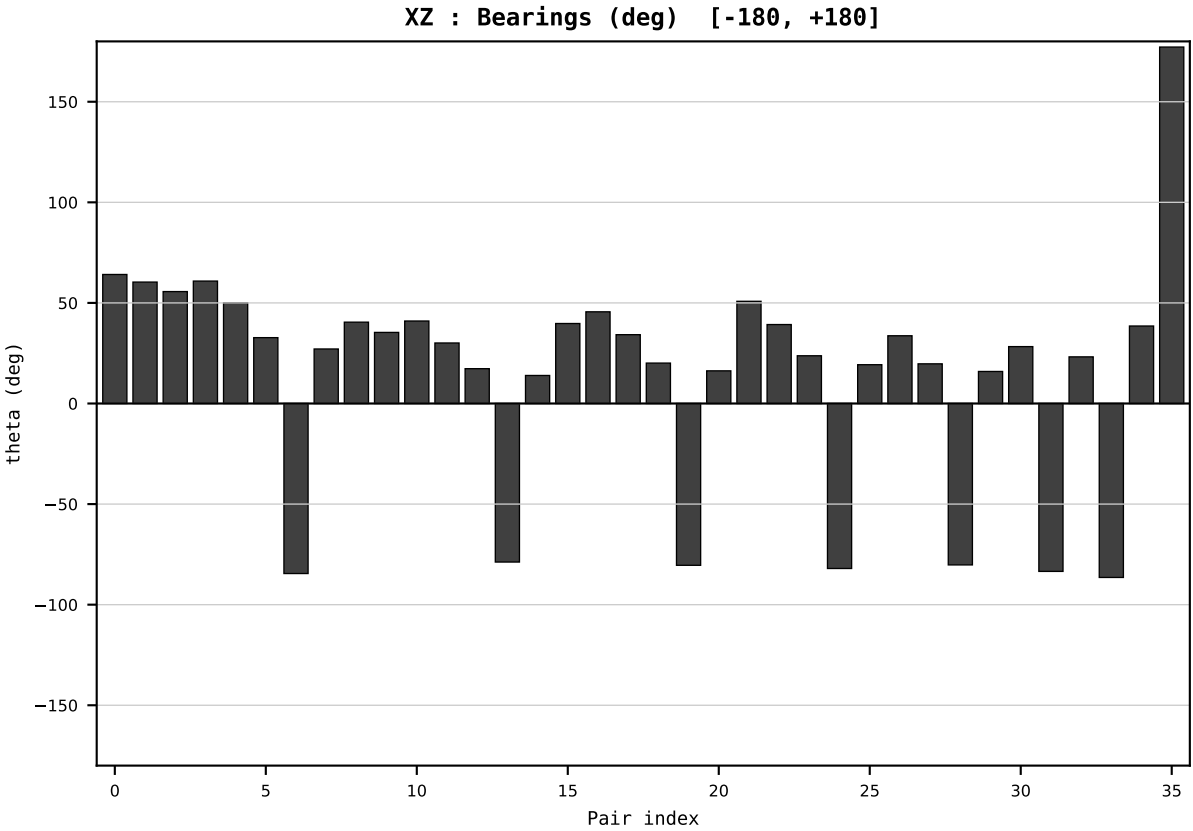
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2002
t : 1 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.383719
Ring = Hs-3
E_metric= 895.5128
kappa_HS= 193315000000000000.00
omega = 0.7941 deg
d_A = 0.522906
Helm = Wind
Helm d = +0.359790
DR = 42.106 HLR
DR ratio= 1933149998657515008.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=2 | Year 2003 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

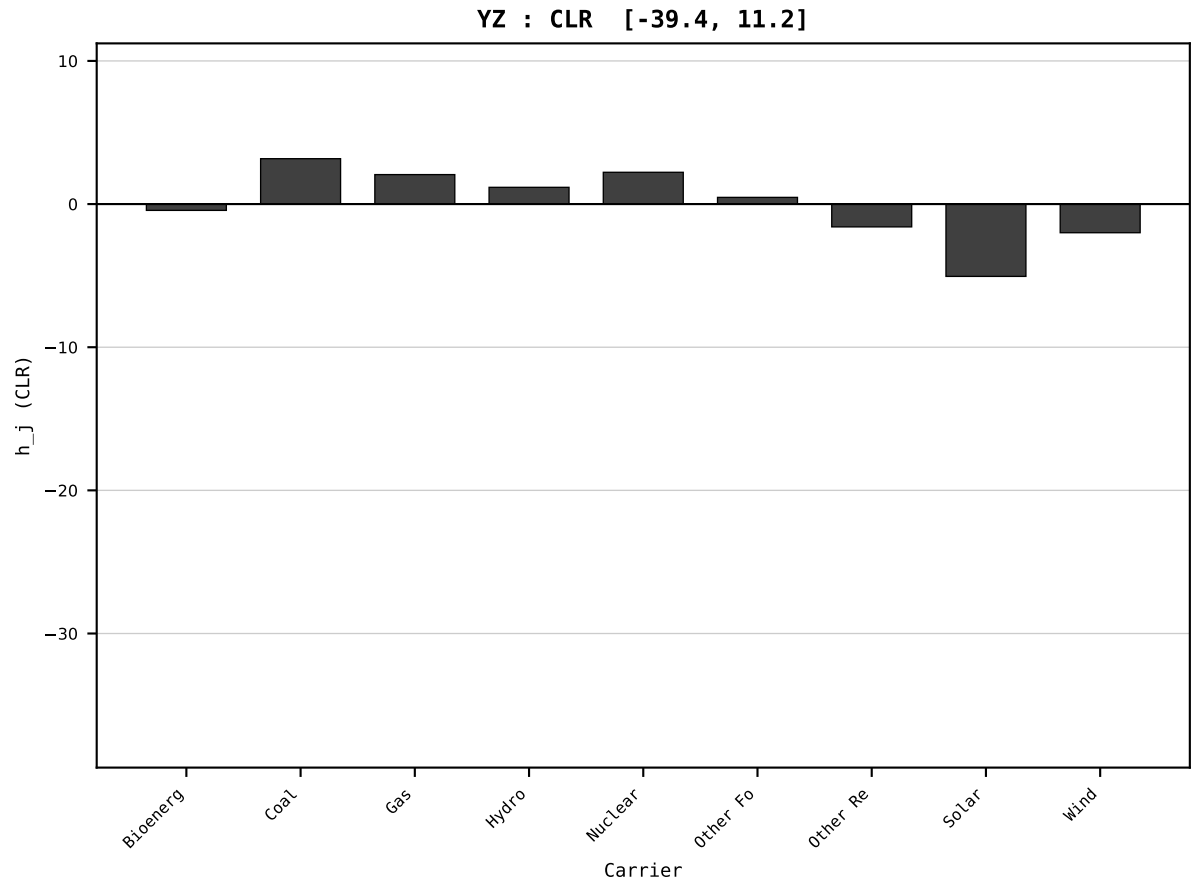
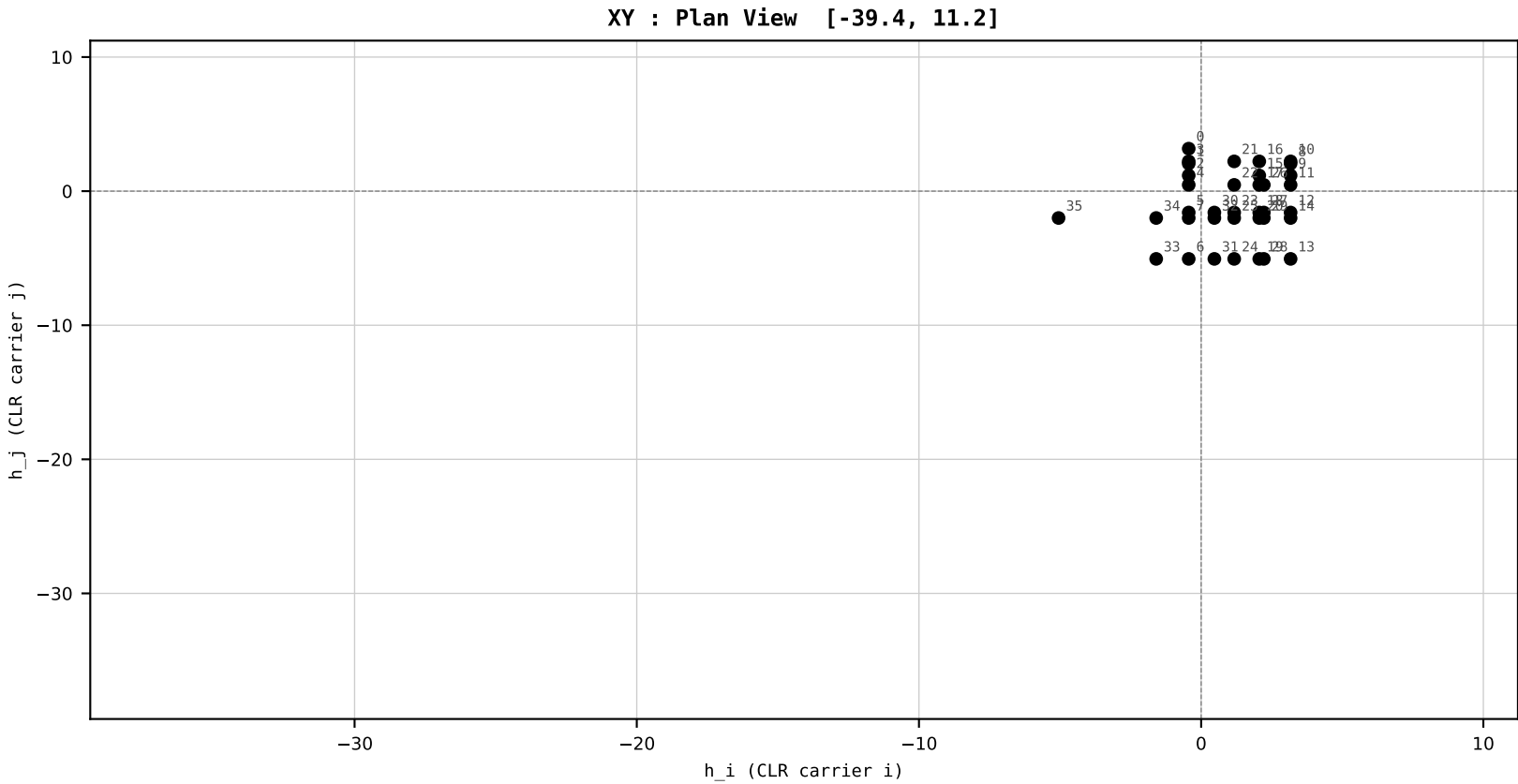
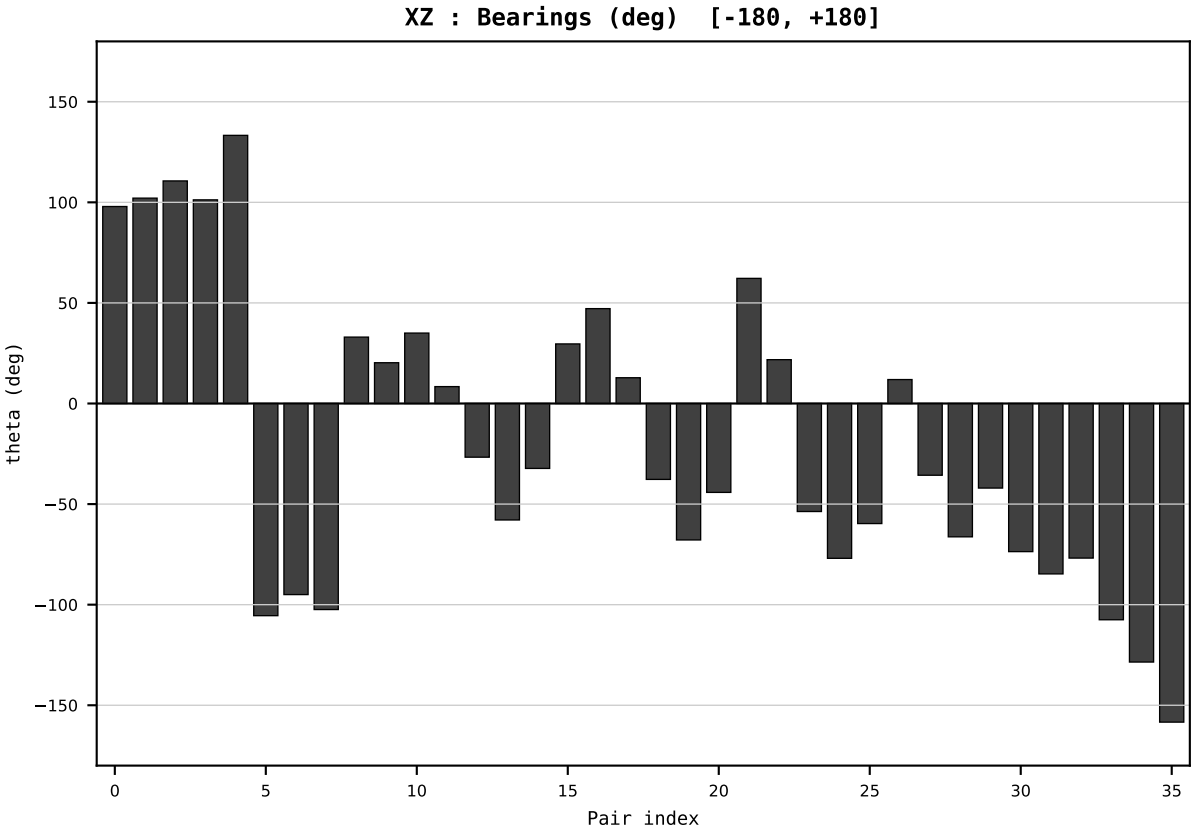
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2003
t : 2 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.379089
Ring = Hs-3
E_metric= 53.0775
kappa_HS= 3724.00
omega = 34.9884 deg
d_A = 31.925473
Helm = Solar
Helm d = +30.098741
DR = 8.223 HLR
DR ratio= 3724.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=3 | Year 2004 | D=9 N=25 pairs=36

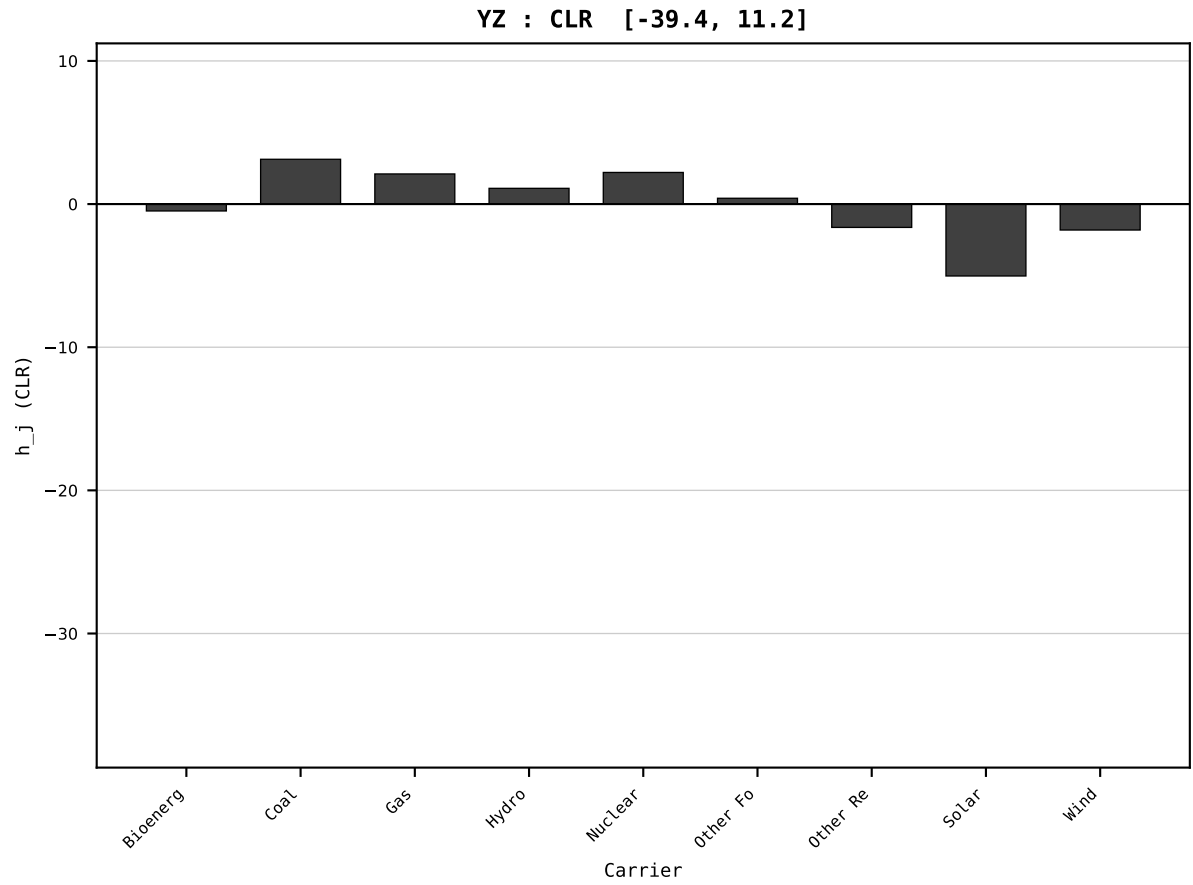
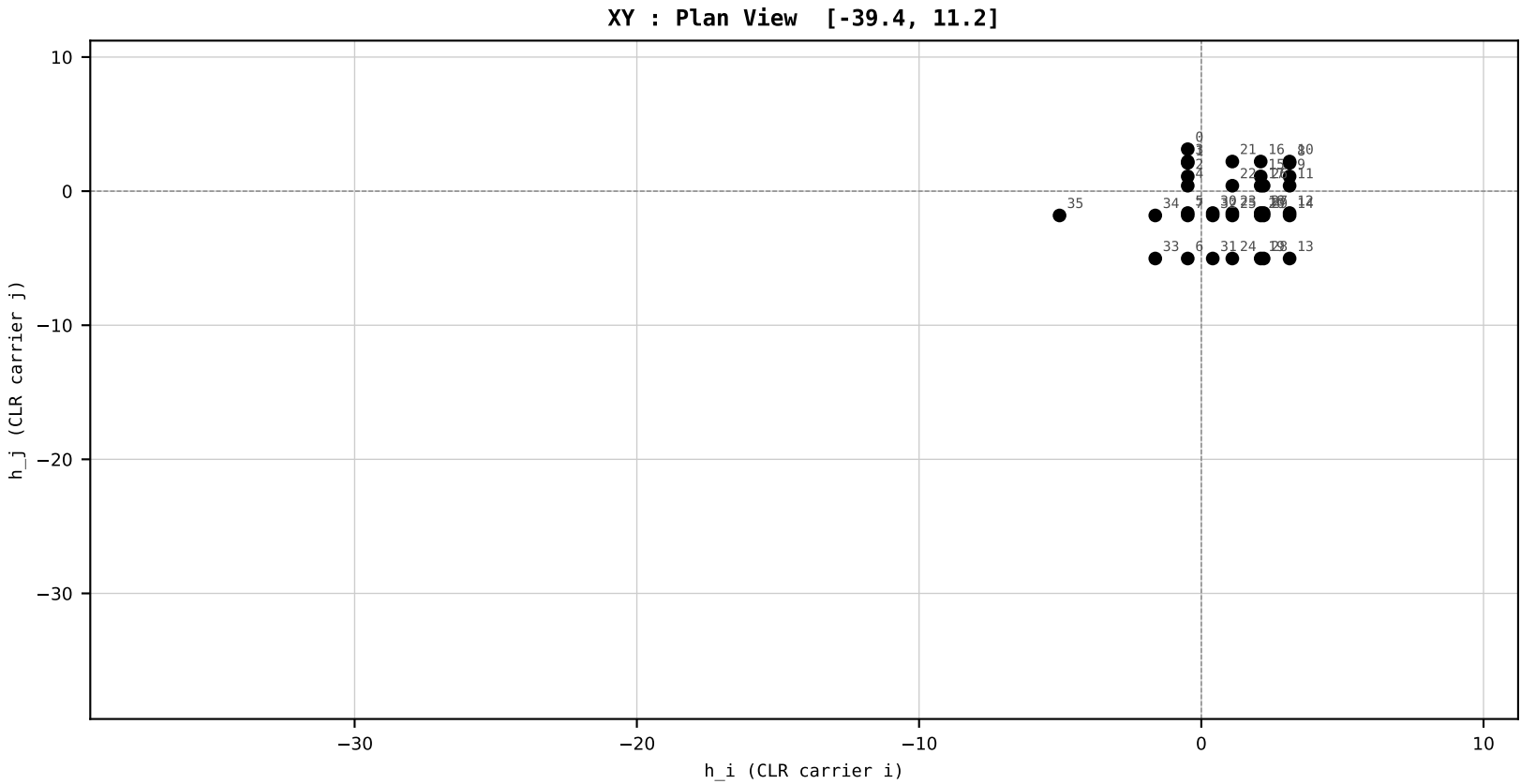
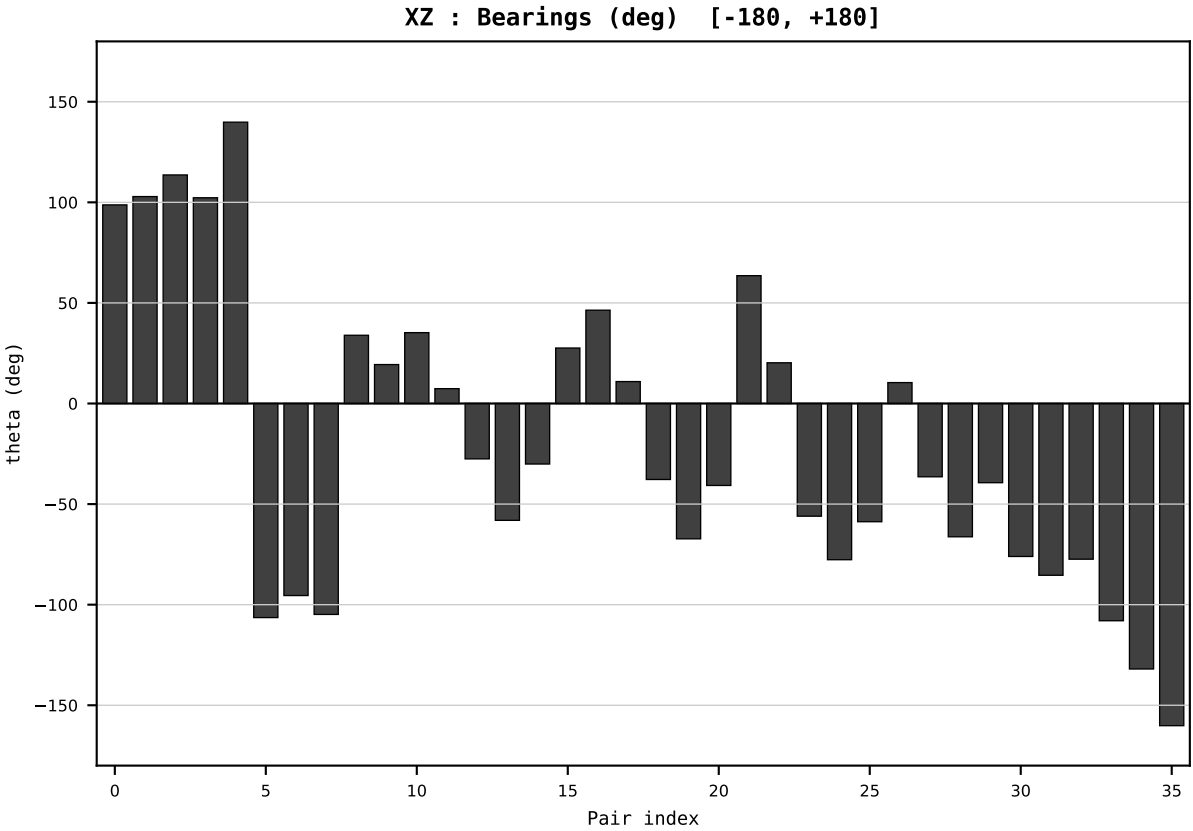
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2004
t : 3 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.376148
Ring = Hs-3
E_metric= 51.8804
kappa_HS= 3470.70
omega = 1.6887 deg
d_A = 0.228934
Helm = Wind
Helm d = +0.188988
DR = 8.152 HLR
DR ratio= 3470.7x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=4 | Year 2005 | D=9 N=25 pairs=36

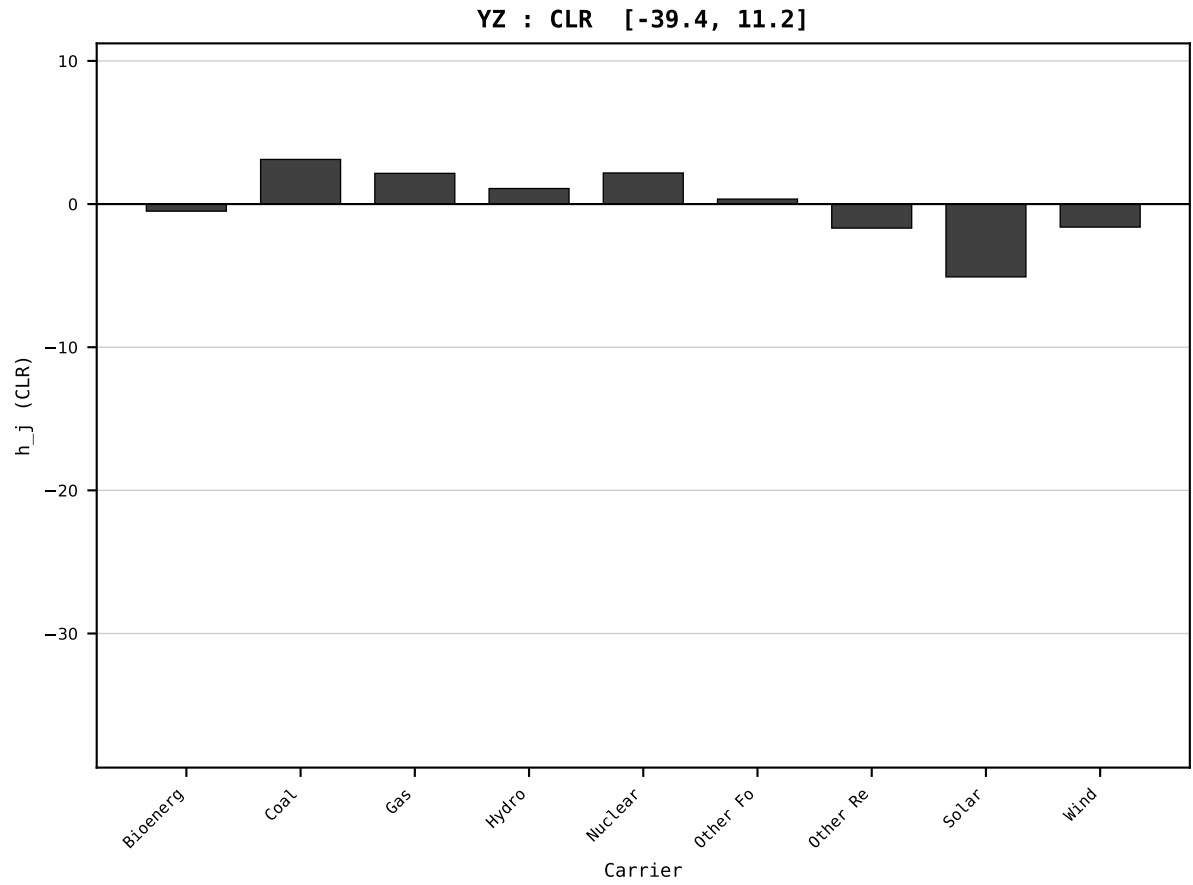
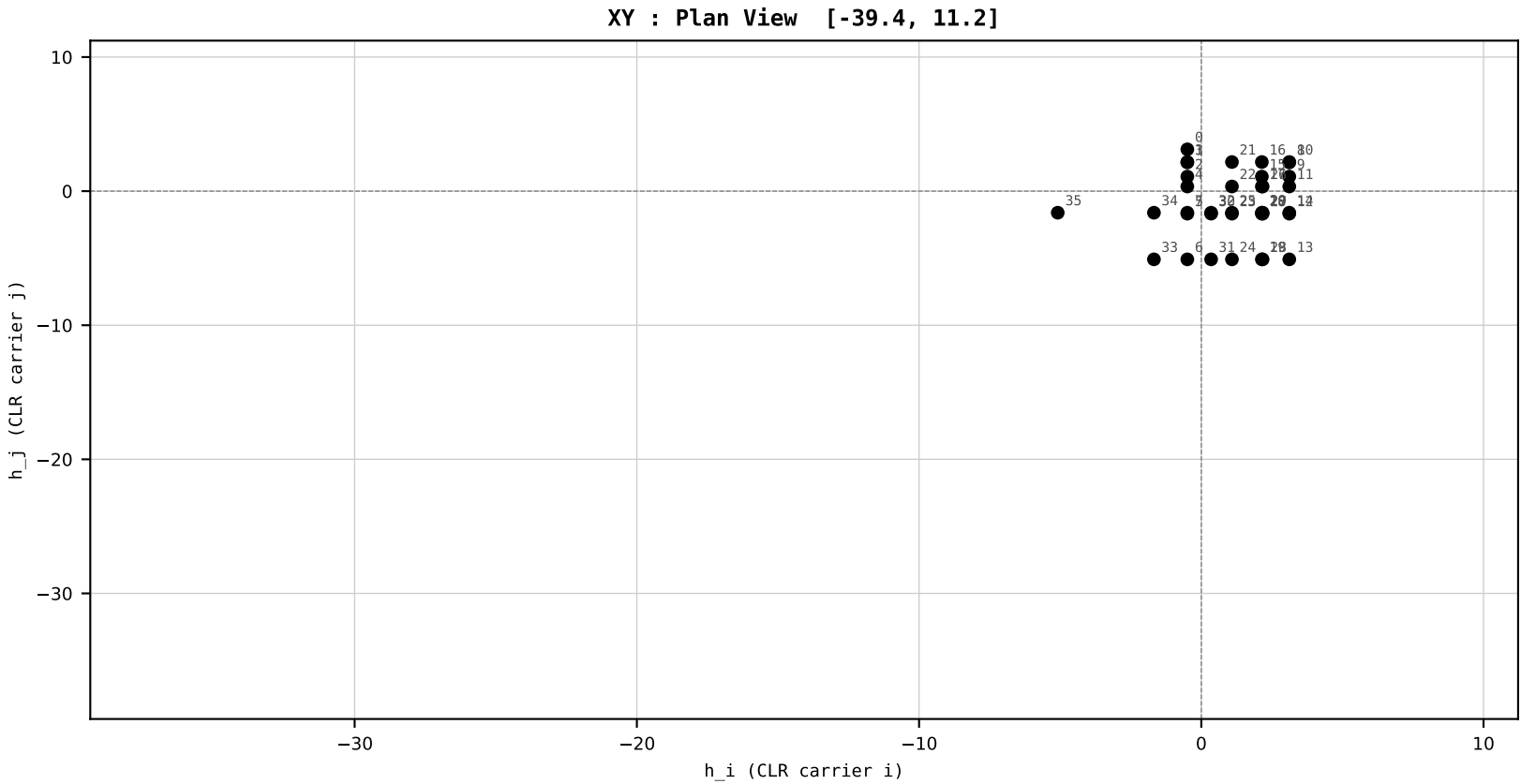
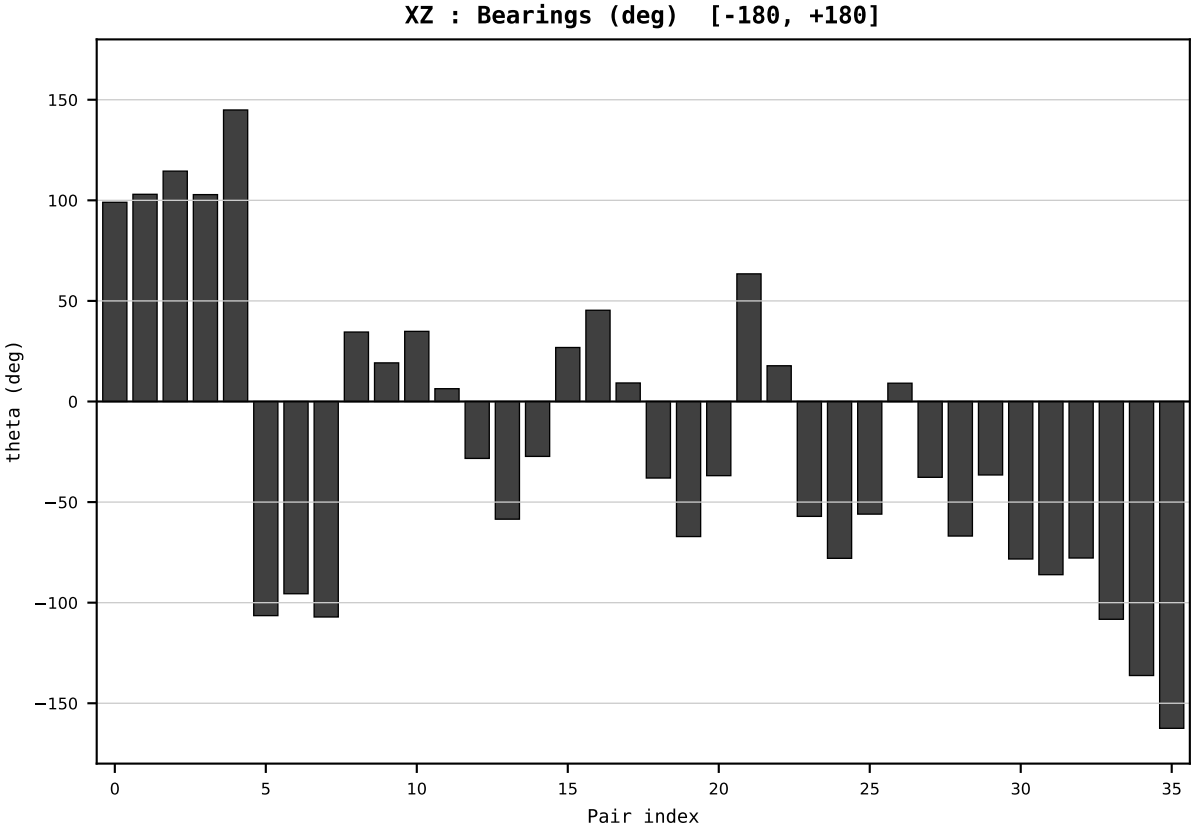
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2005
t : 4 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.375606
Ring = Hs-3
E_metric= 51.8713
kappa_HS= 3659.76
omega = 1.8643 deg
d_A = 0.234341
Helm = Wind
Helm d = +0.204228
DR = 8.205 HLR
DR ratio= 3659.8x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=5 | Year 2006 | D=9 N=25 pairs=36

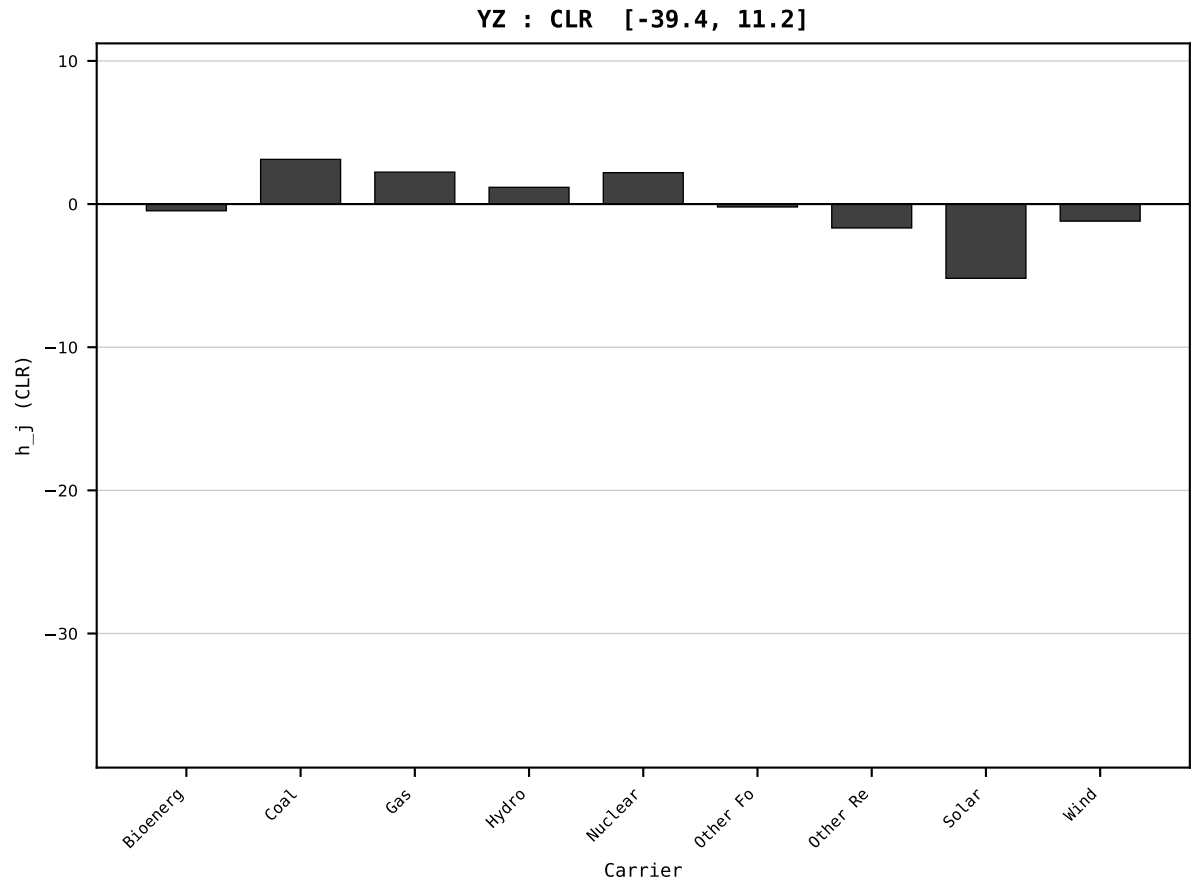
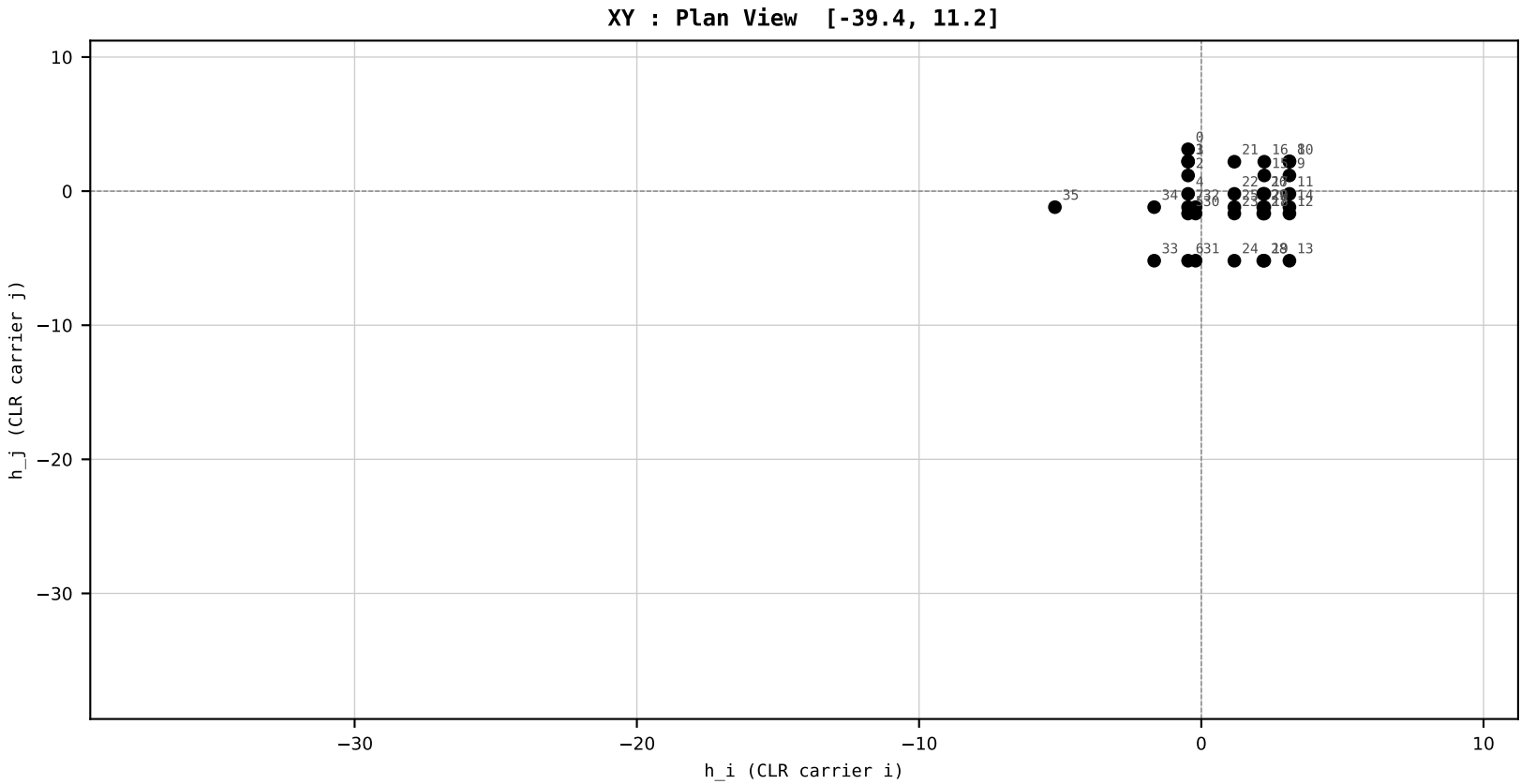
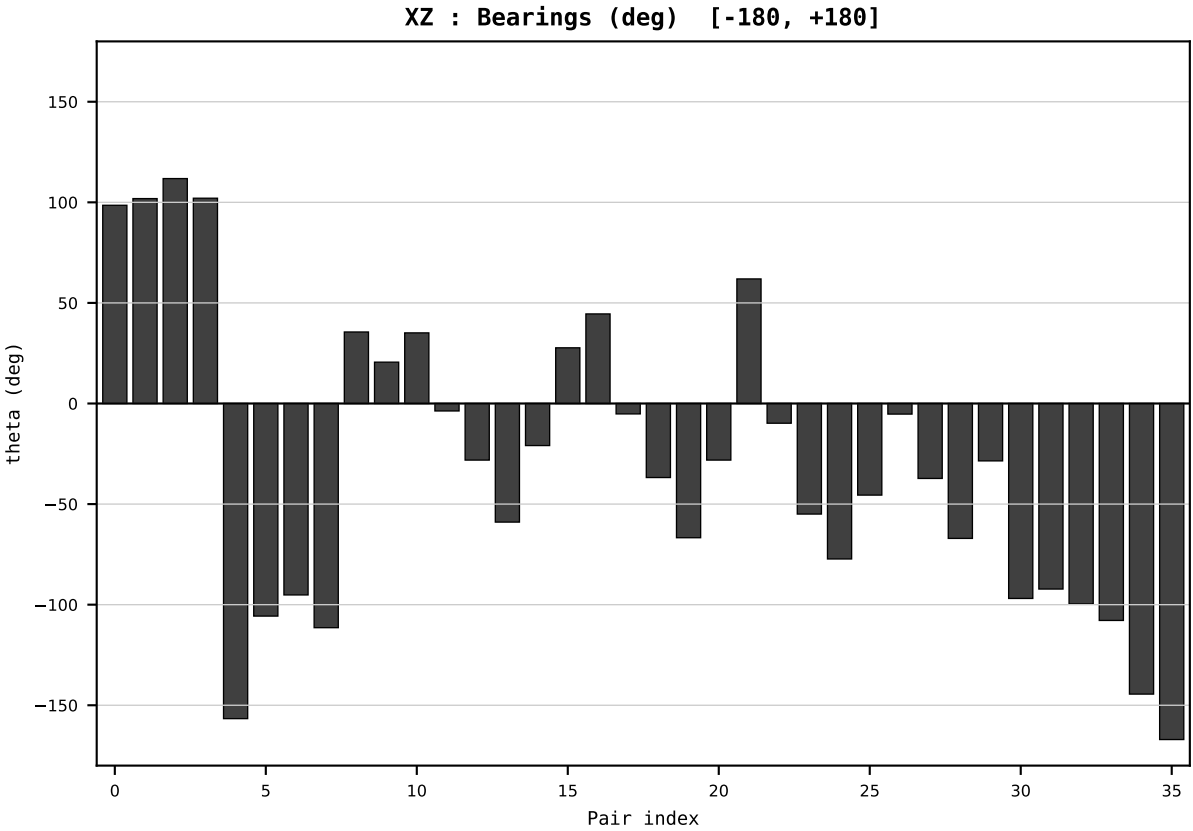
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2006
t : 5 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.379864
Ring = Hs-3
E_metric= 52.2936
kappa_HS= 4062.29
omega = 5.6182 deg
d_A = 0.707976
Helm = Other Fossil
Helm d = -0.550093
DR = 8.310 HLR
DR ratio= 4062.3x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=6 | Year 2007 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

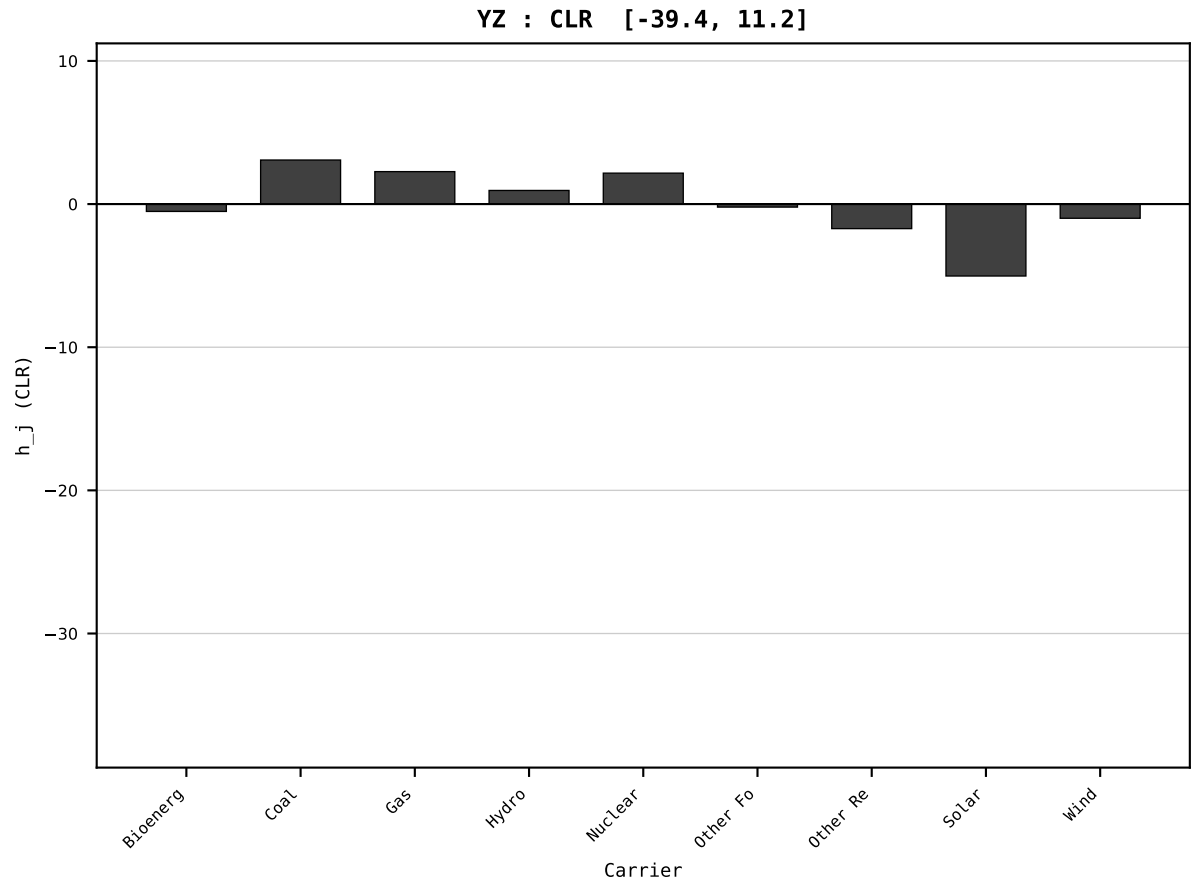
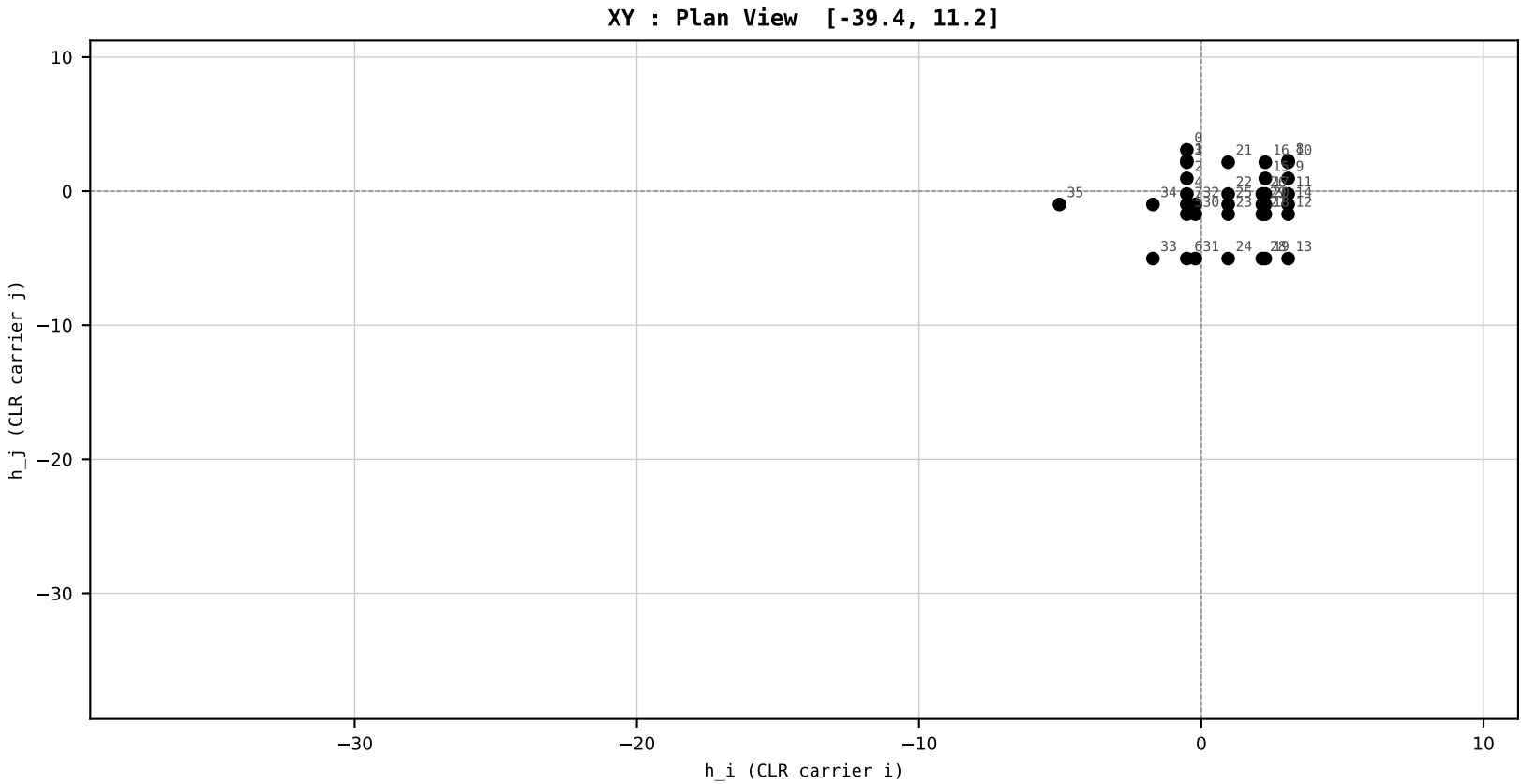
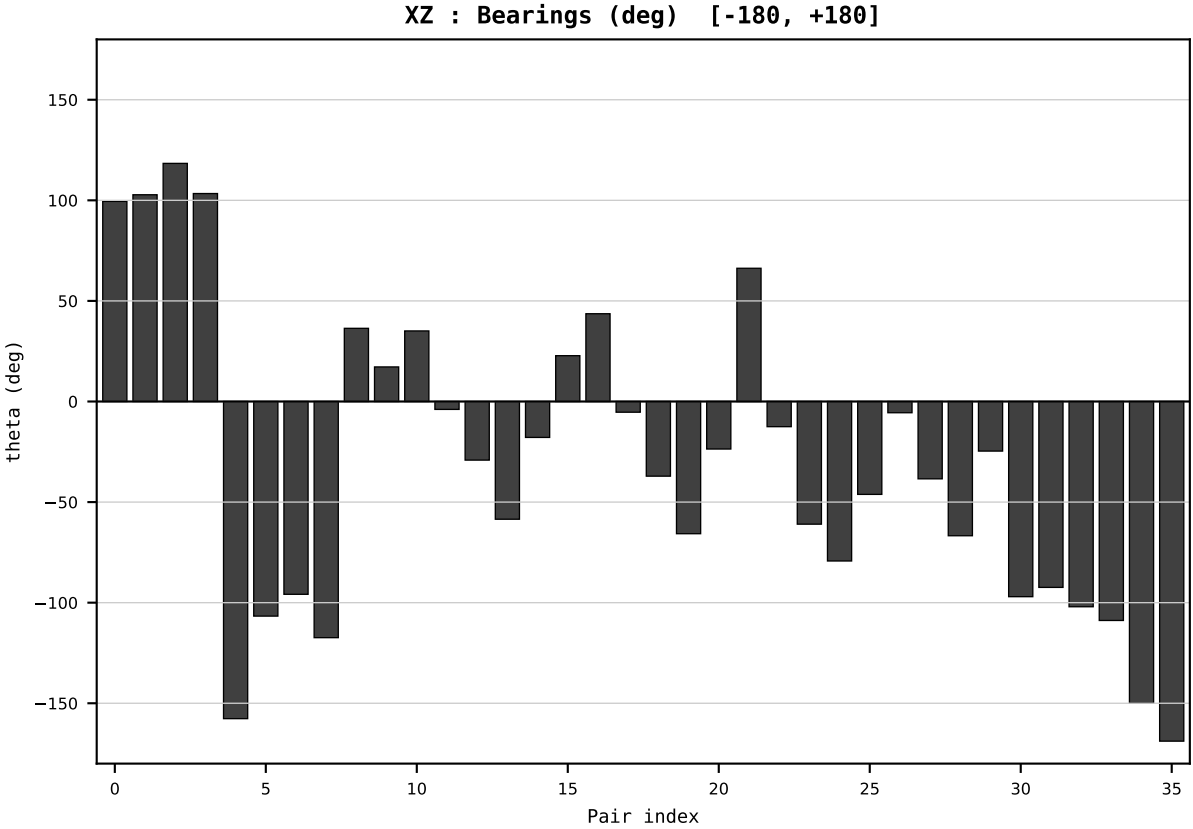
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2007
t : 6 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.381244
Ring = Hs-3
E_metric= 49.6813
kappa_HS= 3305.62
omega = 2.4037 deg
d_A = 0.350949
Helm = Hydro
Helm d = -0.219022
DR = 8.103 HLR
DR ratio= 3305.6x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=7 | Year 2008 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

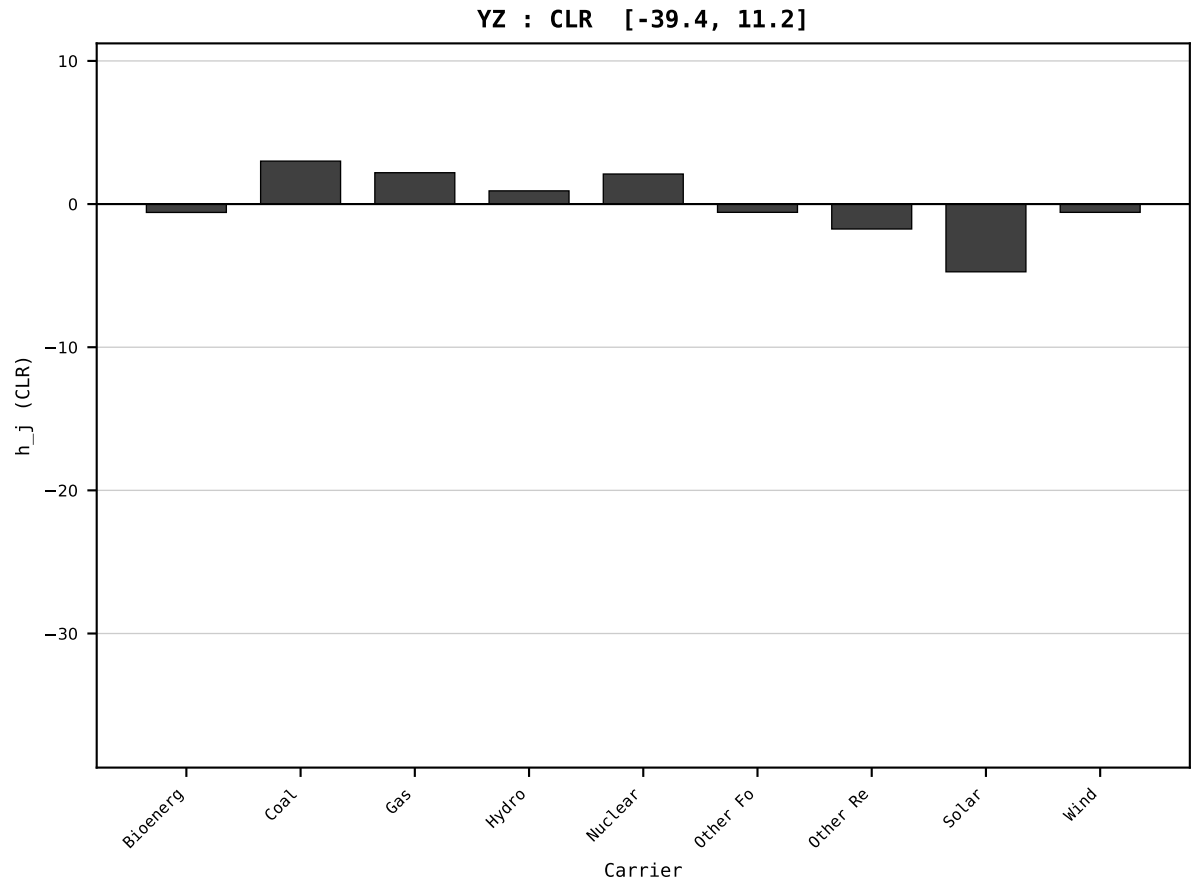
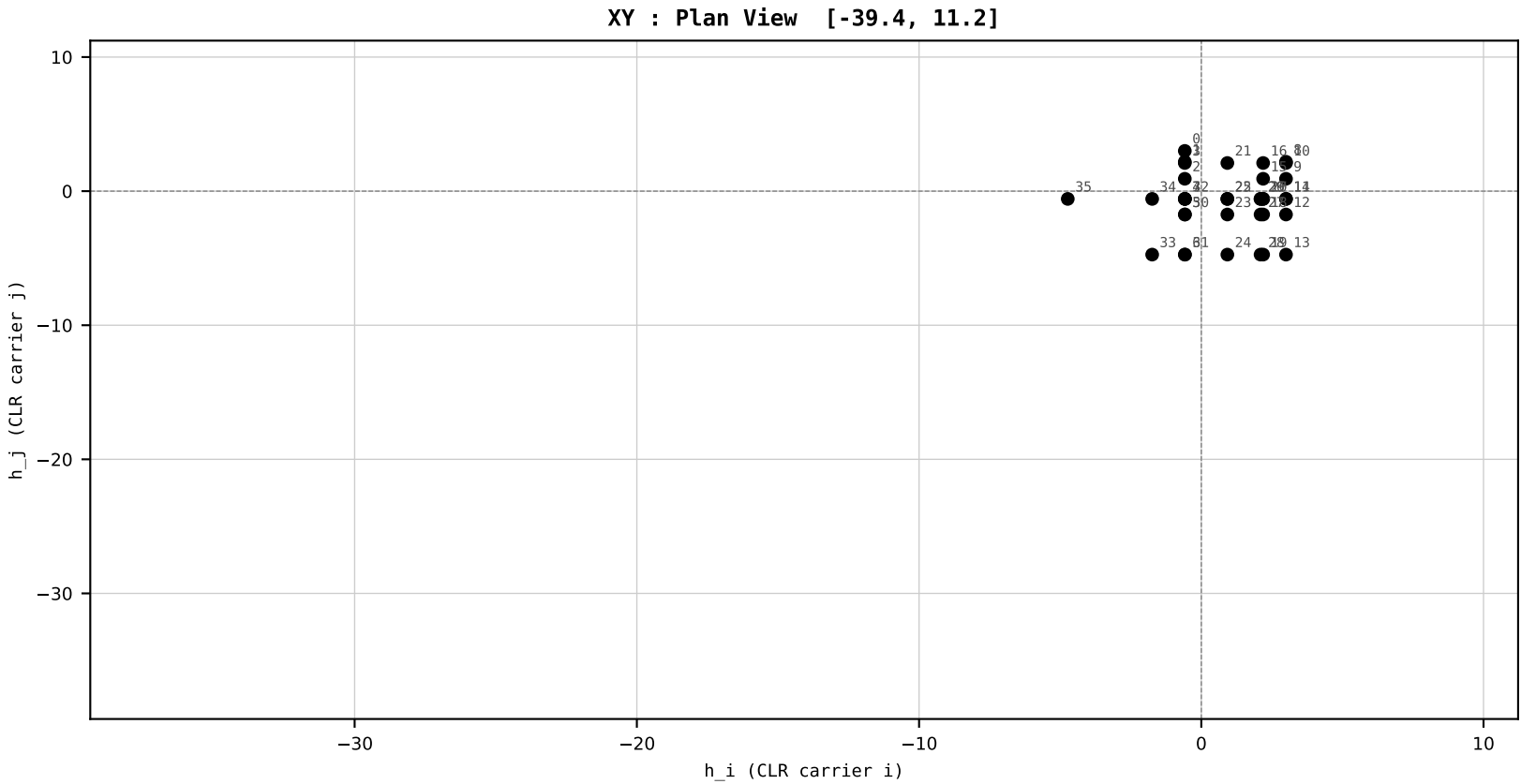
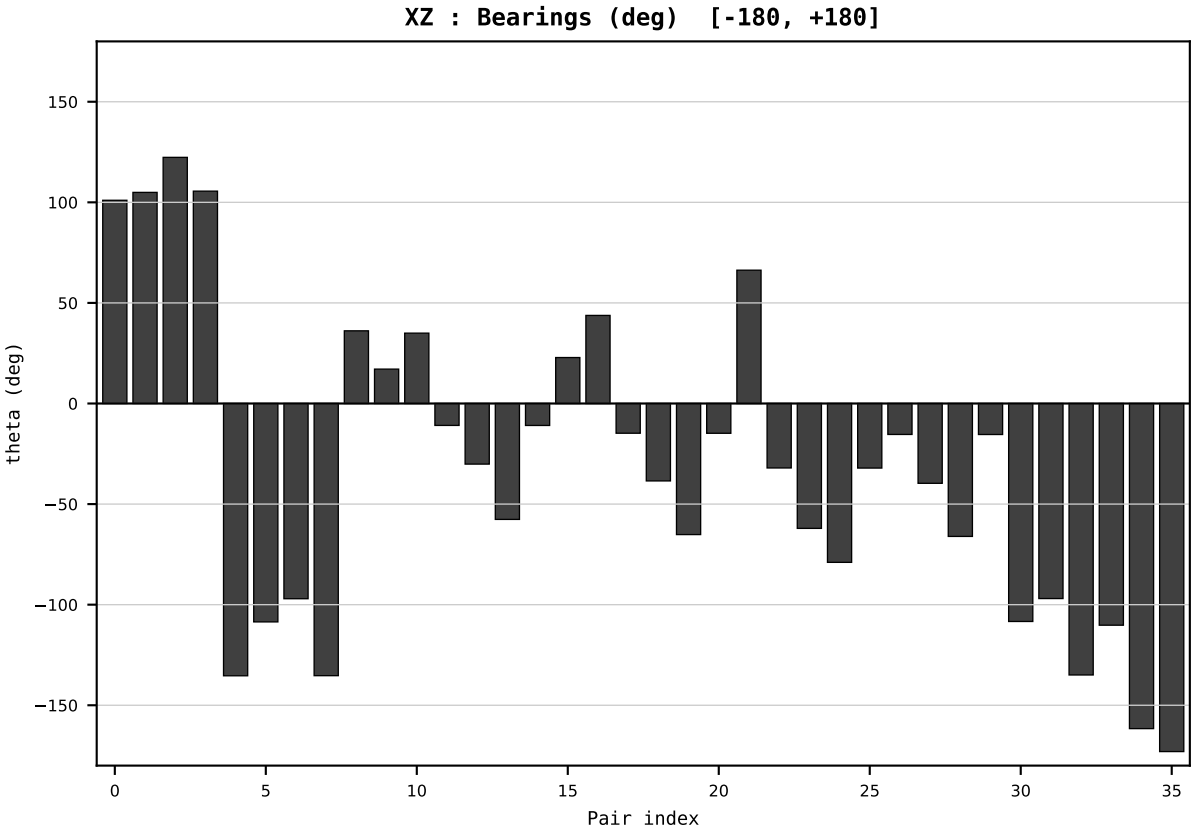
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2008
t : 7 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.376532
Ring = Hs-3
E_metric= 45.5003
kappa_HS= 2282.54
omega = 4.7104 deg
d_A = 0.642683
Helm = Wind
Helm d = +0.412825
DR = 7.733 HLR
DR ratio= 2282.5x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=8 | Year 2009 | D=9 N=25 pairs=36

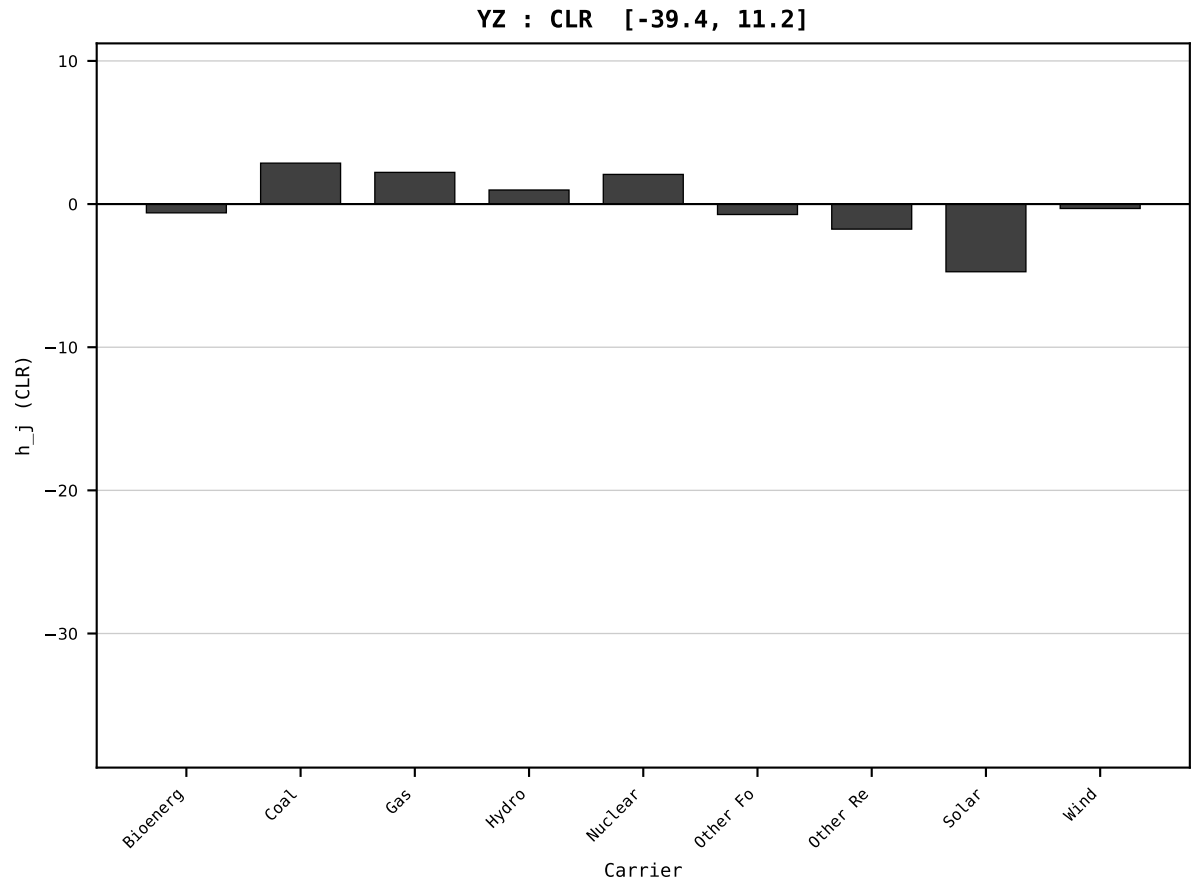
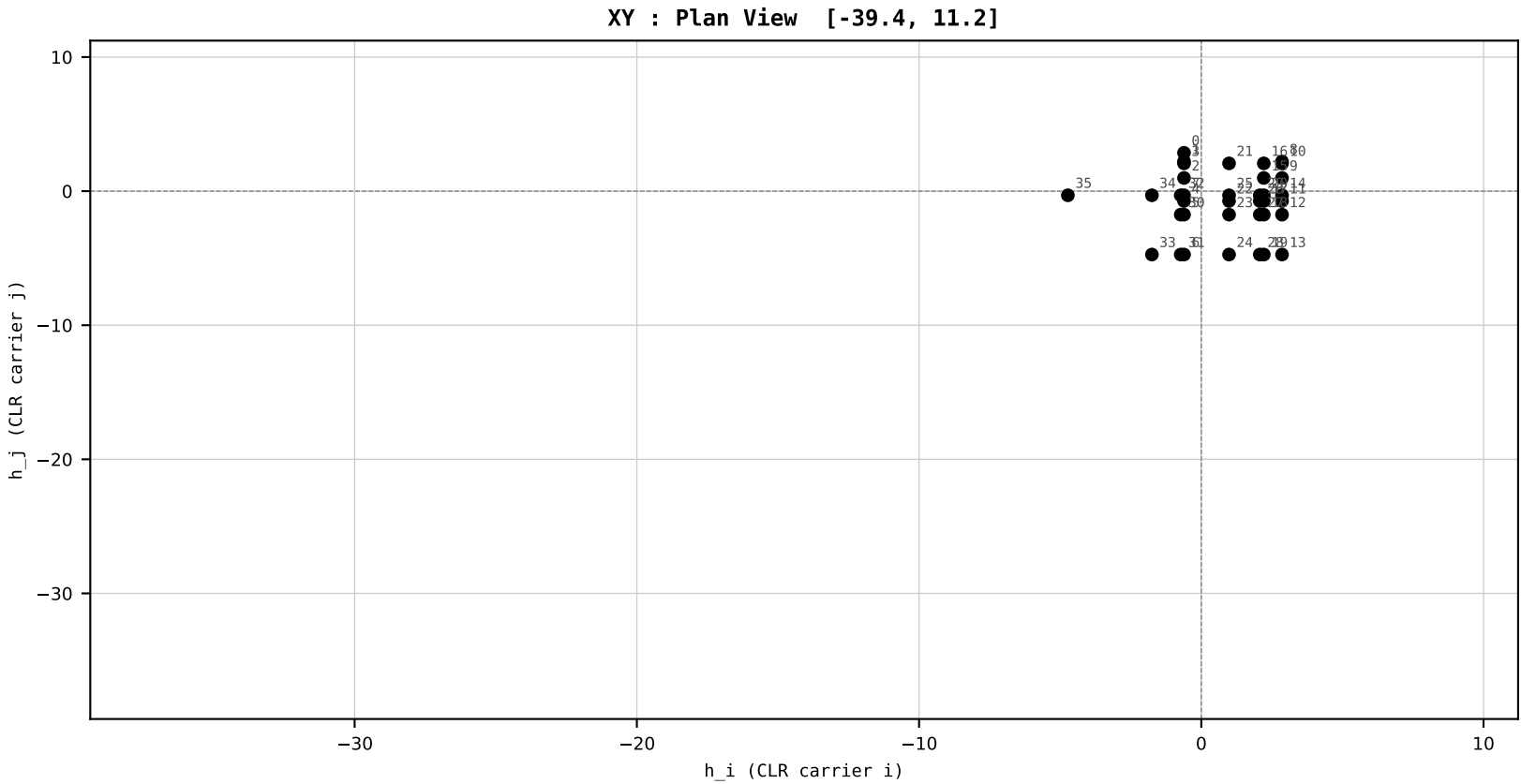
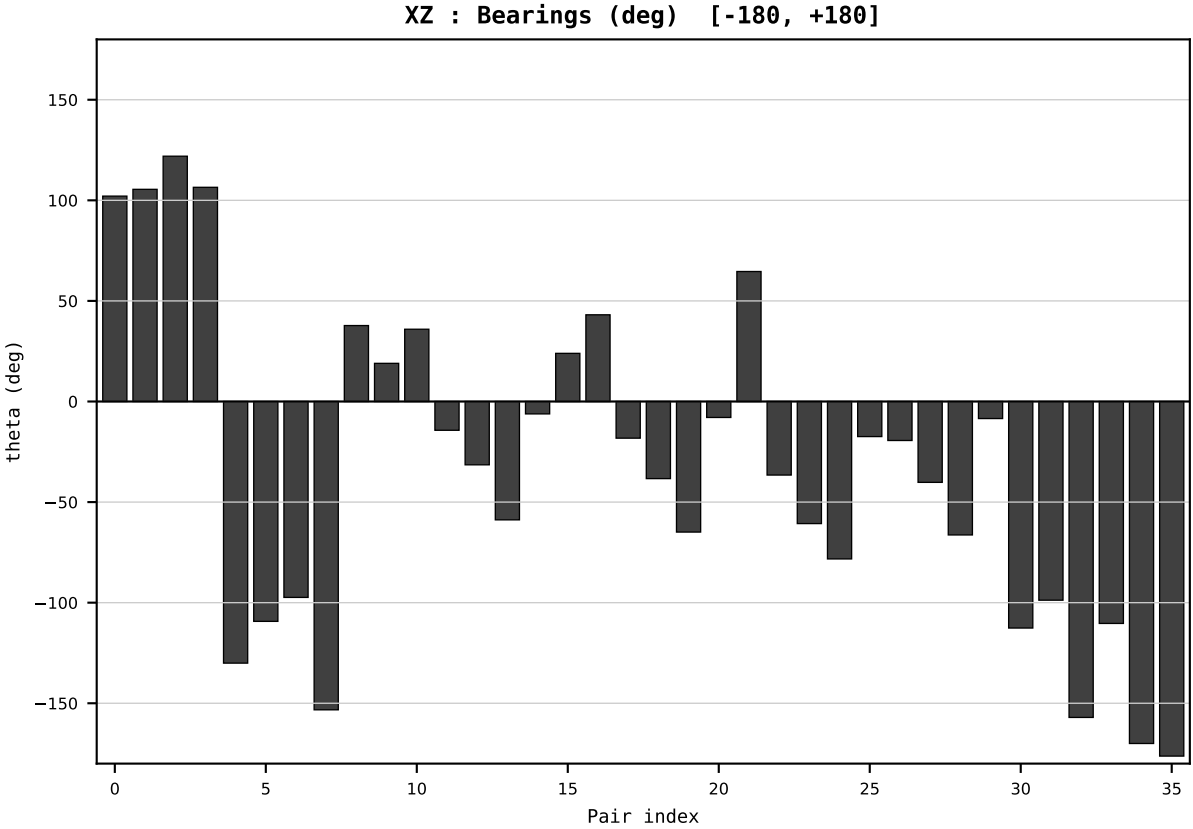
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2009
t : 8 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.353477
Ring = Hs-3
E_metric= 44.7578
kappa_HS= 1972.93
omega = 2.9378 deg
d_A = 0.348811
Helm = Wind
Helm d = +0.270064
DR = 7.587 HLR
DR ratio= 1972.9x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=9 | Year 2010 | D=9 N=25 pairs=36

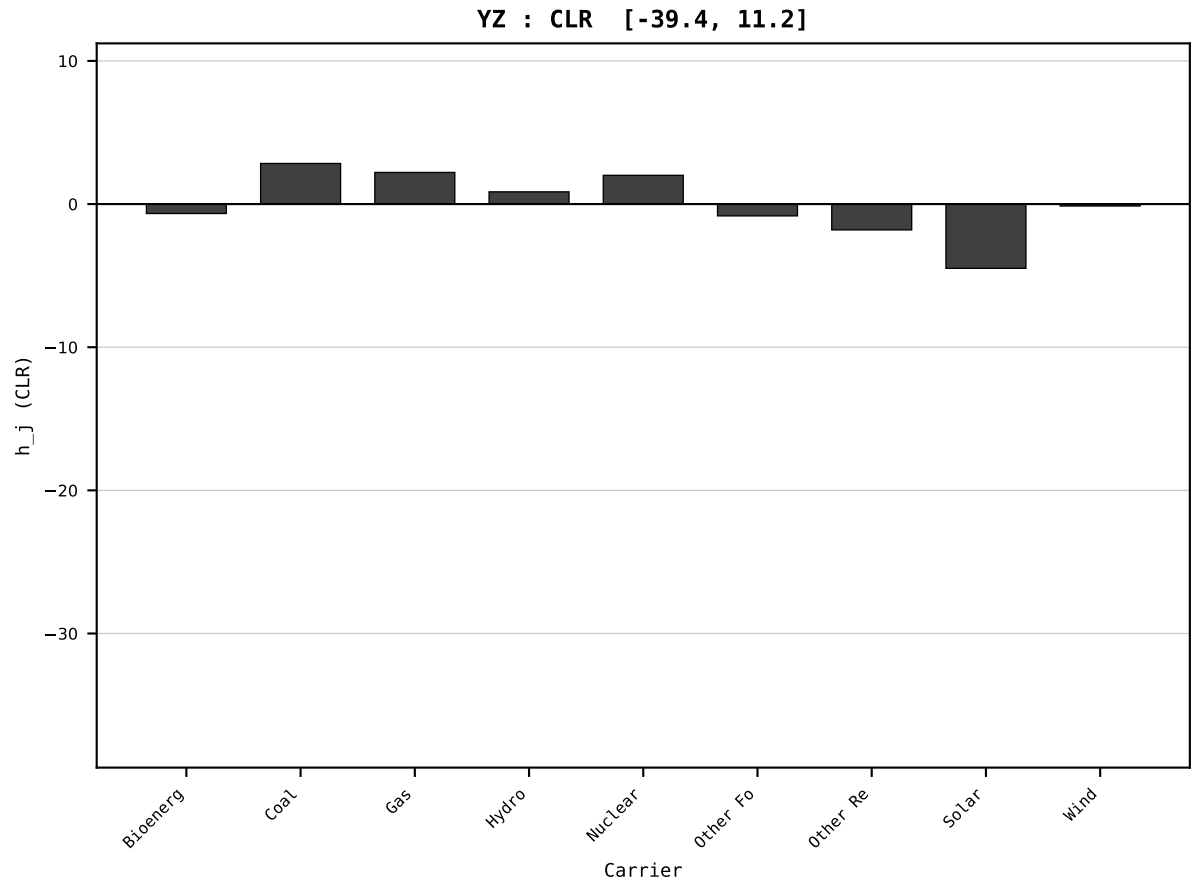
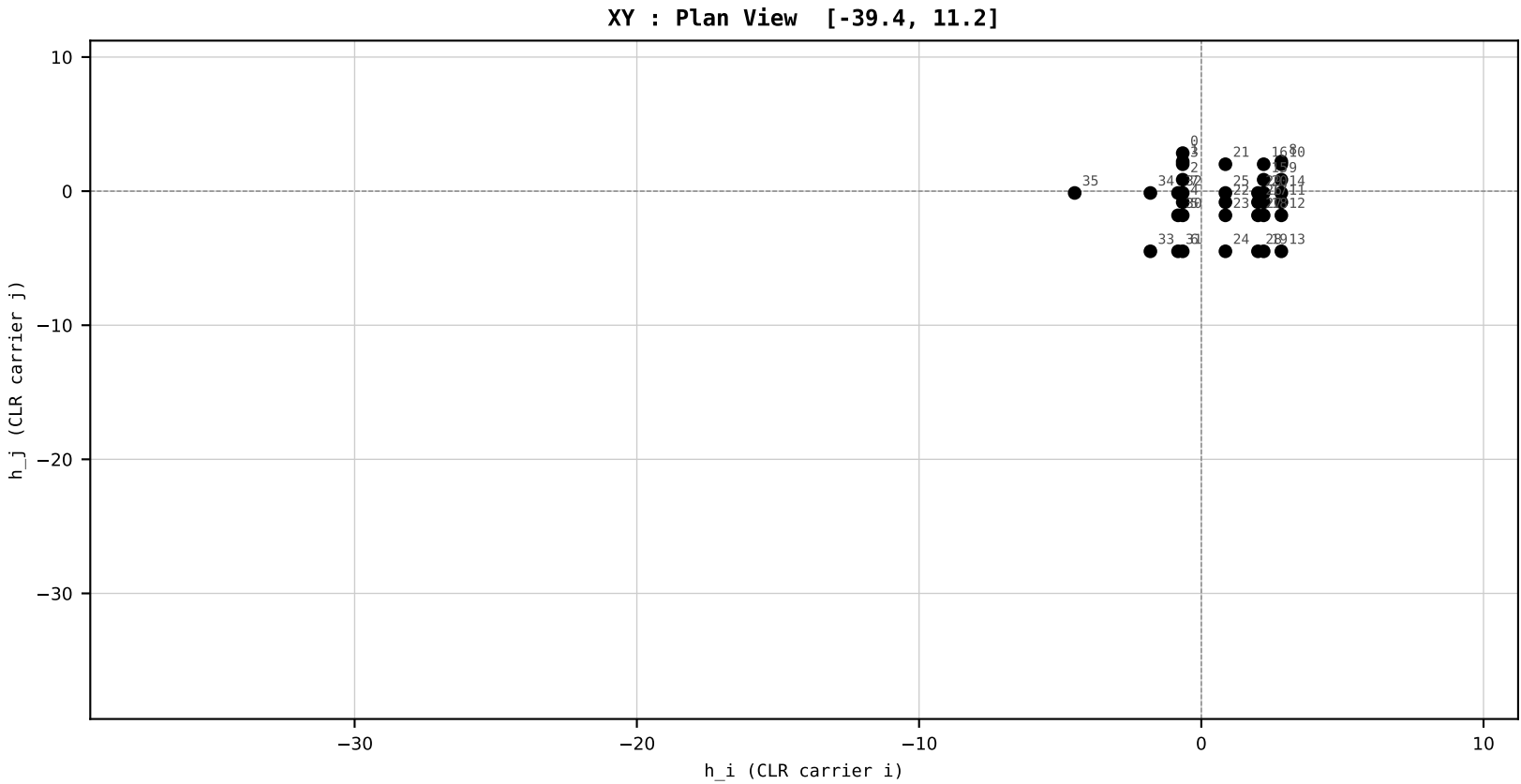
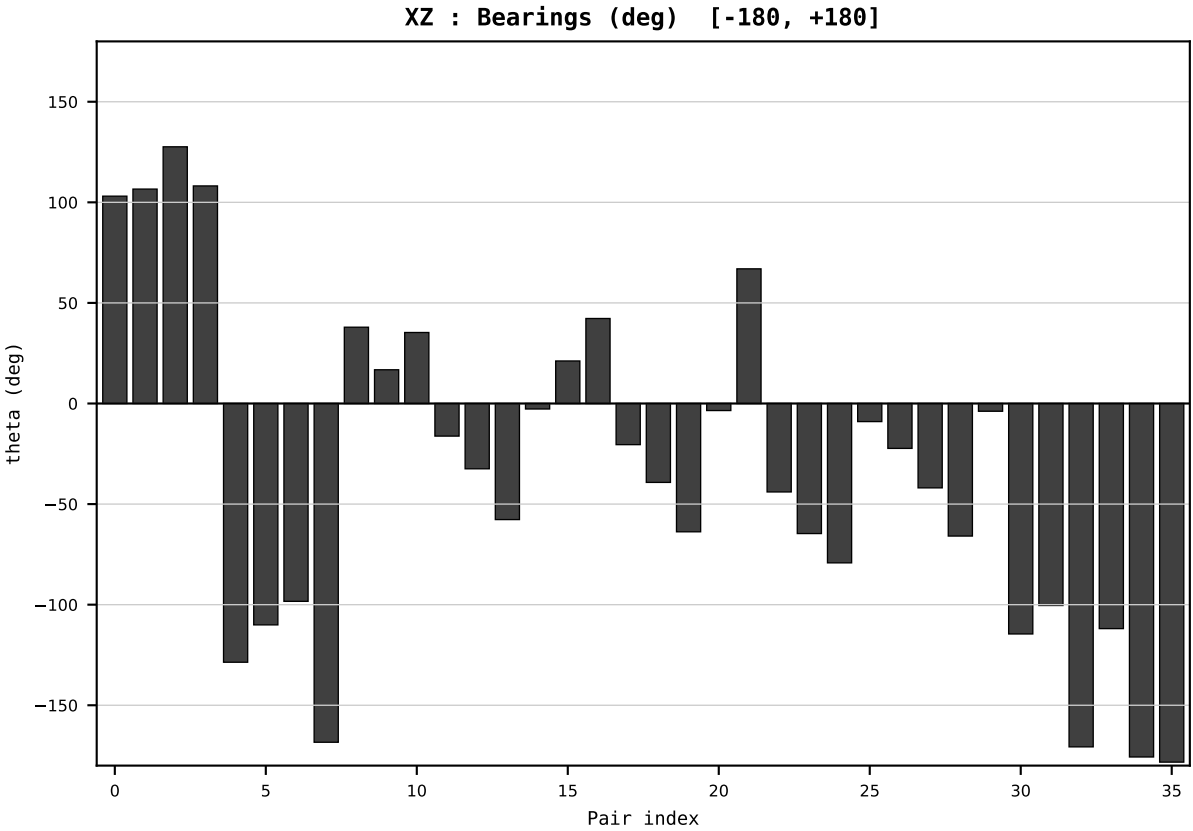
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2010
t : 9 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.355235
Ring = Hs-3
E_metric= 42.2037
kappa_HS= 1514.16
omega = 2.5467 deg
d_A = 0.351238
Helm = Solar
Helm d = +0.241100
DR = 7.323 HLR
DR ratio= 1514.2x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=10 | Year 2011 | D=9 N=25 pairs=36

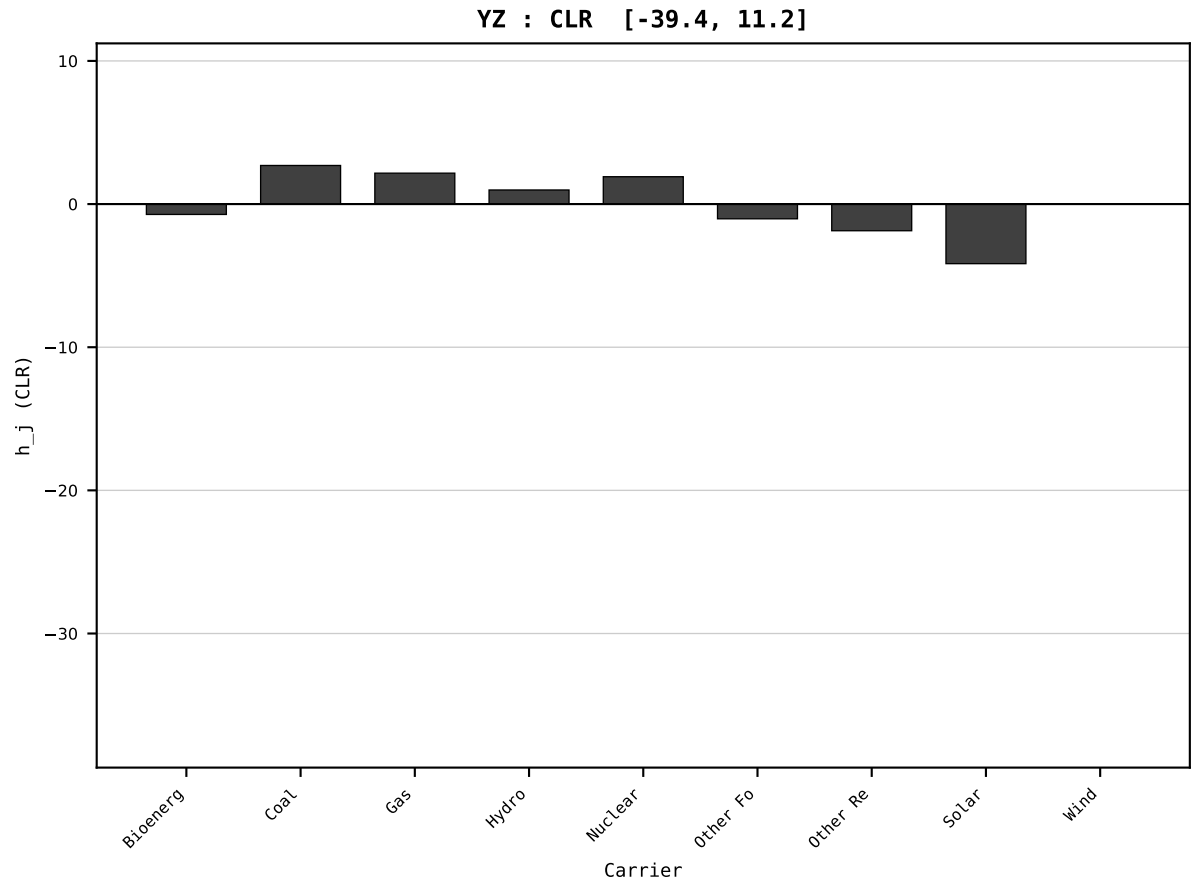
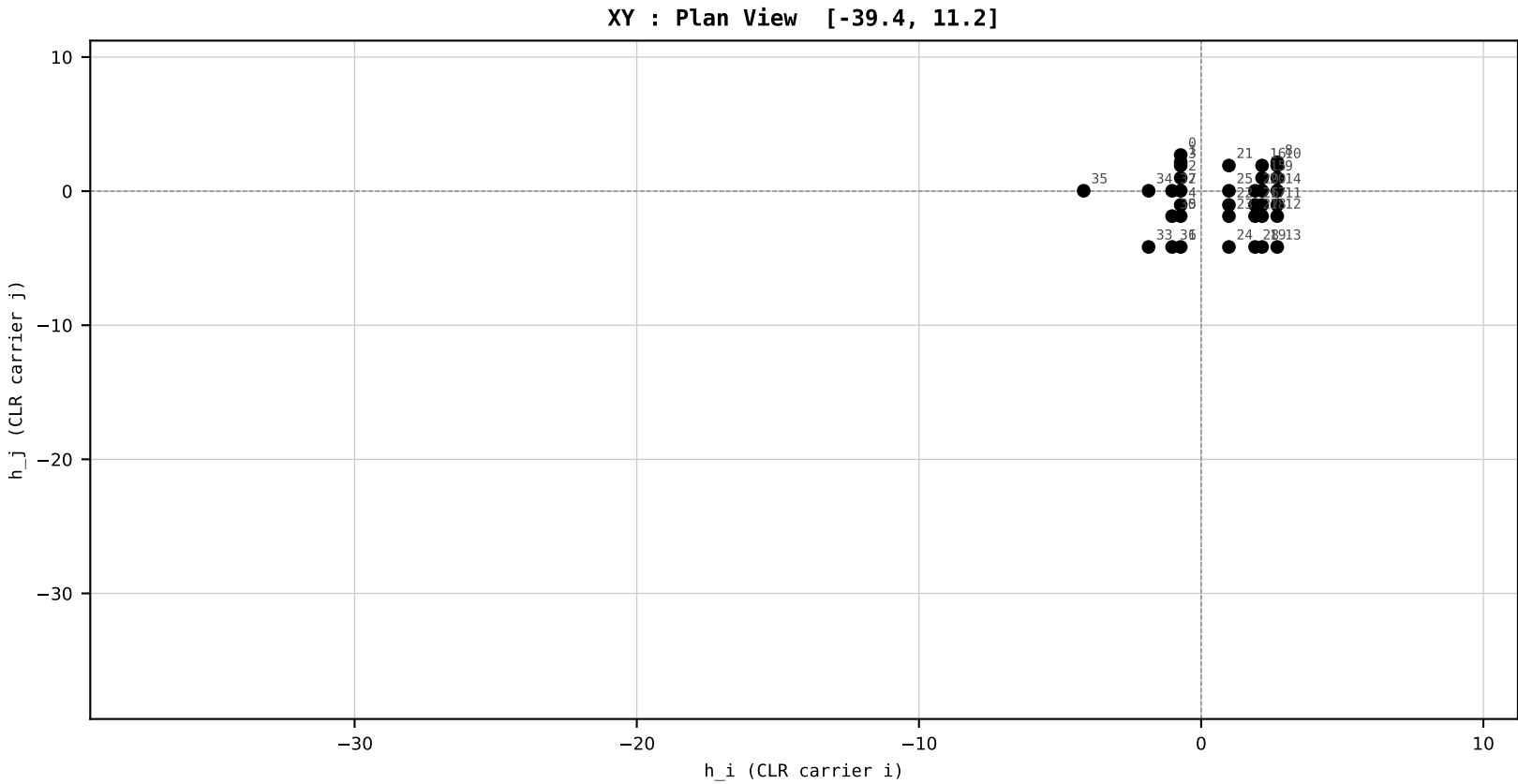
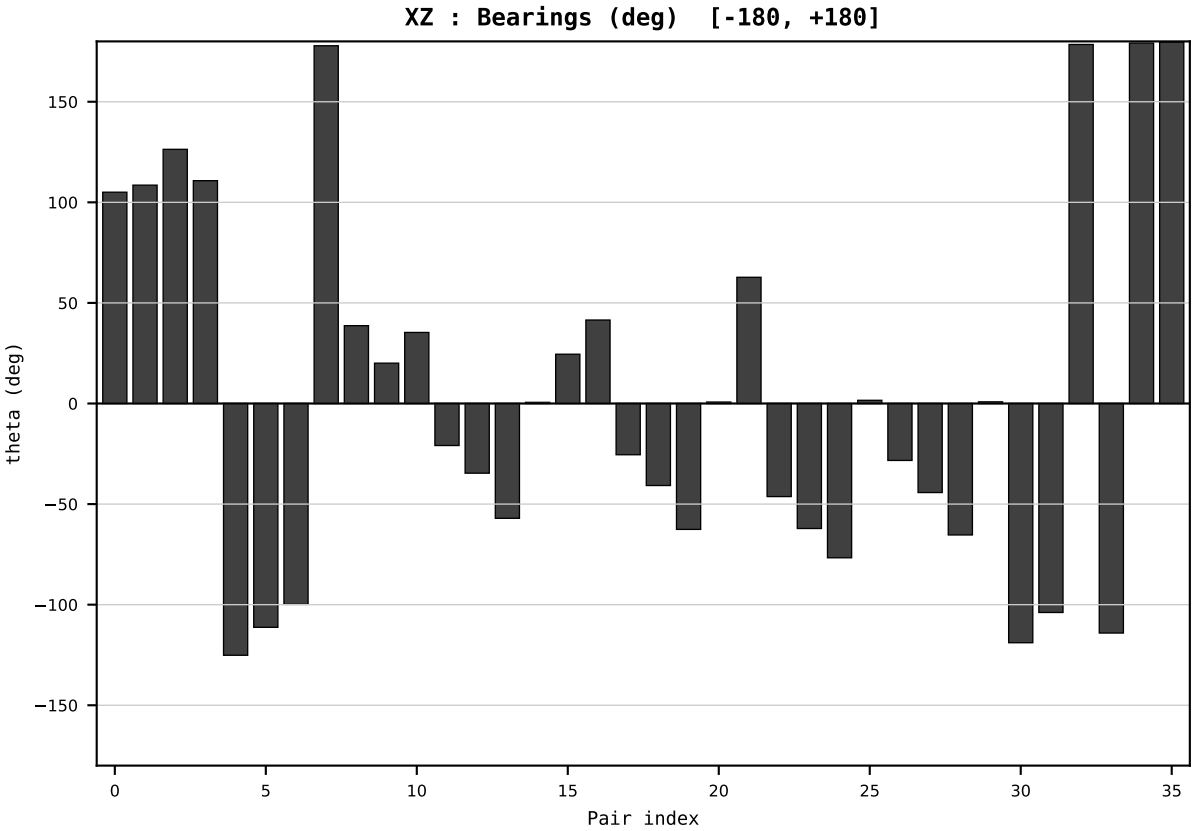
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2011
t : 10 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.334969
Ring = Hs-3
E_metric= 38.9355
kappa_HS= 952.43
omega = 3.6391 deg
d_A = 0.478871
Helm = Solar
Helm d = +0.323844
DR = 6.859 HLR
DR ratio= 952.4x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=11 | Year 2012 | D=9 N=25 pairs=36

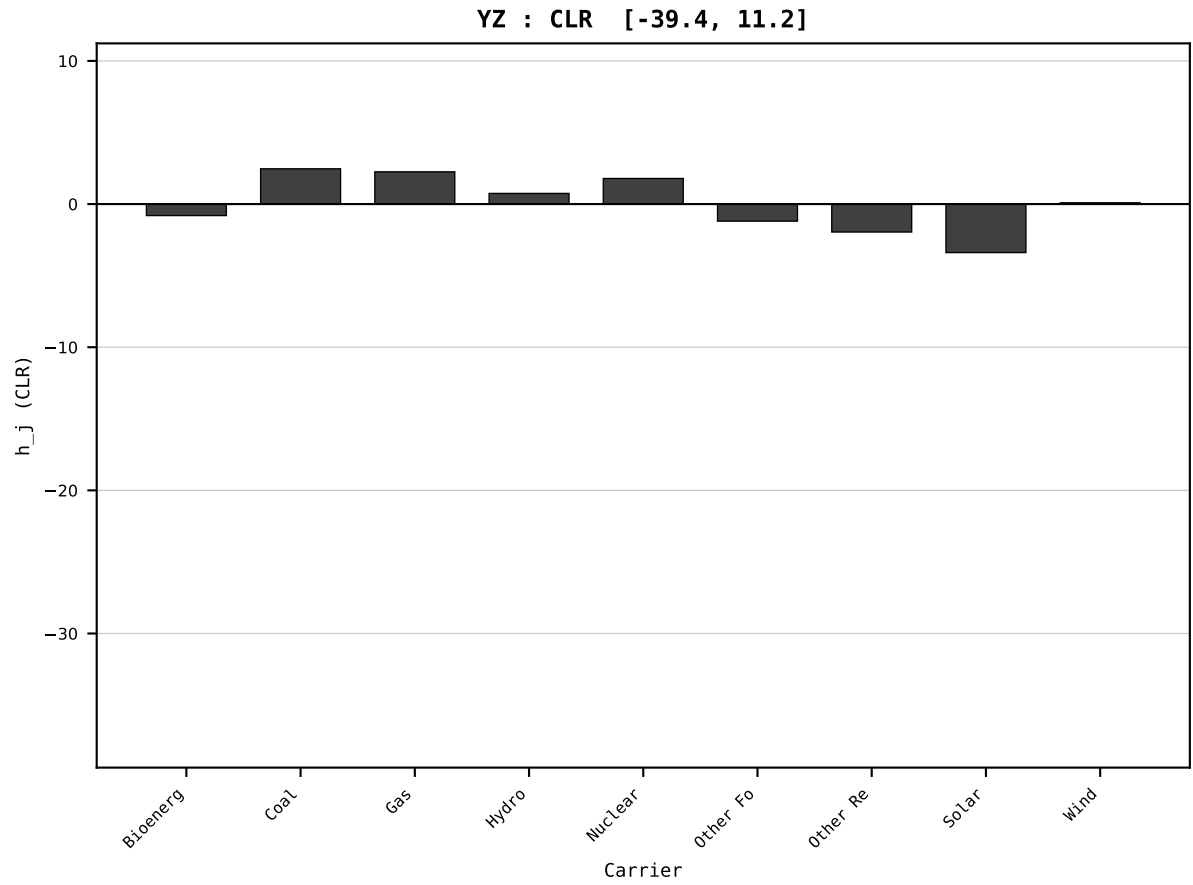
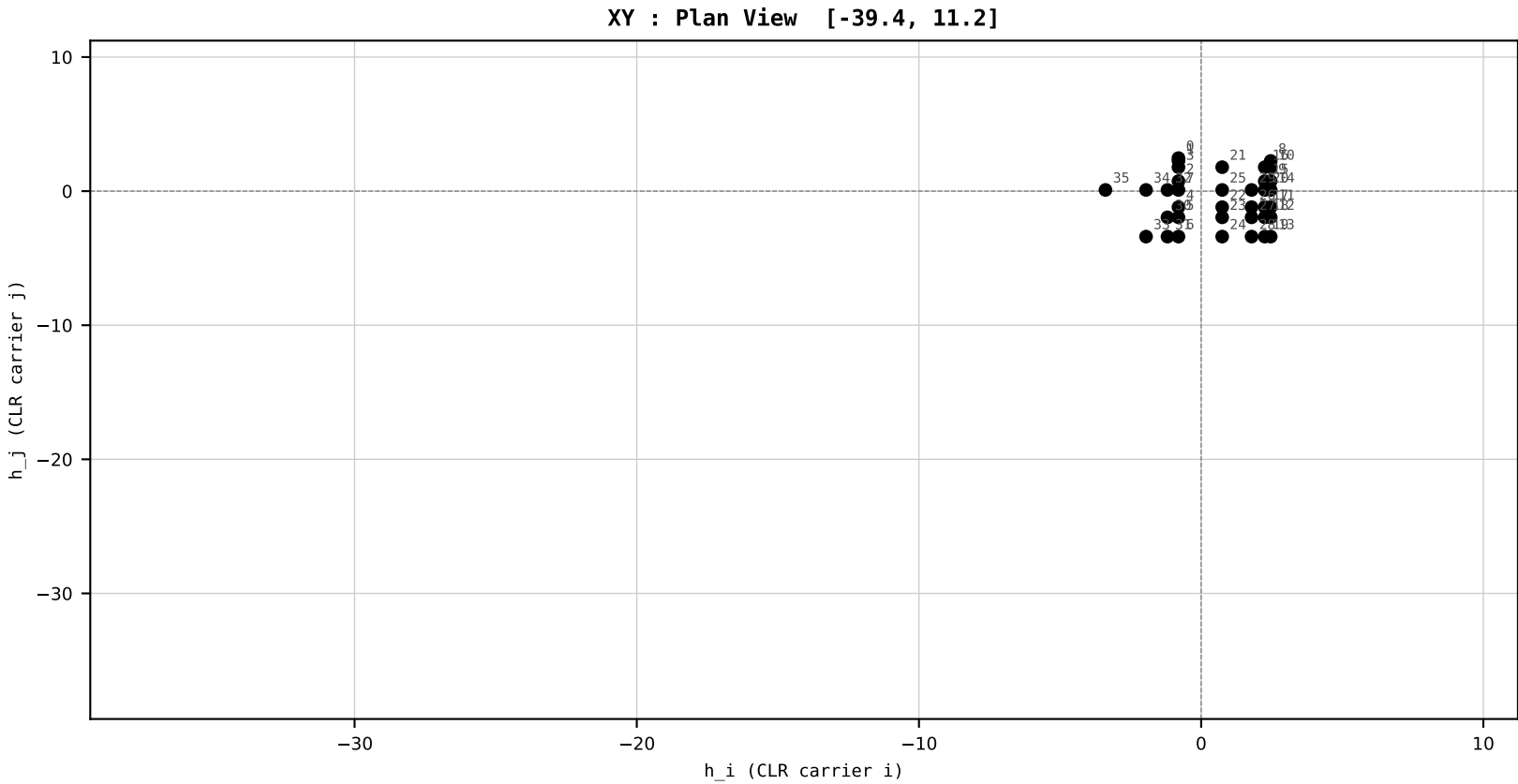
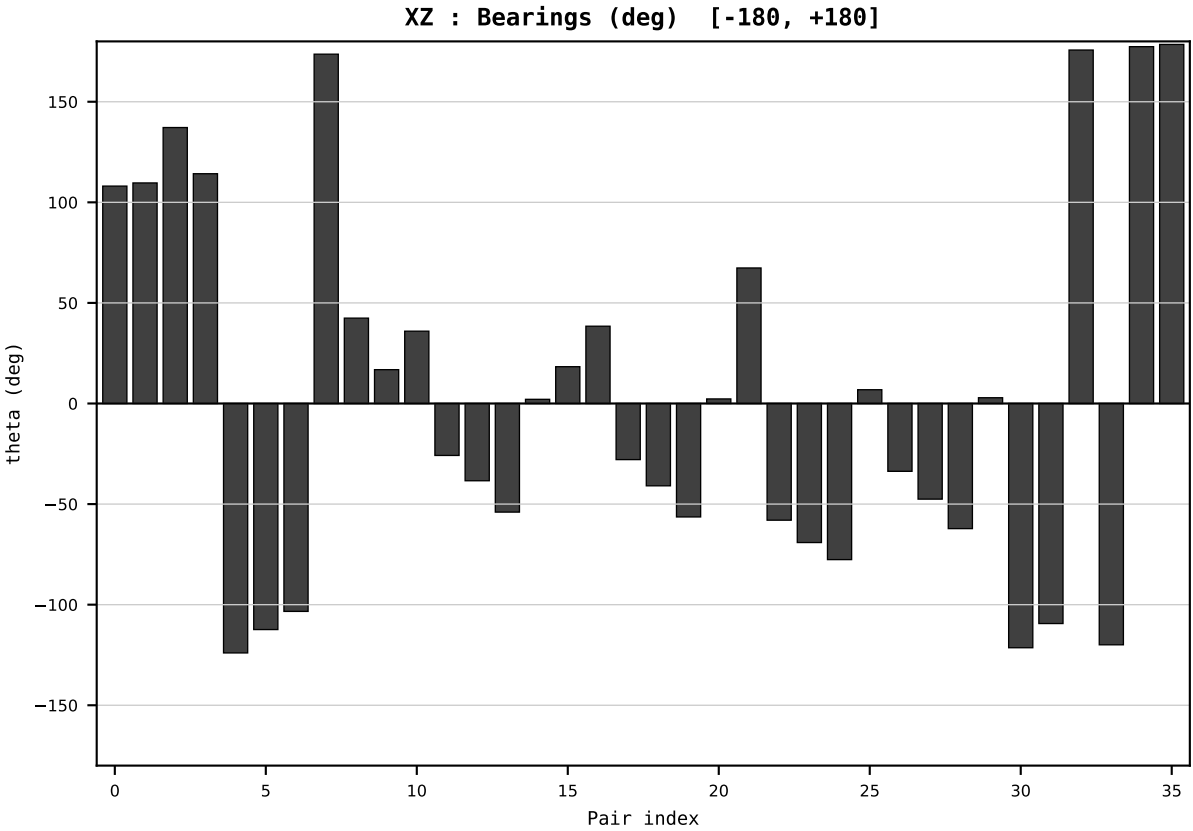
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2012
t : 11 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.325964
Ring = Hs-3
E_metric= 32.2877
kappa_HS= 348.86
omega = 6.5734 deg
d_A = 0.881530
Helm = Solar
Helm d = +0.772628
DR = 5.855 HLR
DR ratio= 348.9x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=12 | Year 2013 | D=9 N=25 pairs=36

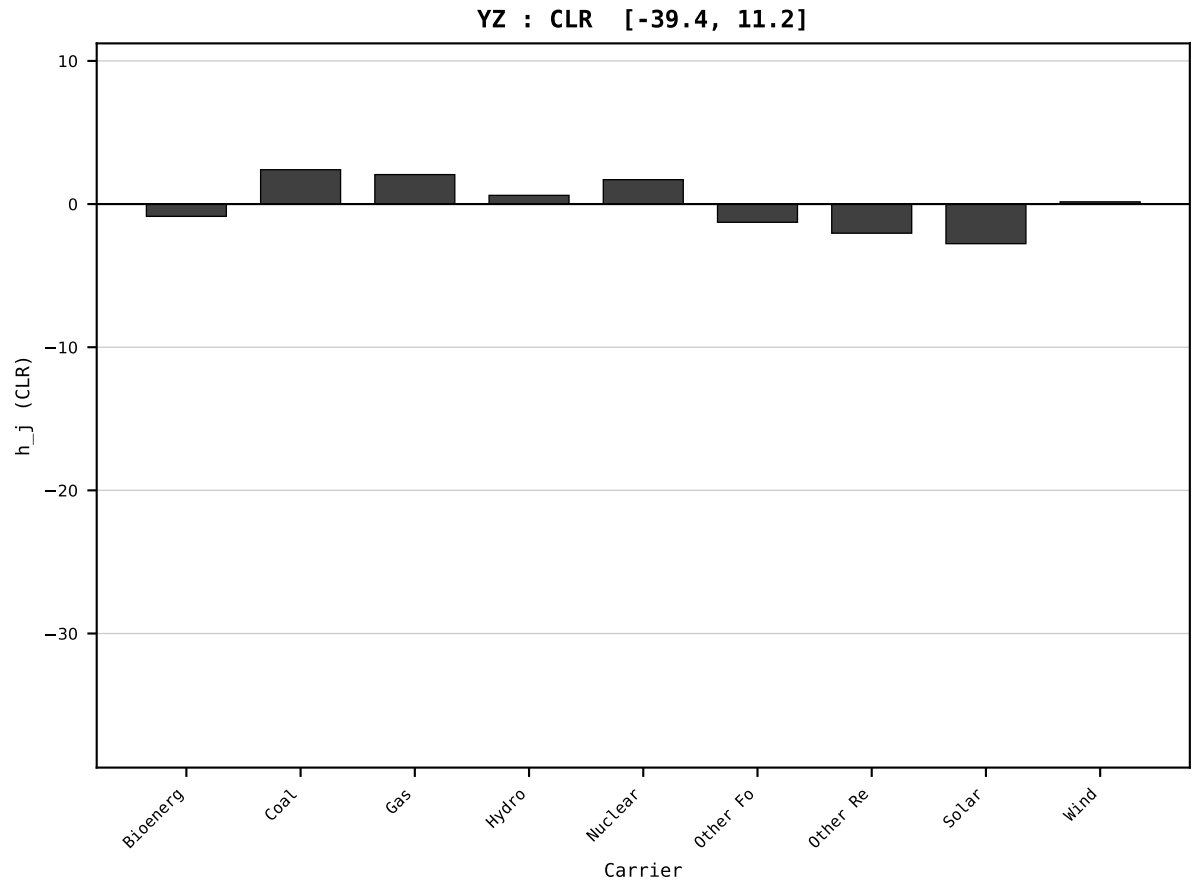
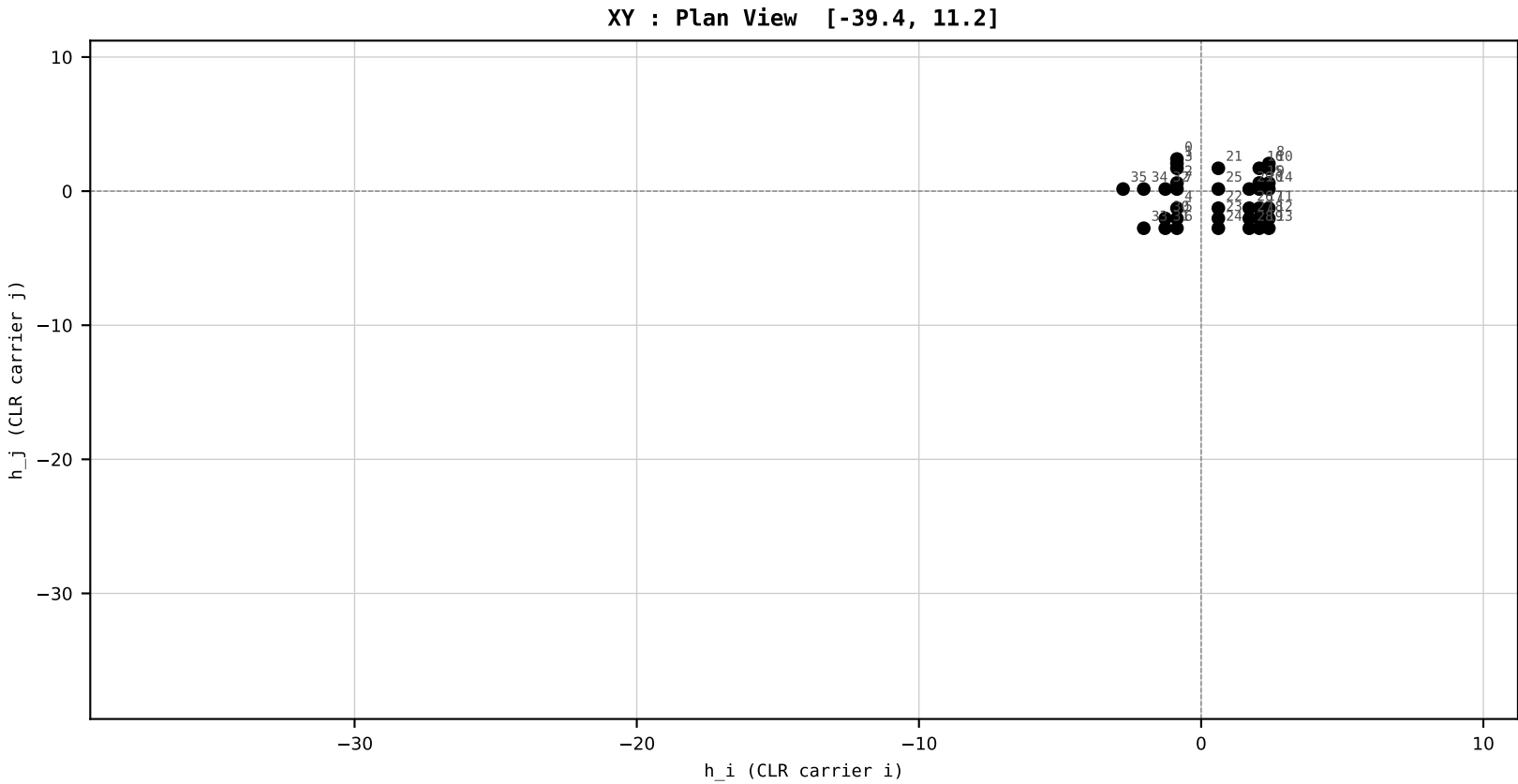
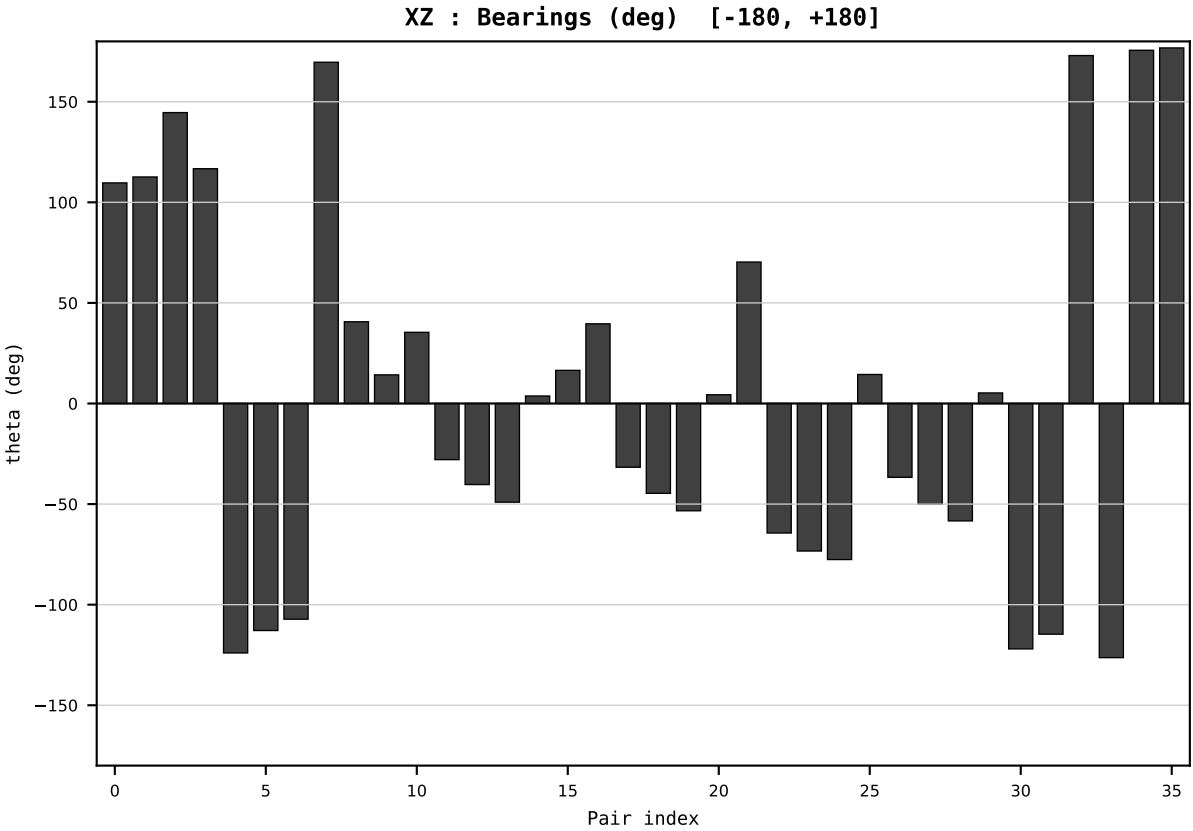
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2013
t : 12 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.318102
Ring = Hs-3
E_metric= 27.4388
kappa_HS= 175.10
omega = 5.5612 deg
d_A = 0.690898
Helm = Solar
Helm d = +0.624351
DR = 5.165 HLR
DR ratio= 175.1x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=13 | Year 2014 | D=9 N=25 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

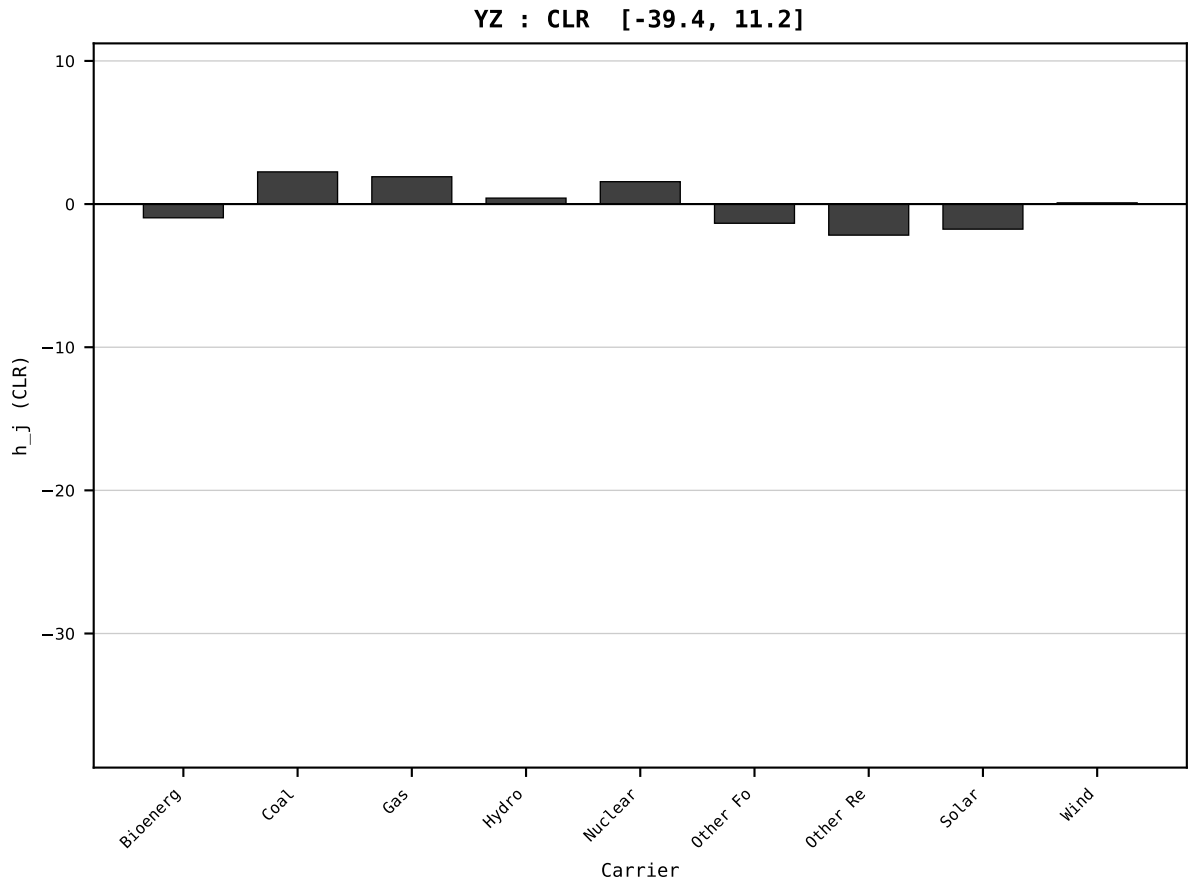
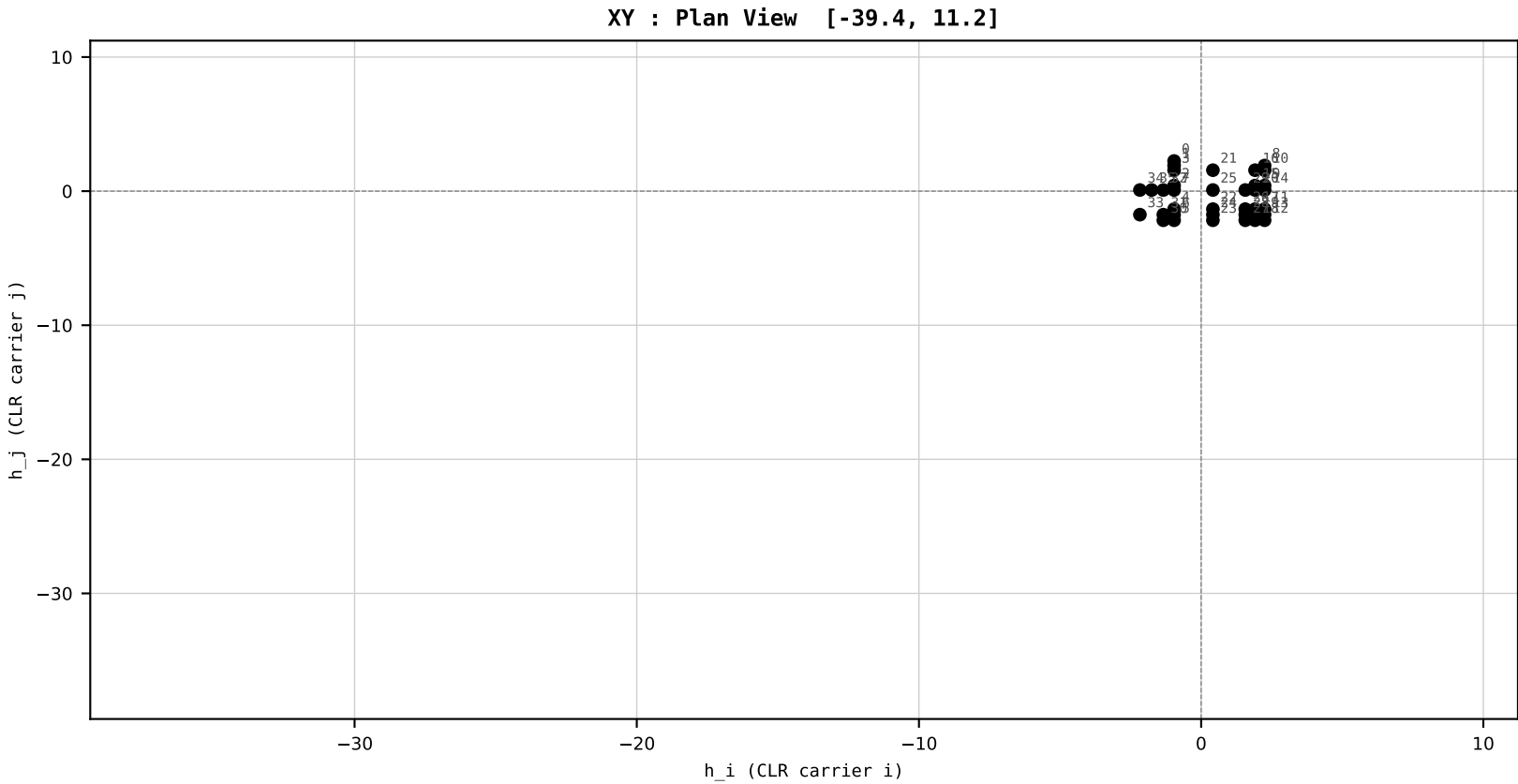
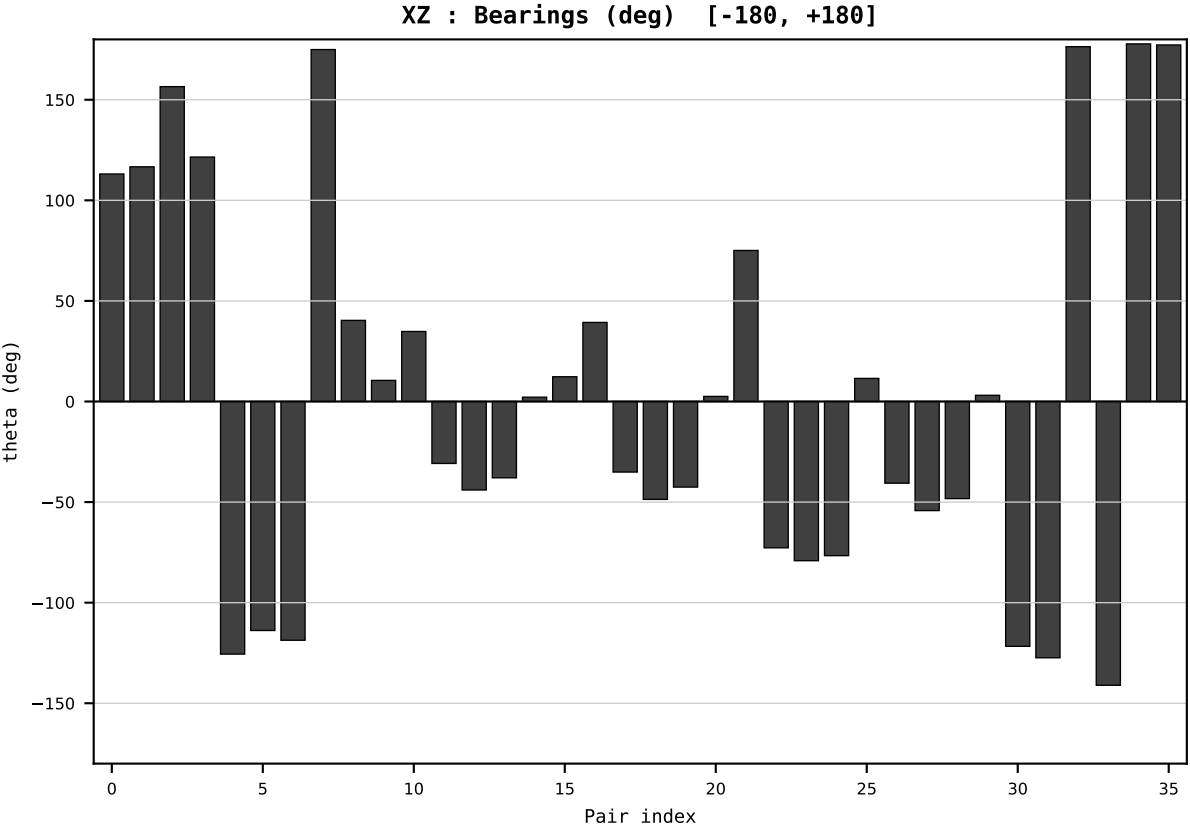
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2014
t : 13 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.306116
Ring = Hs-3
E_metric= 21.8268
kappa_HS= 83.03
omega = 10.6716 deg
d_A = 1.080374
Helm = Solar
Helm d = +1.013049
DR = 4.419 HLR
DR ratio= 83.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=14 | Year 2015 | D=9 N=25 pairs=36

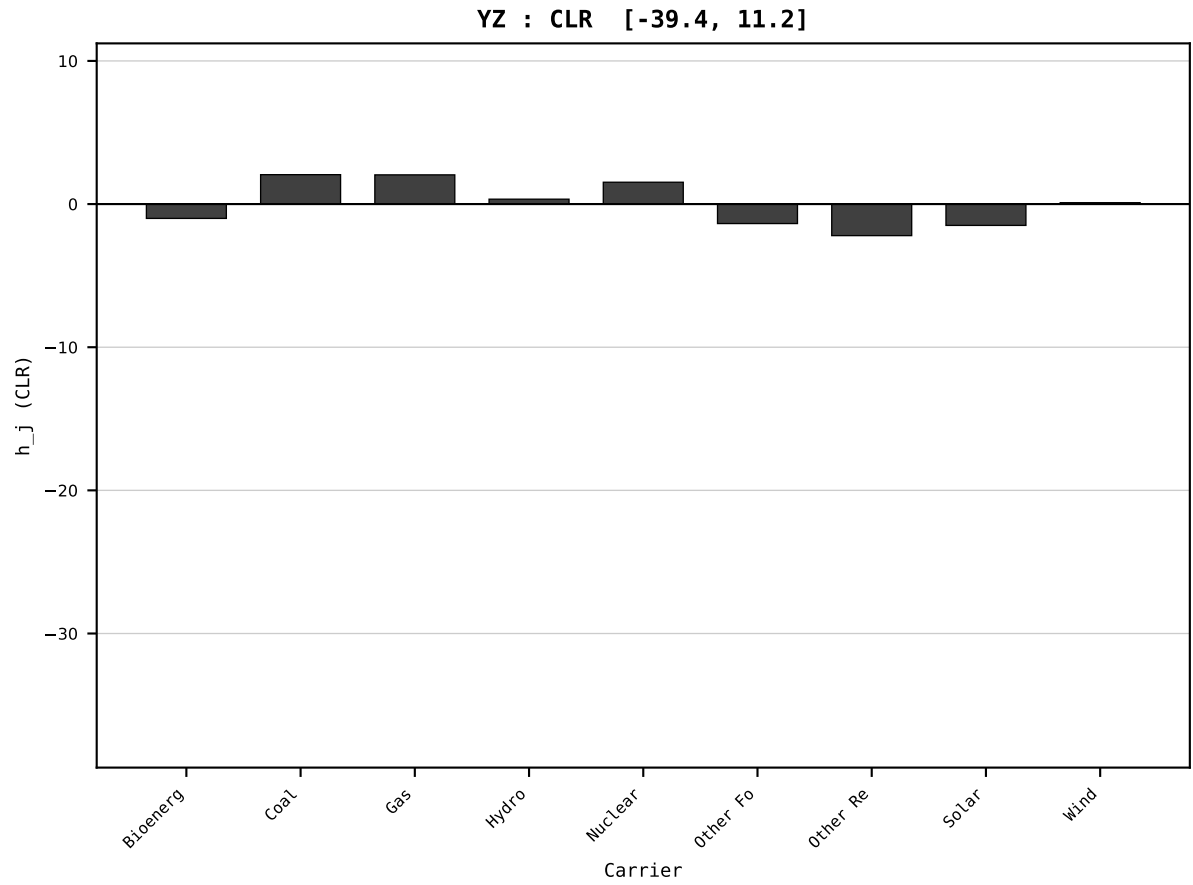
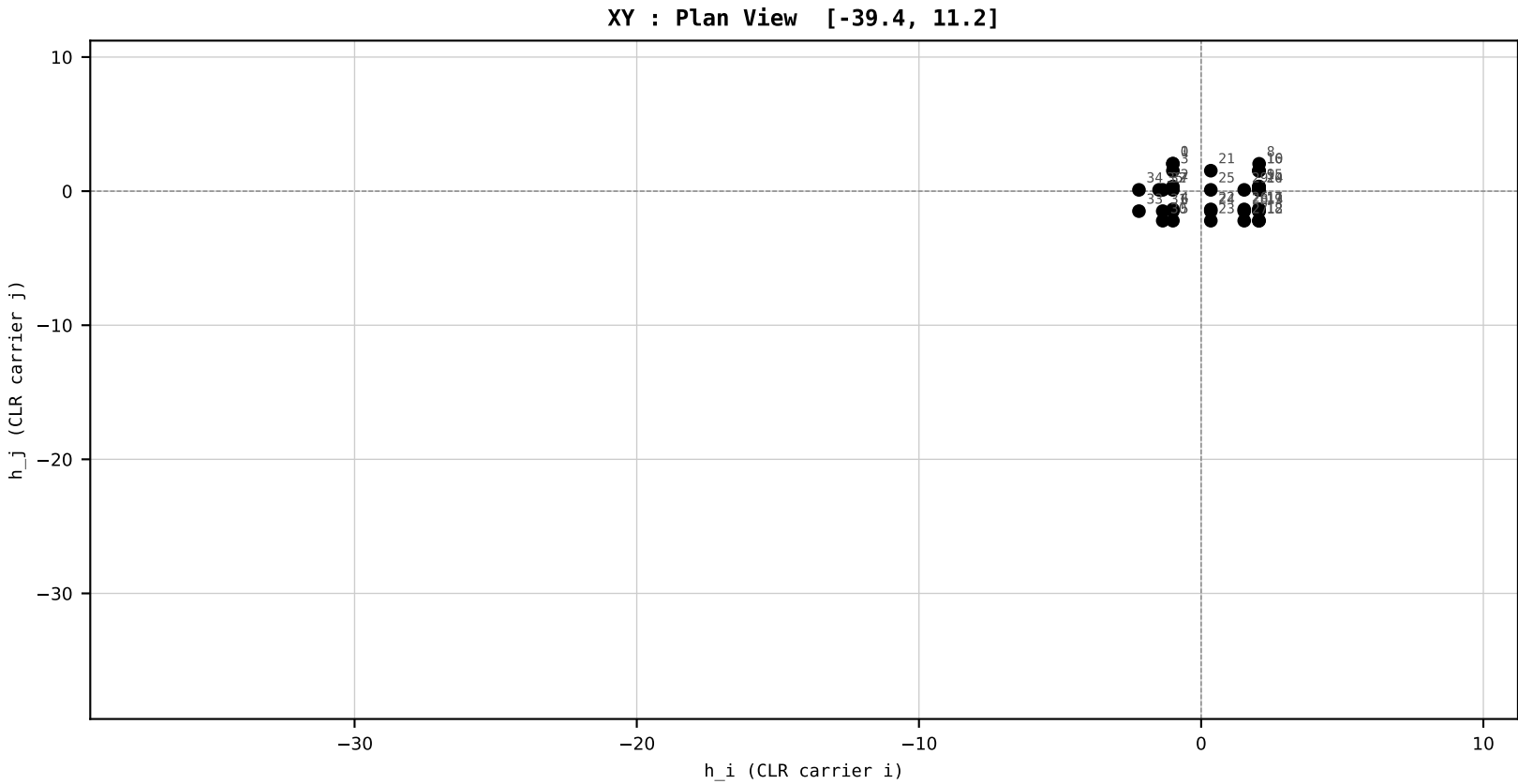
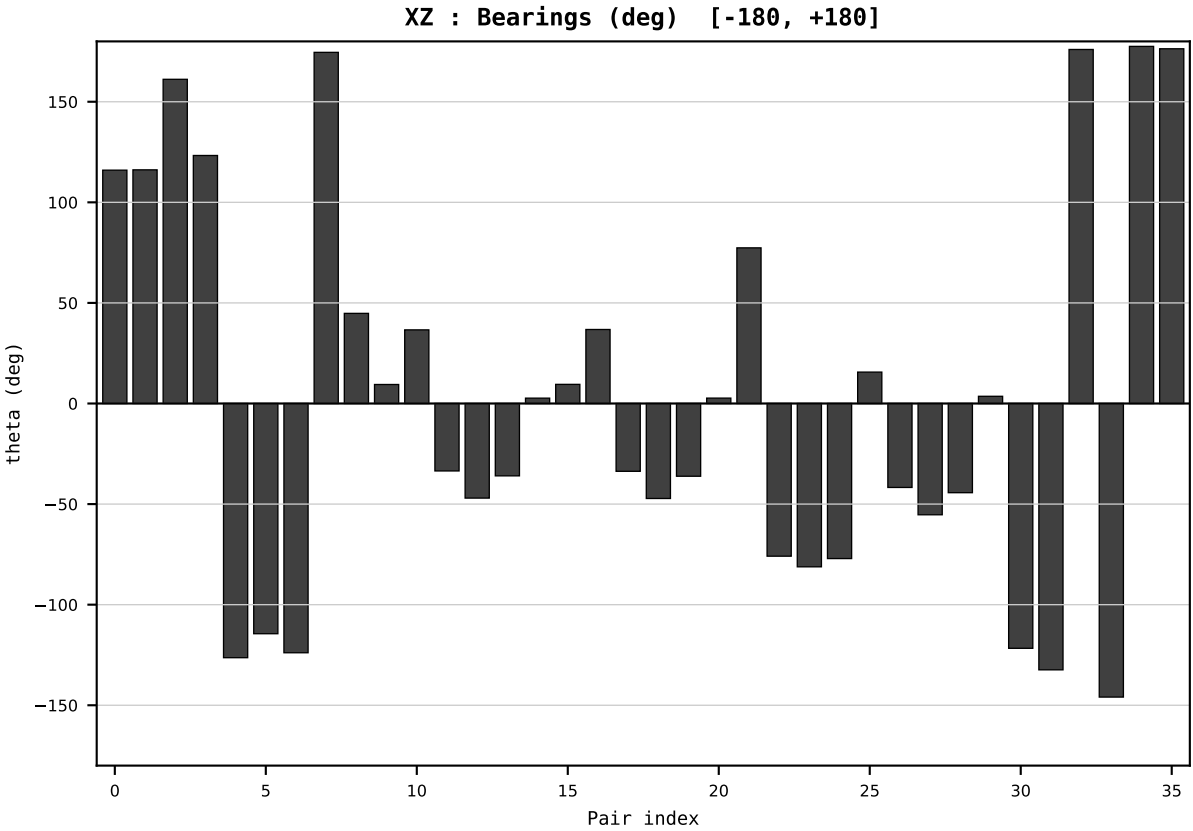
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2015
t : 14 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.296660
Ring = Hs-2
E_metric= 20.7750
kappa_HS= 70.73
omega = 4.3215 deg
d_A = 0.366154
Helm = Solar
Helm d = +0.262026
DR = 4.259 HLR
DR ratio= 70.7x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=15 | Year 2016 | D=9 N=25 pairs=36

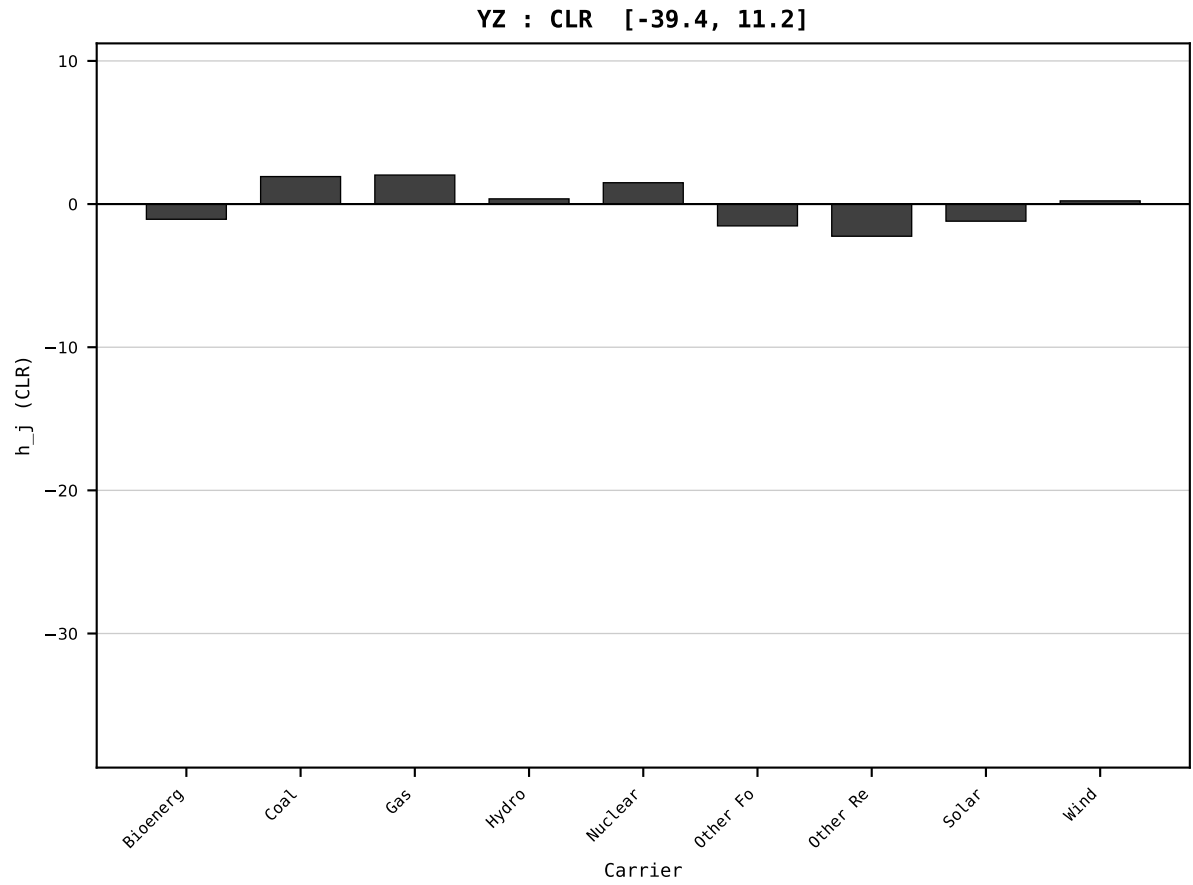
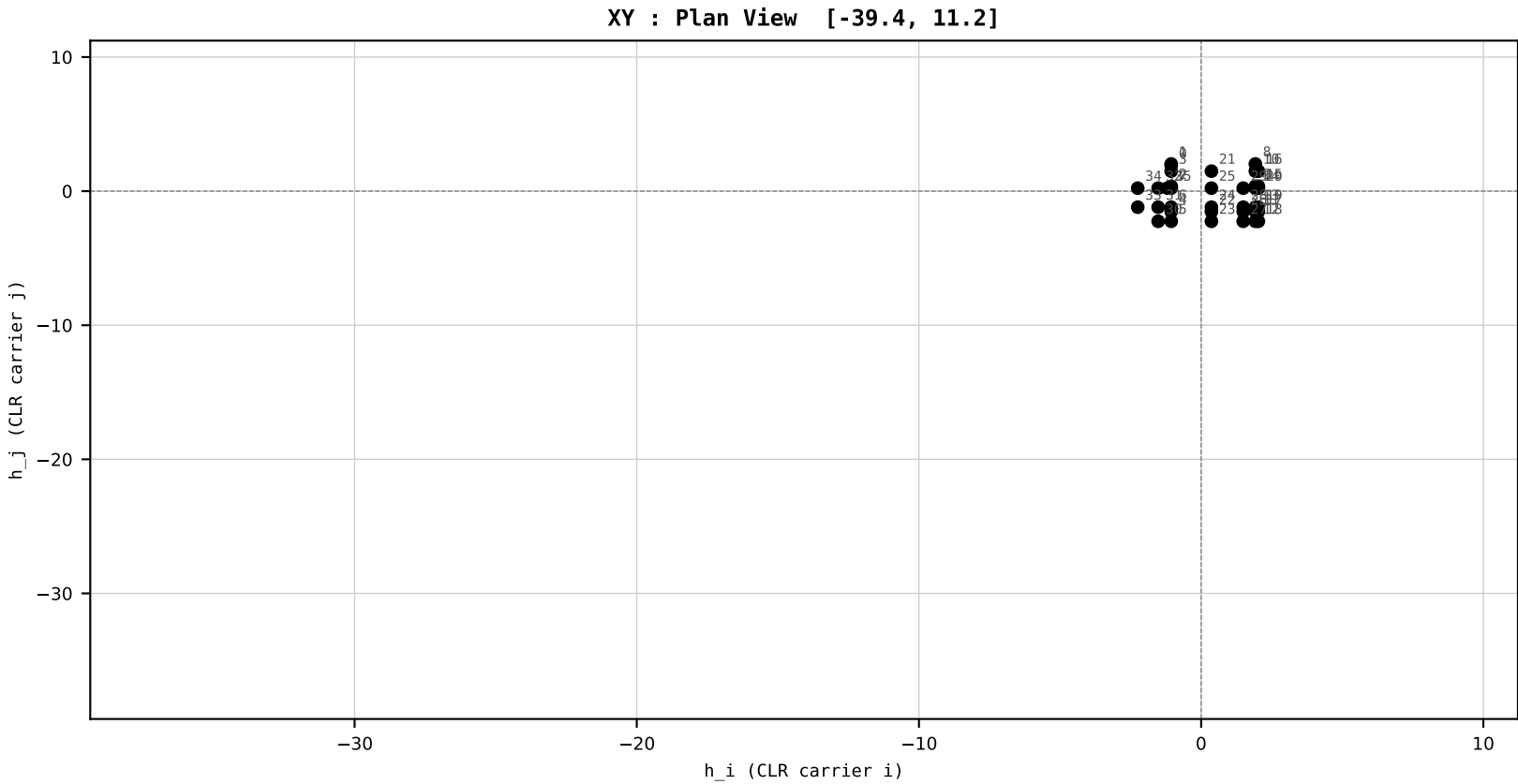
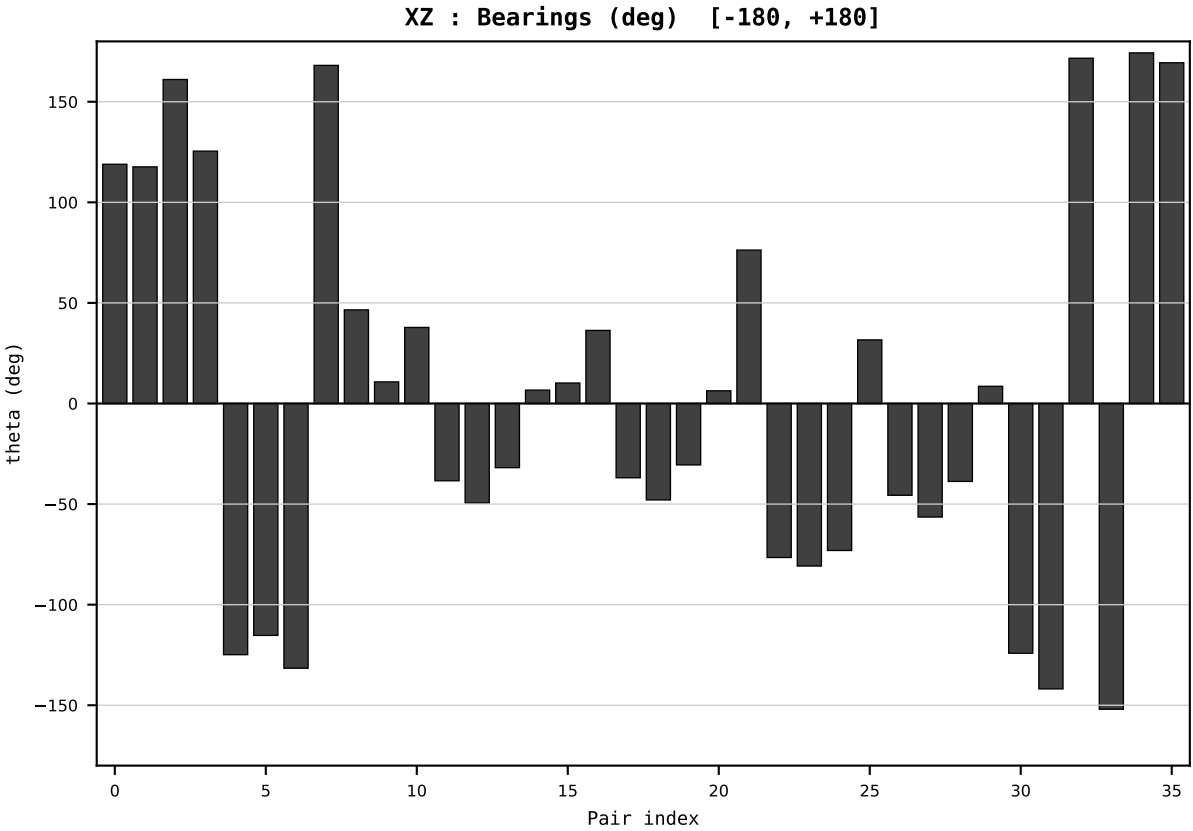
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2016
t : 15 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.282403
Ring = Hs-2
E_metric= 20.1405
kappa_HS= 71.90
omega = 4.9089 deg
d_A = 0.393670
Helm = Solar
Helm d = +0.294744
DR = 4.275 HLR
DR ratio= 71.9x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



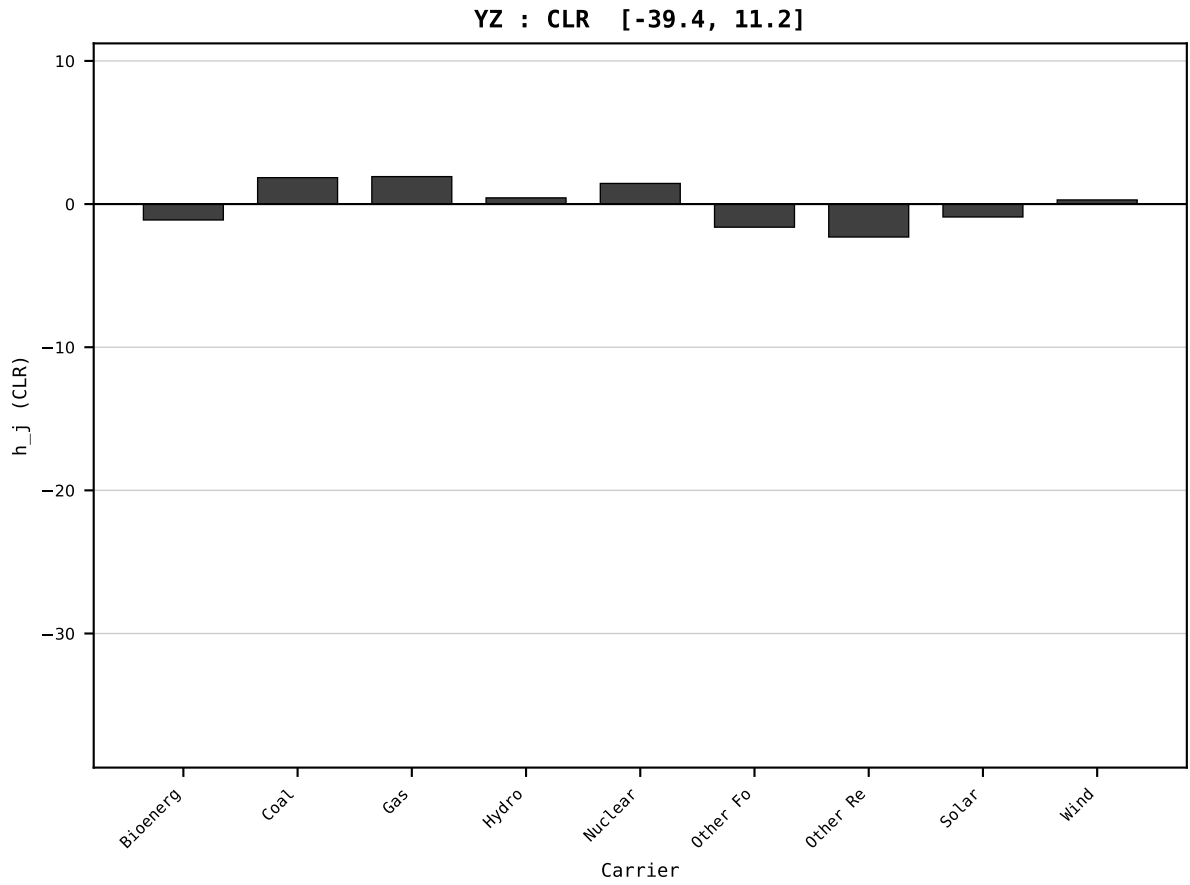
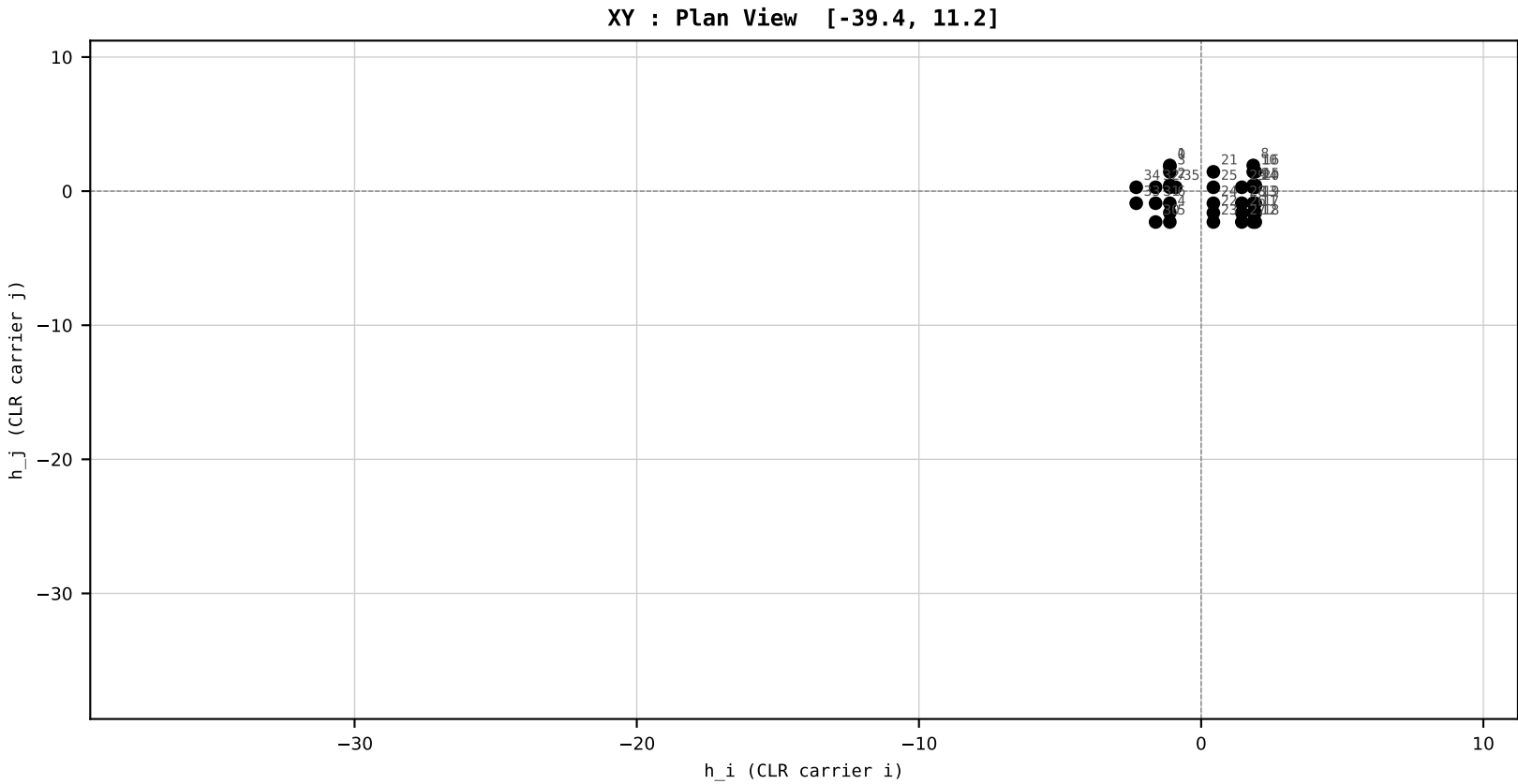
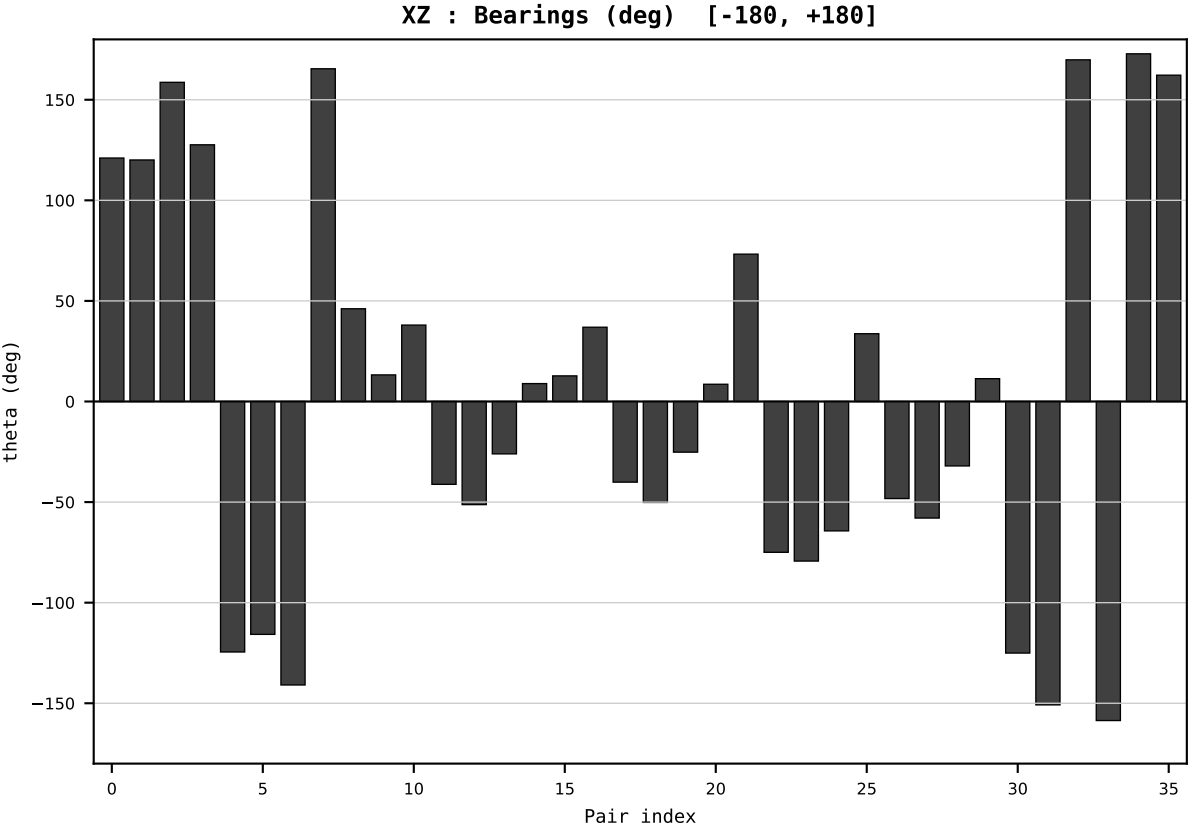
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2017
t : 16 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.263028
Ring = Hs-2
E_metric= 19.3839
kappa_HS= 67.98
omega = 4.5042 deg
d_A = 0.359563
Helm = Solar
Helm d = +0.294082
DR = 4.219 HLR
DR ratio= 68.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



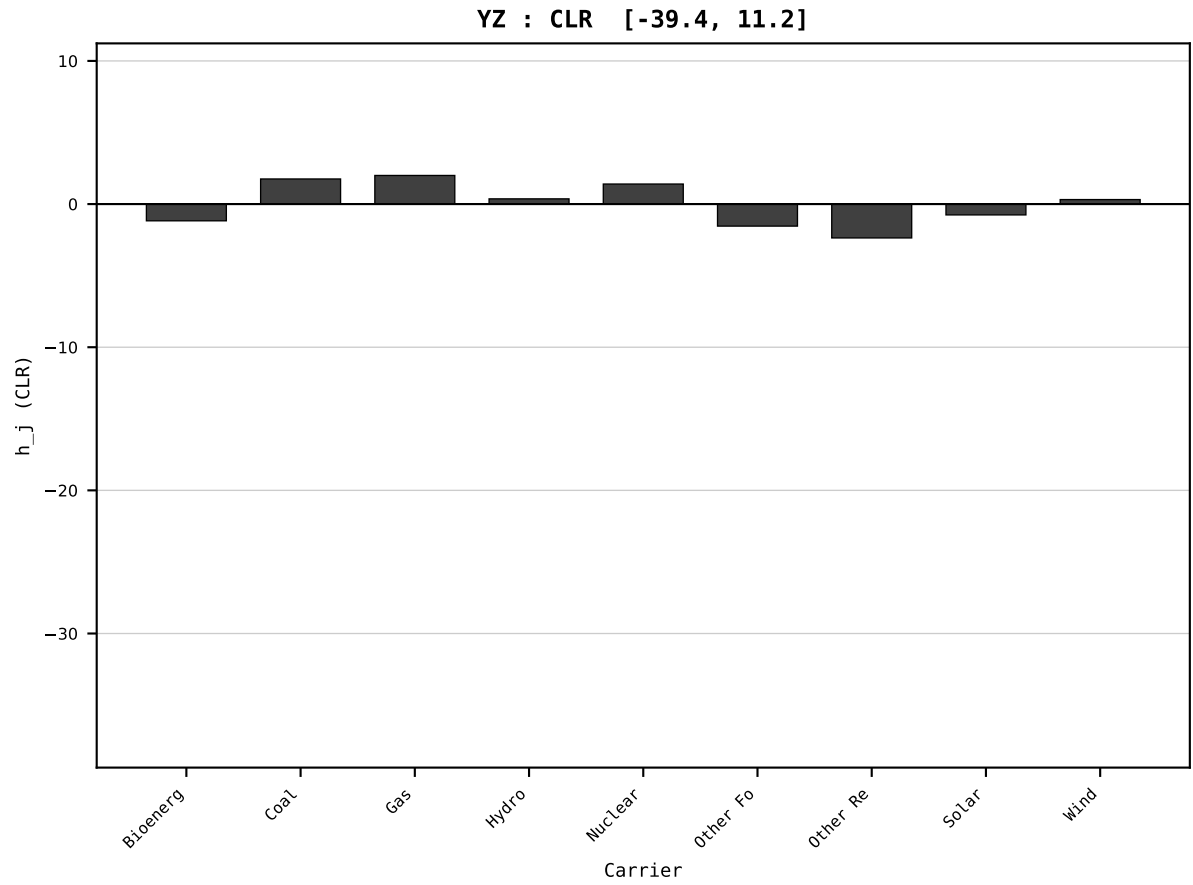
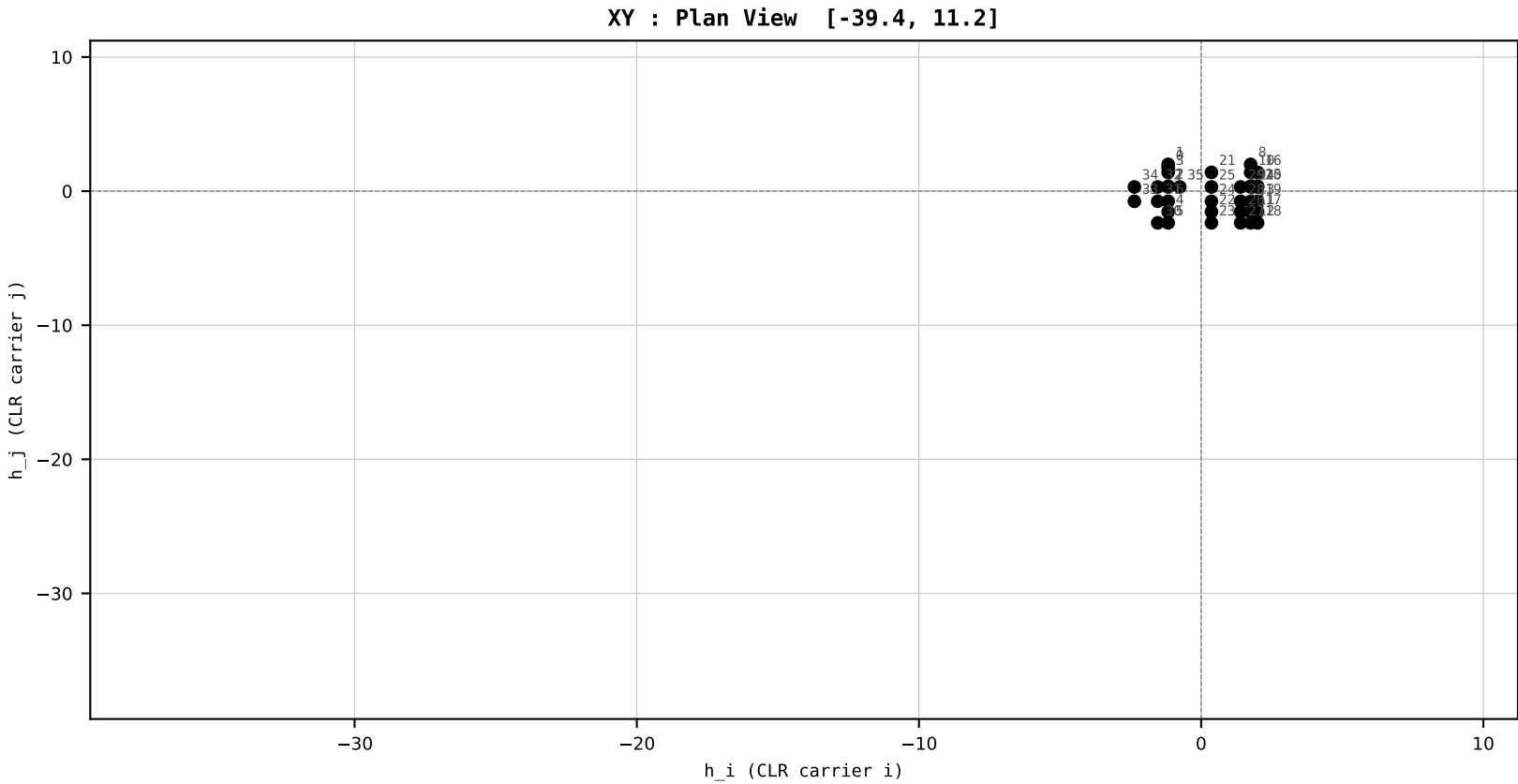
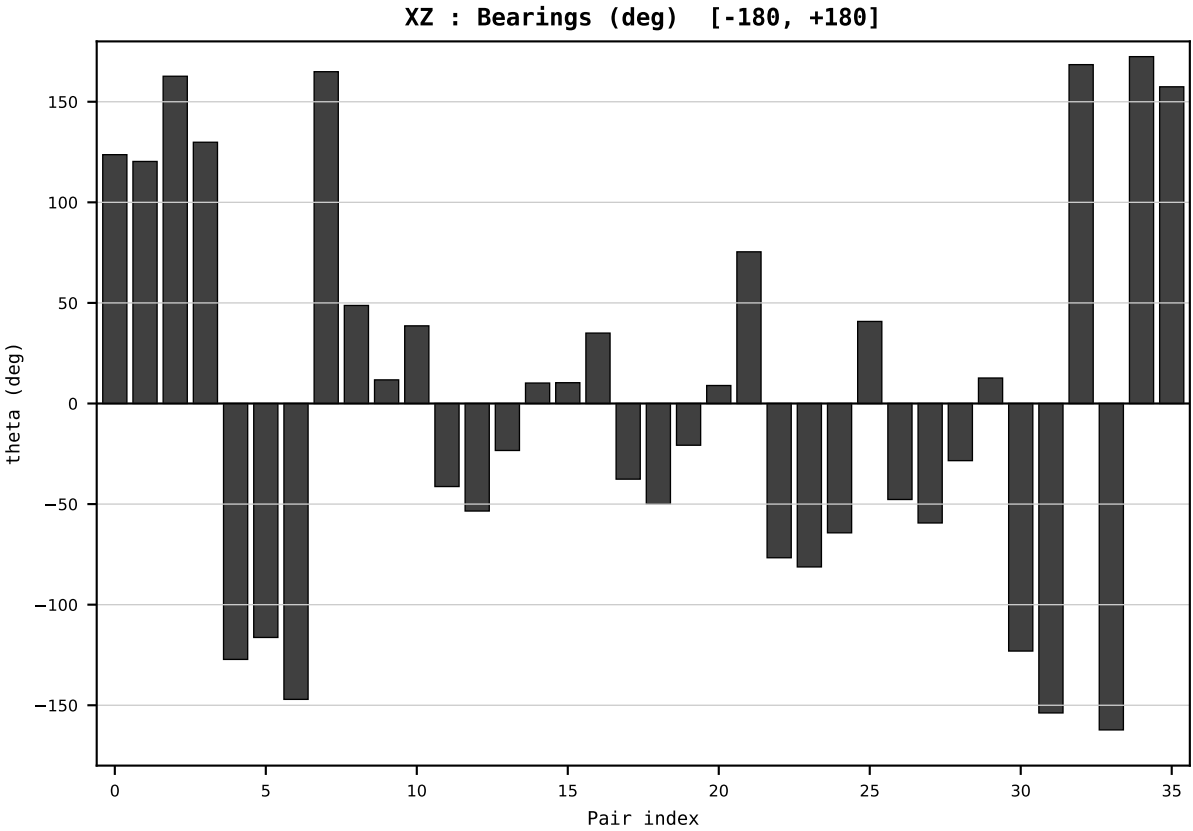
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2018
t : 17 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.264078
Ring = Hs-2
E_metric= 19.1683
kappa_HS= 78.61
omega = 3.0814 deg
d_A = 0.237367
Helm = Solar
Helm d = +0.144753
DR = 4.364 HLR
DR ratio= 78.6x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=18 | Year 2019 | D=9 N=25 pairs=36

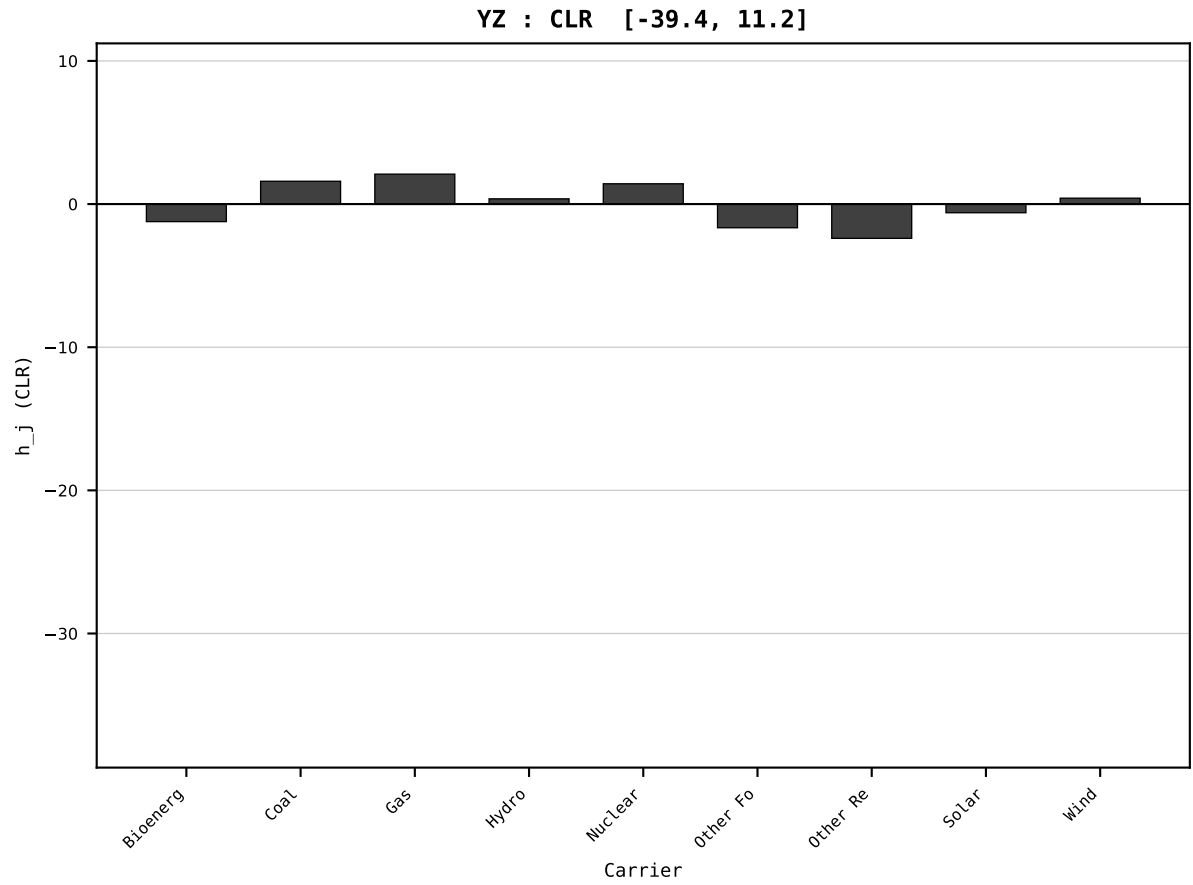
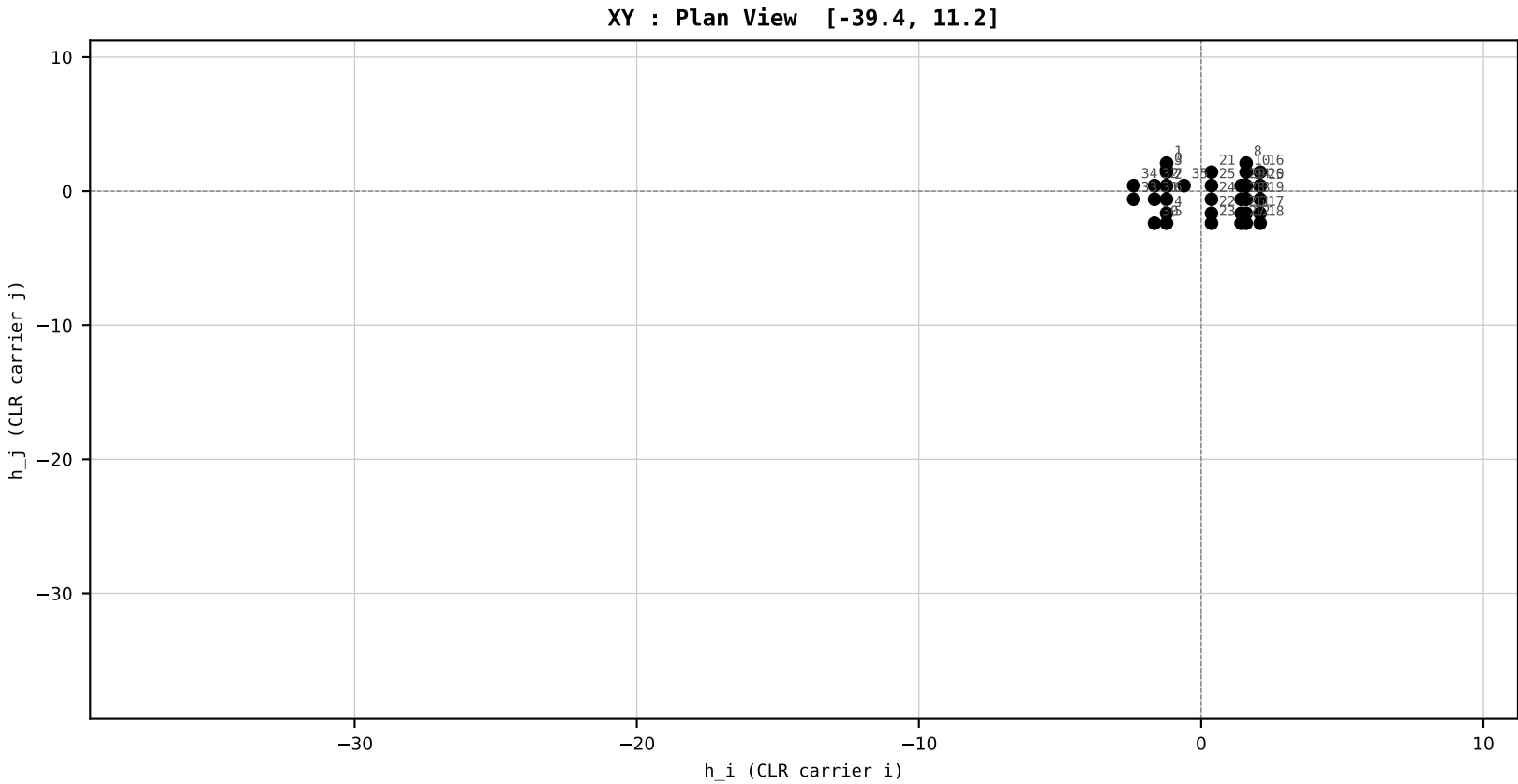
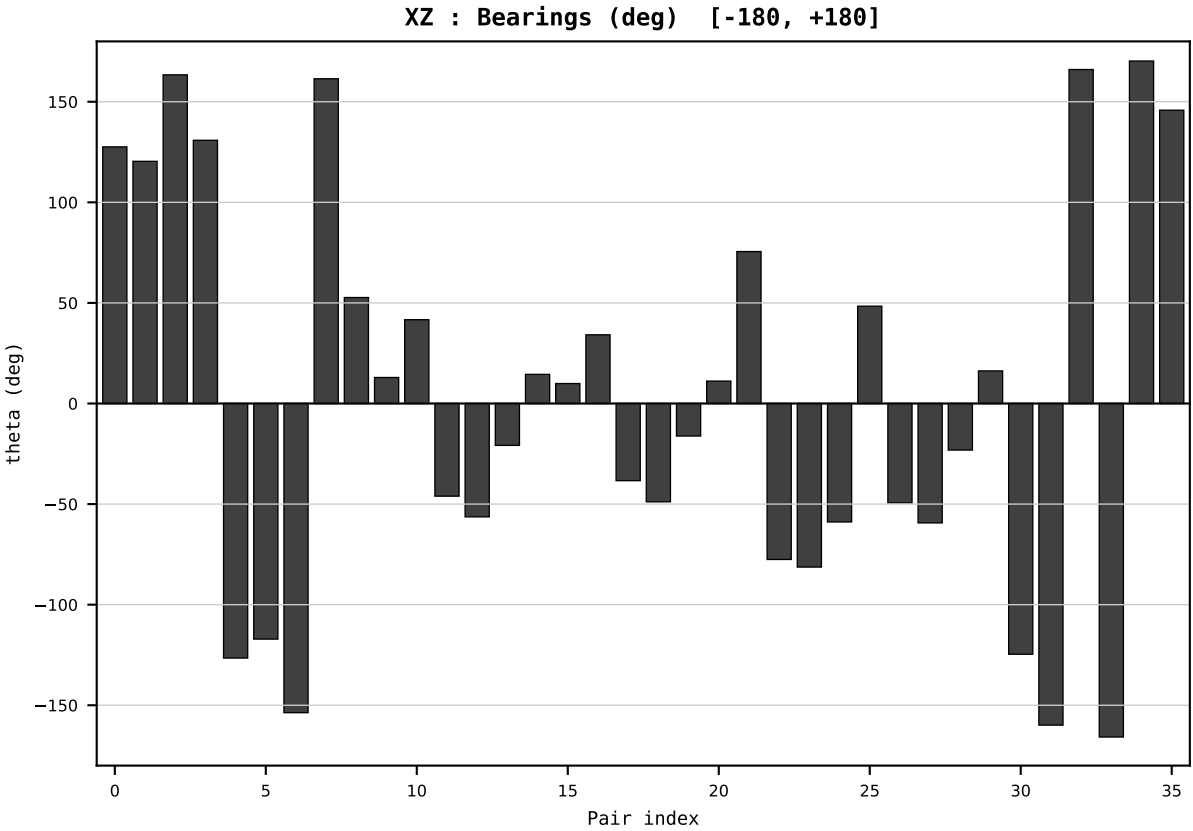
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2019
t : 18 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.264929
Ring = Hs-2
E_metric= 19.5686
kappa_HS= 88.69
omega = 3.7204 deg
d_A = 0.289310
Helm = Coal
Helm d = -0.159435
DR = 4.485 HLR
DR ratio= 88.7x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=19 | Year 2020 | D=9 N=25 pairs=36

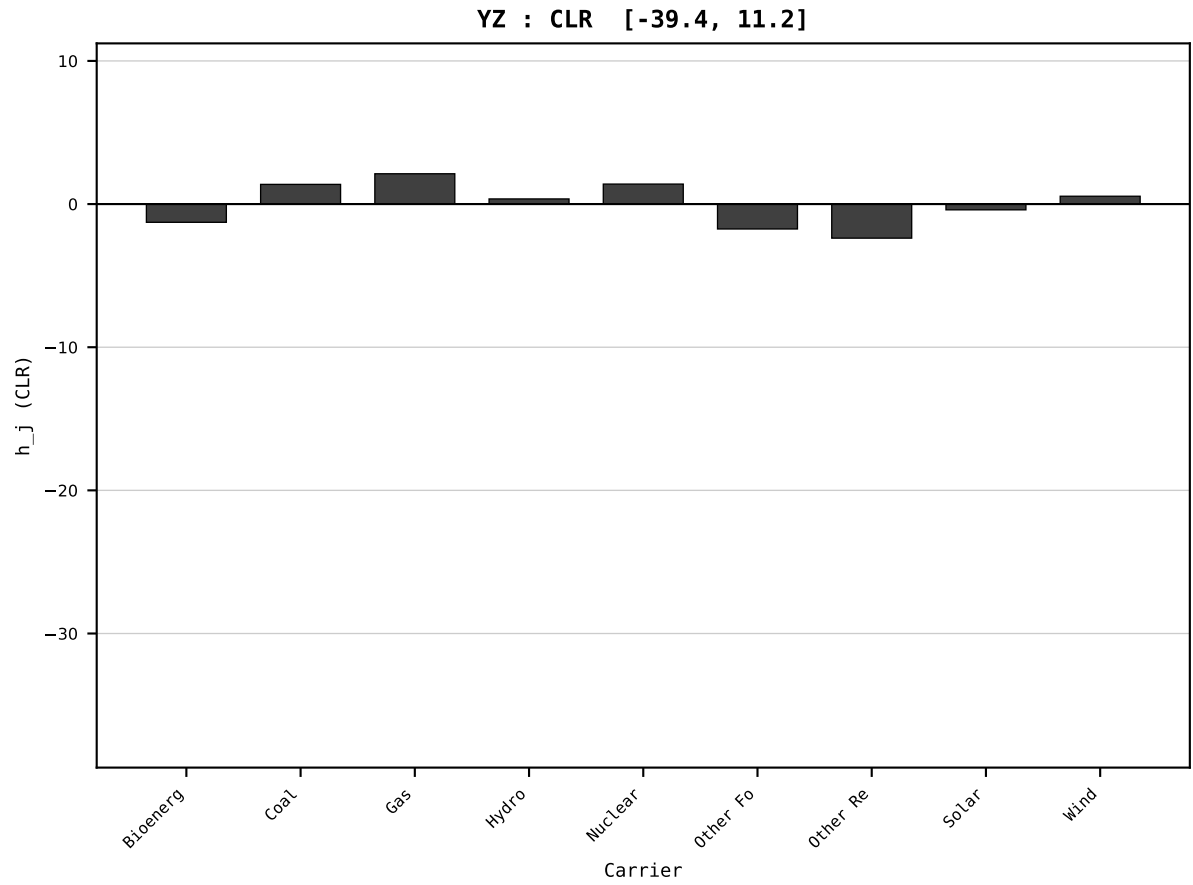
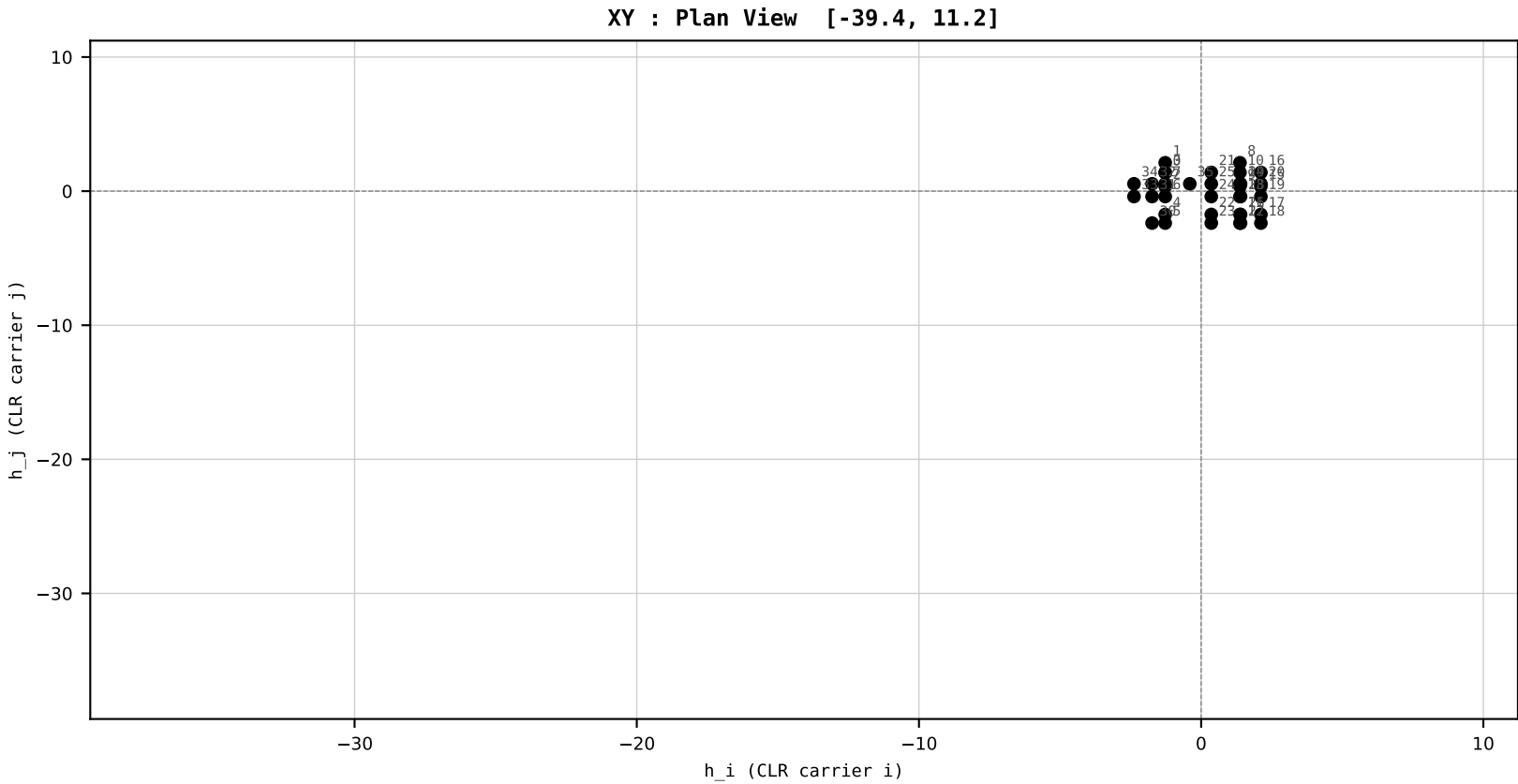
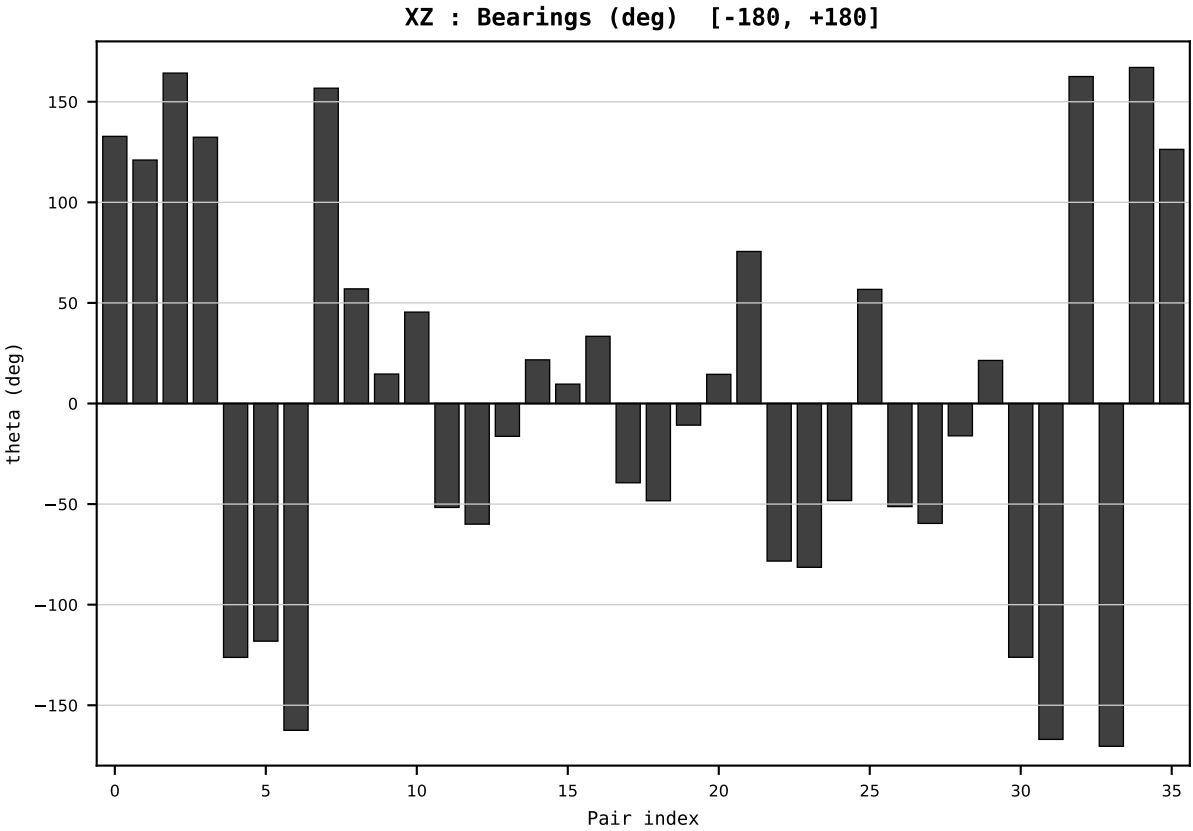
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2020
t : 19 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.259091
Ring = Hs-2
E_metric= 19.2298
kappa_HS= 89.83
omega = 4.4528 deg
d_A = 0.344356
Helm = Coal
Helm d = -0.218685
DR = 4.498 HLR
DR ratio= 89.8x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=20 | Year 2021 | D=9 N=25 pairs=36

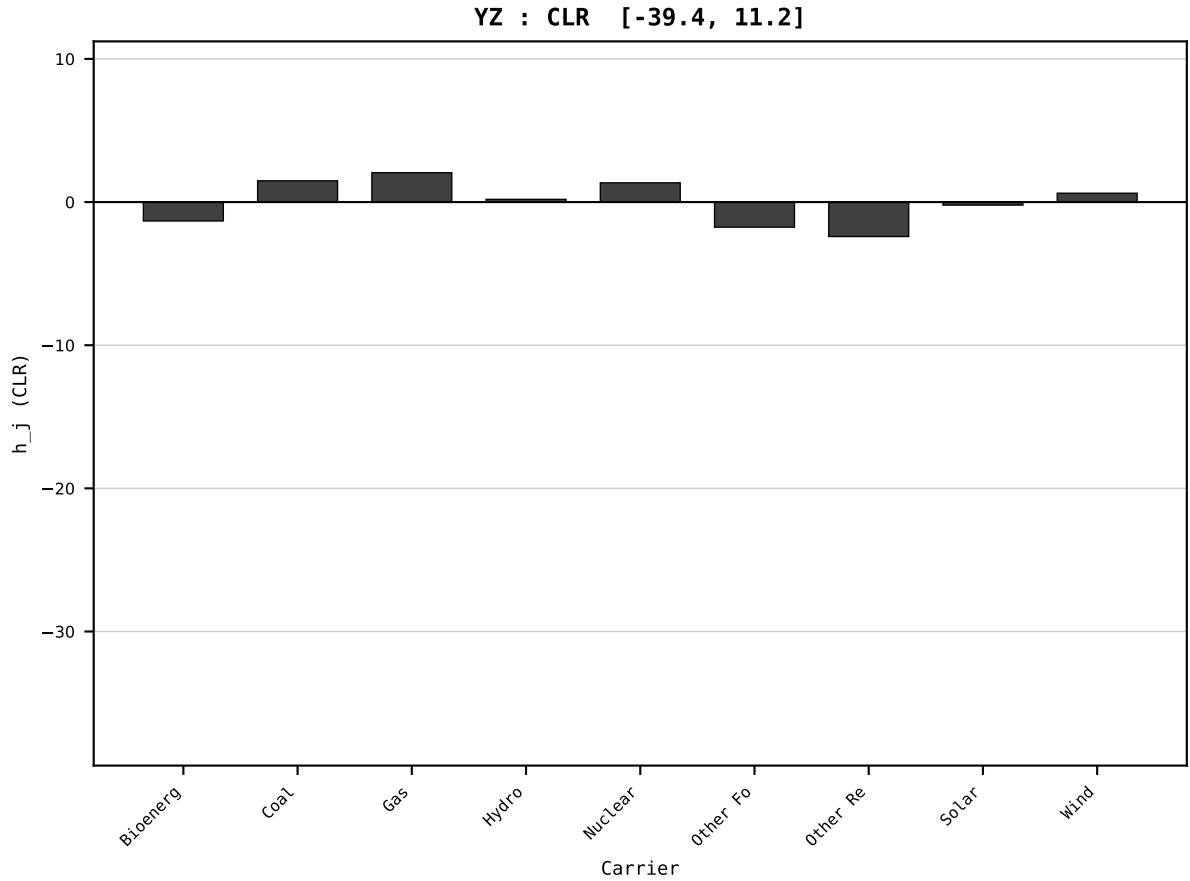
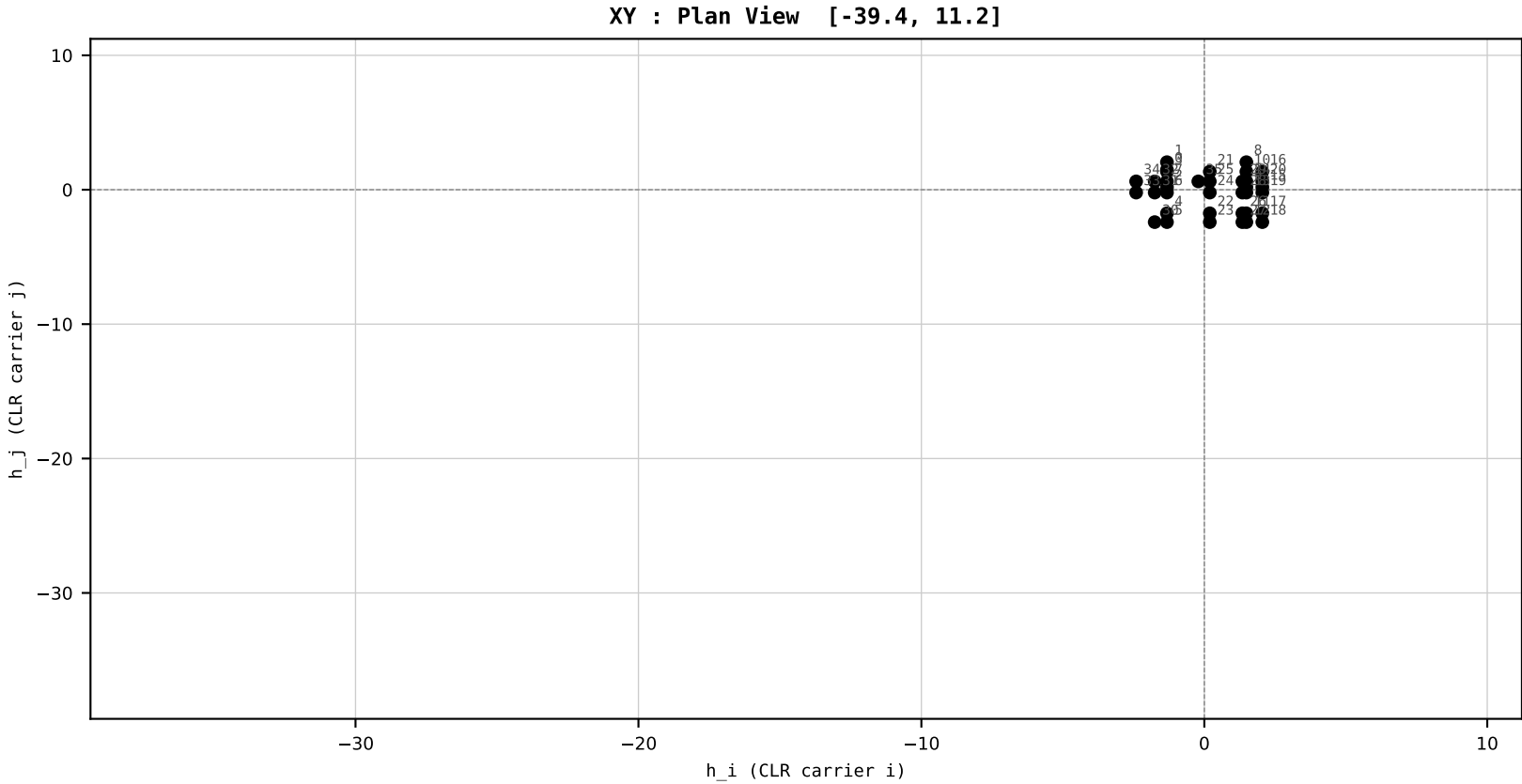
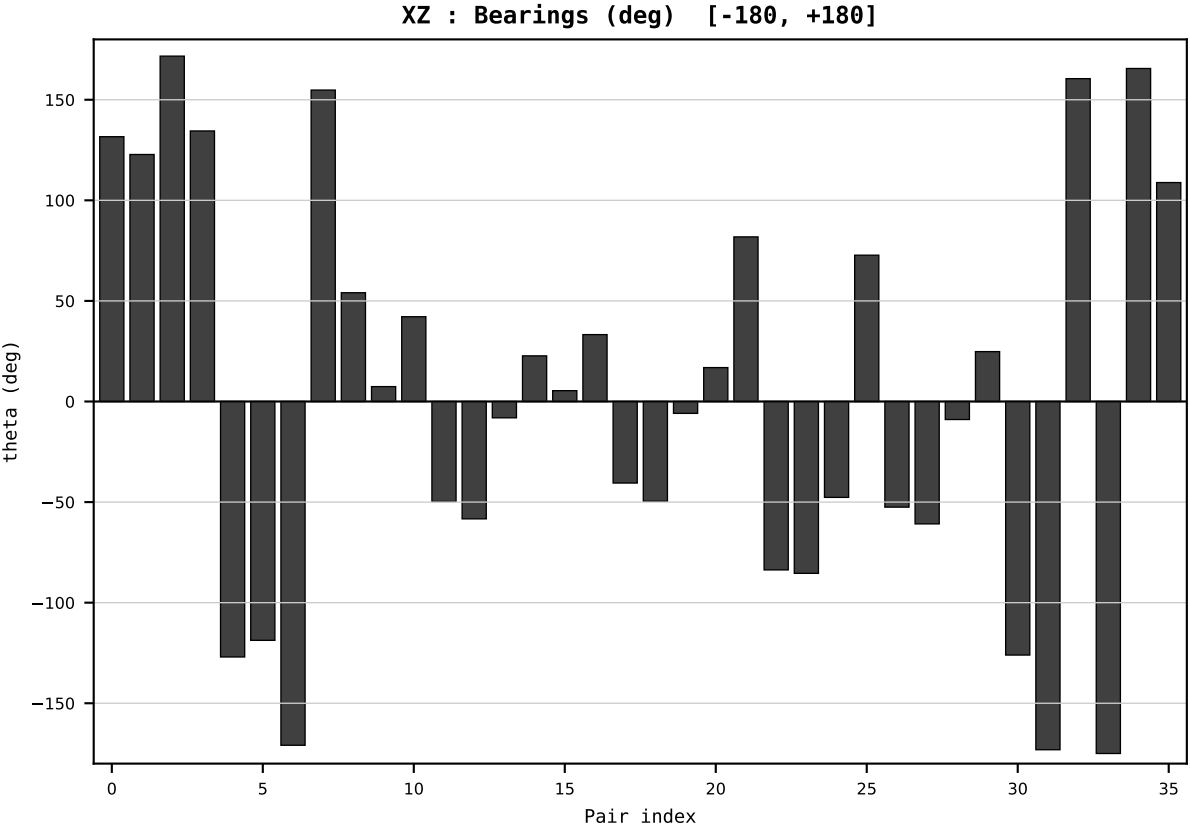
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2021
t : 20 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.250253
Ring = Hs-2
E_metric= 19.3124
kappa_HS= 86.48
omega = 3.9579 deg
d_A = 0.303328
Helm = Solar
Helm d = +0.190528
DR = 4.460 HLR
DR ratio= 86.5x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=21 | Year 2022 | D=9 N=25 pairs=36

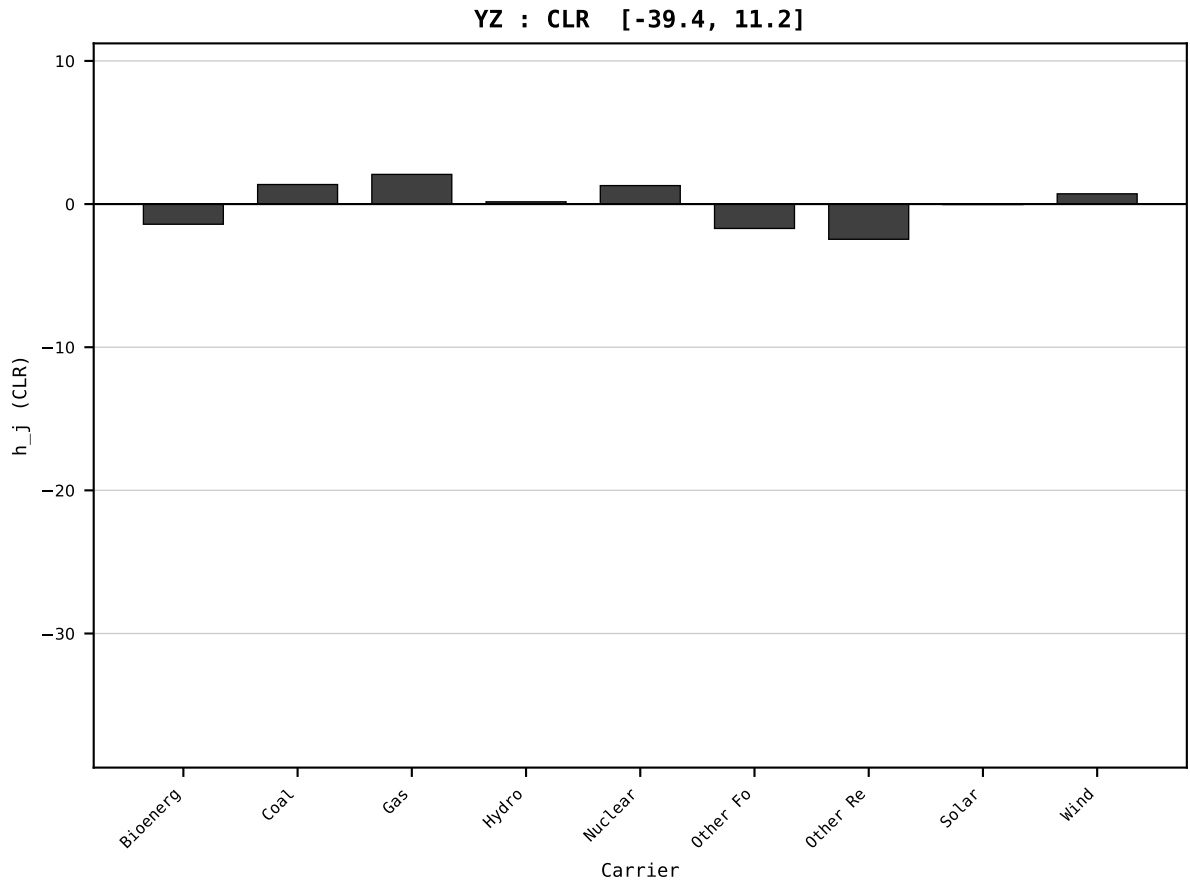
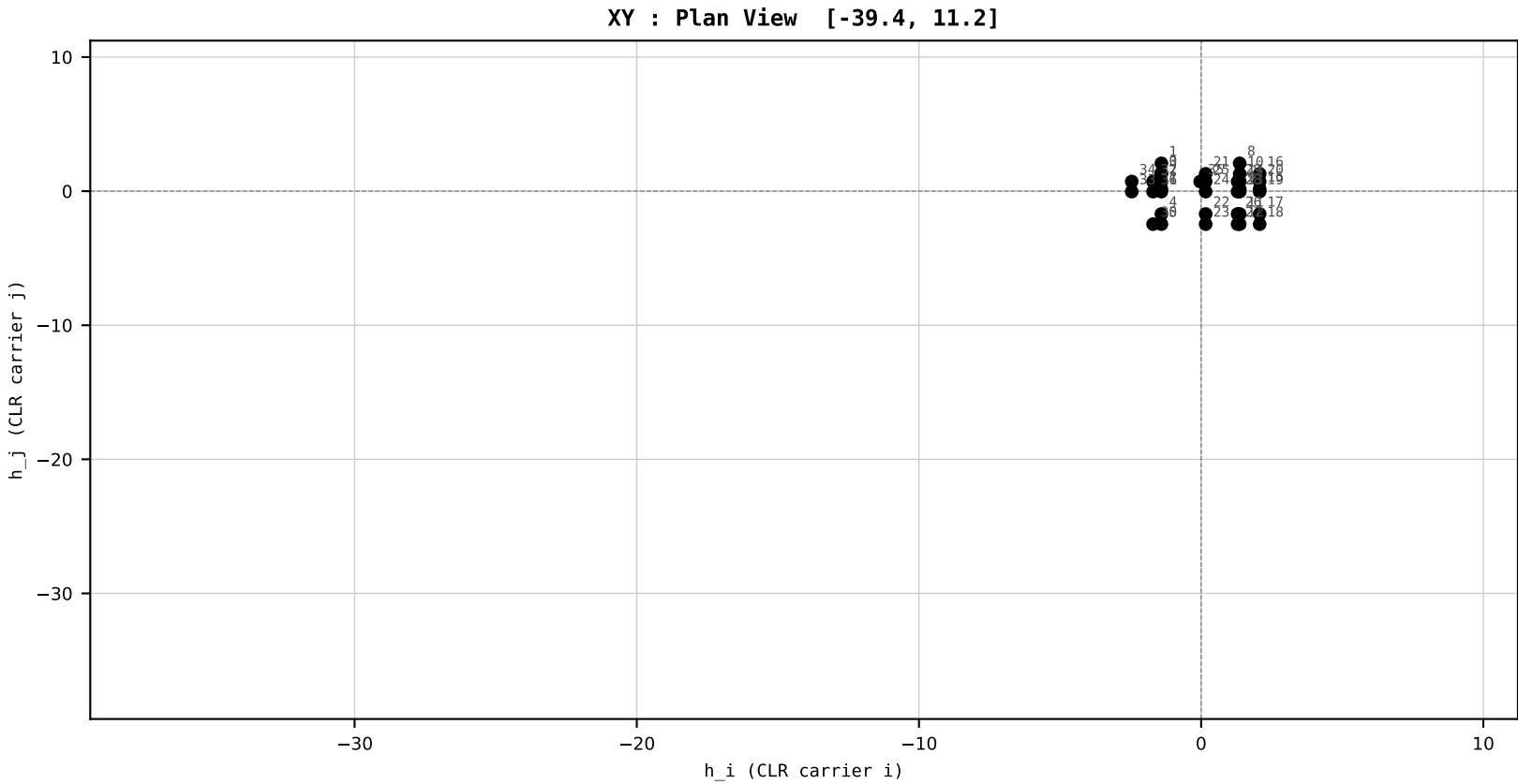
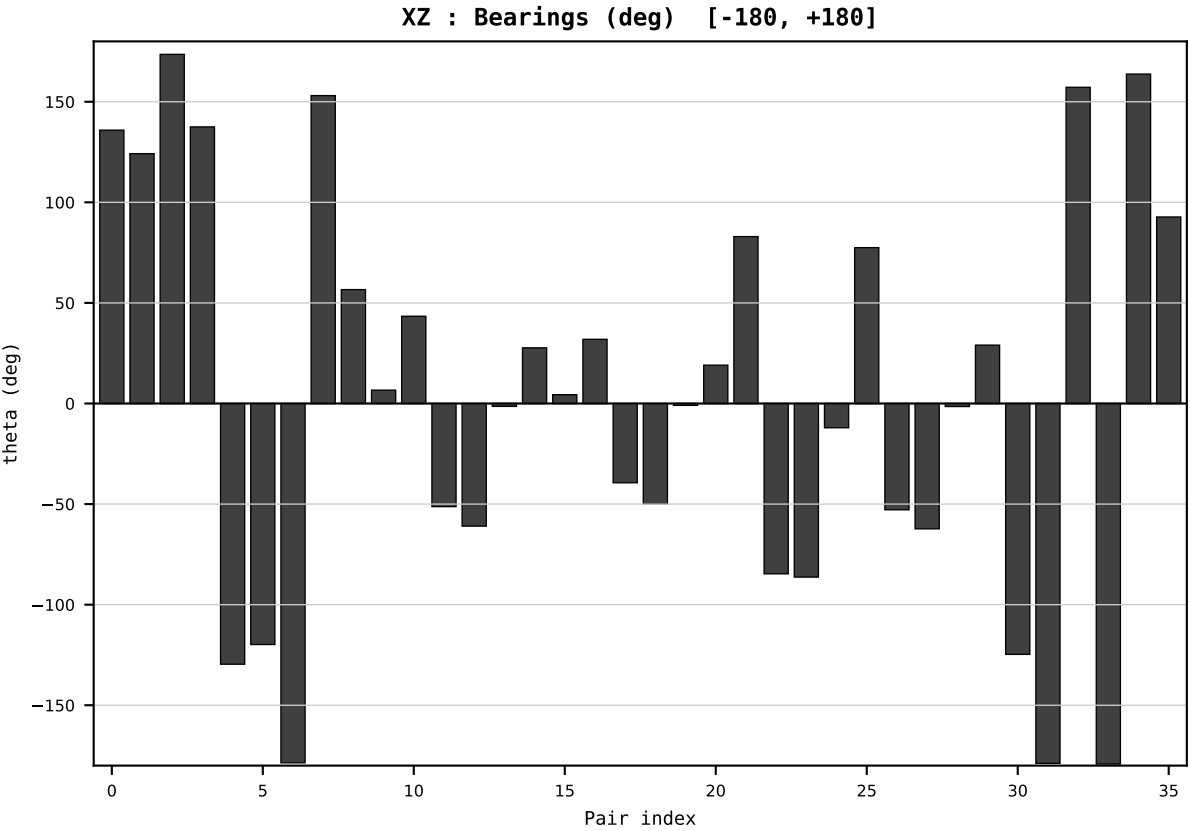
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2022
t : 21 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.246637
Ring = Hs-2
E_metric= 19.3055
kappa_HS= 93.00
omega = 3.5094 deg
d_A = 0.269105
Helm = Solar
Helm d = +0.177922
DR = 4.533 HLR
DR ratio= 93.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=22 | Year 2023 | D=9 N=25 pairs=36

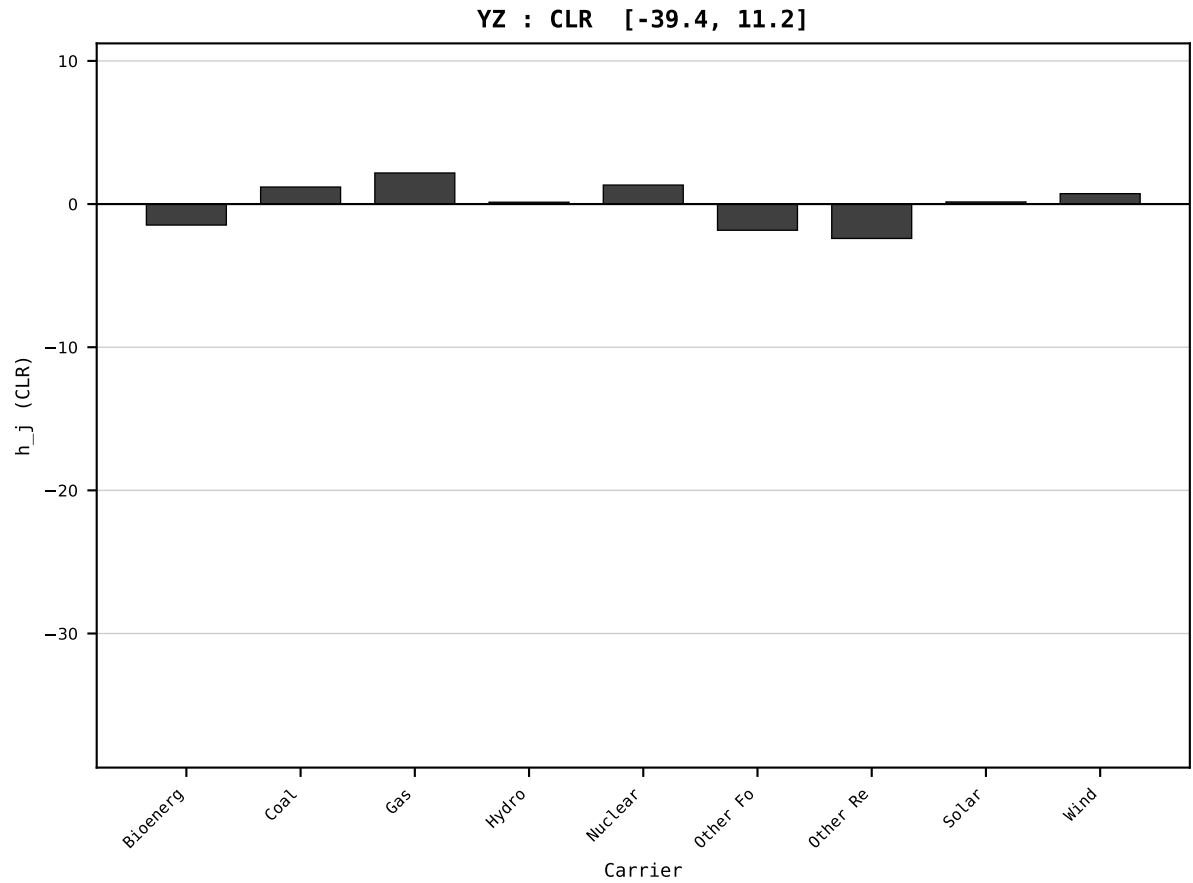
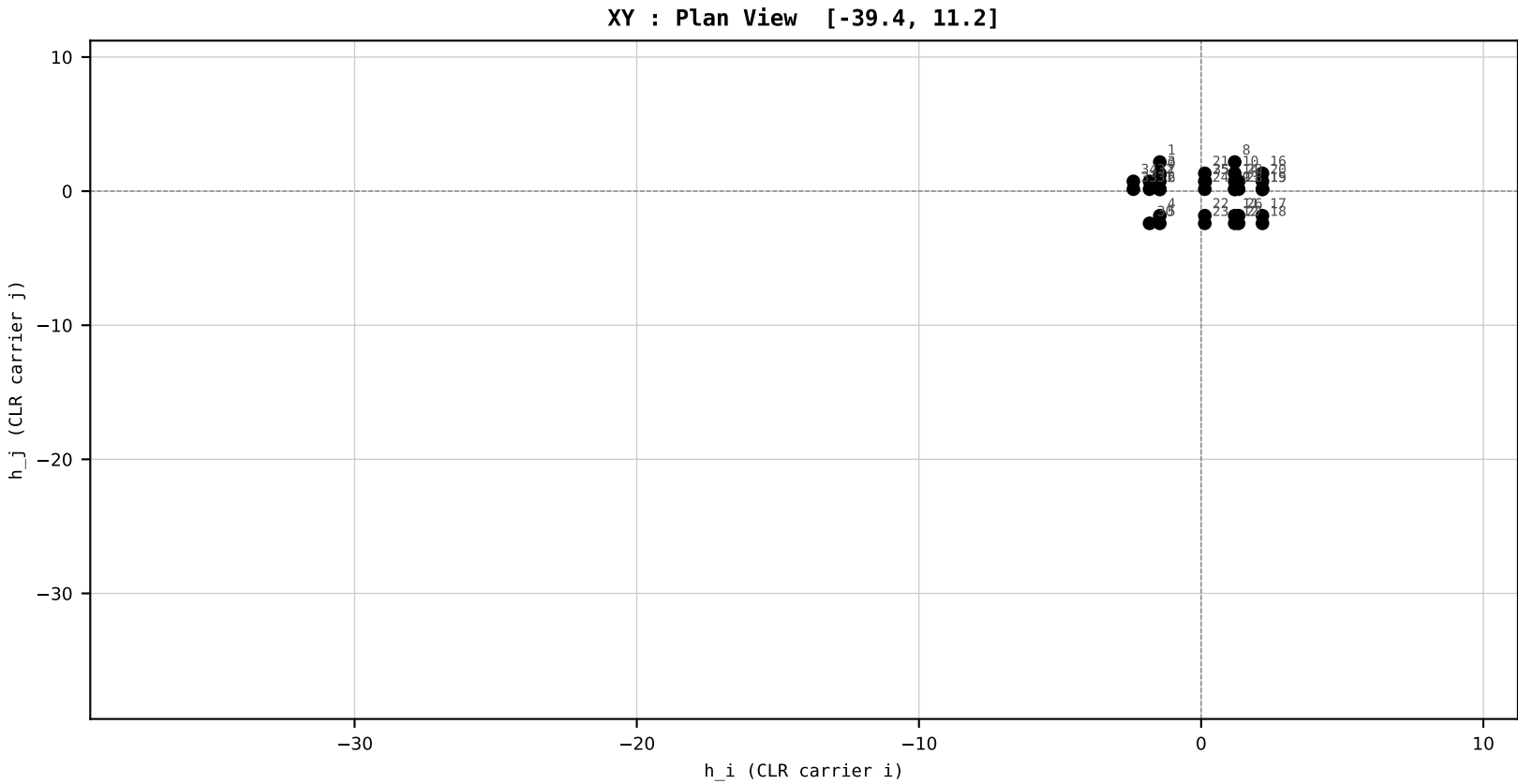
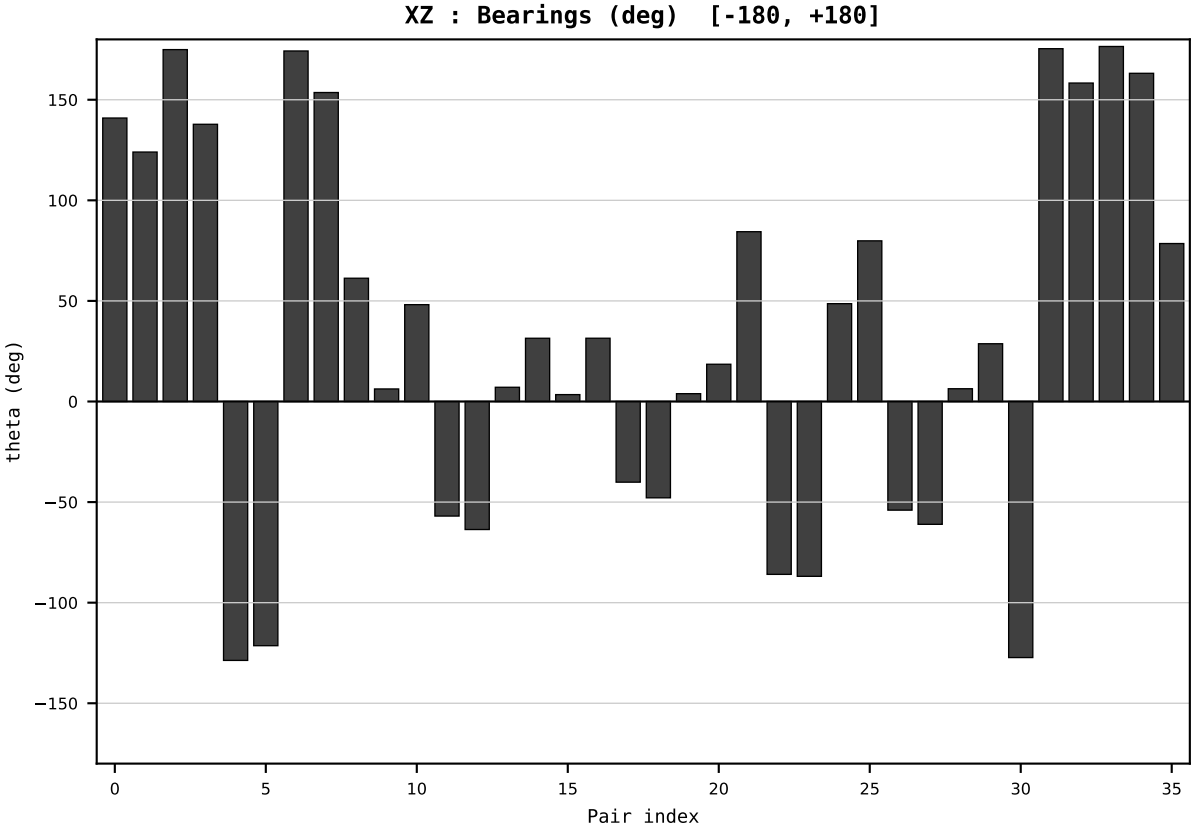
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2023
t : 22 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.258277
Ring = Hs-2
E_metric= 19.7208
kappa_HS= 96.83
omega = 4.0175 deg
d_A = 0.313215
Helm = Solar
Helm d = +0.181686
DR = 4.573 HLR
DR ratio= 96.8x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=23 | Year 2024 | D=9 N=25 pairs=36

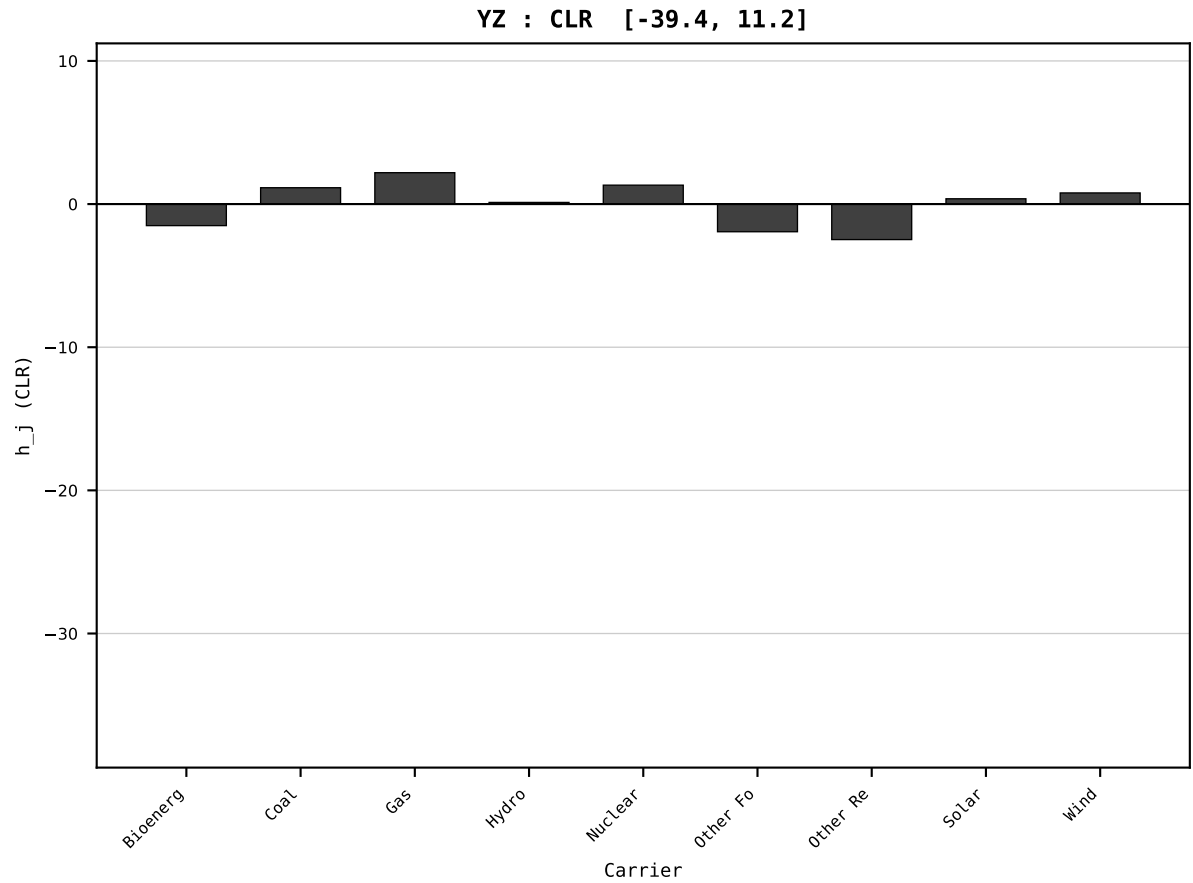
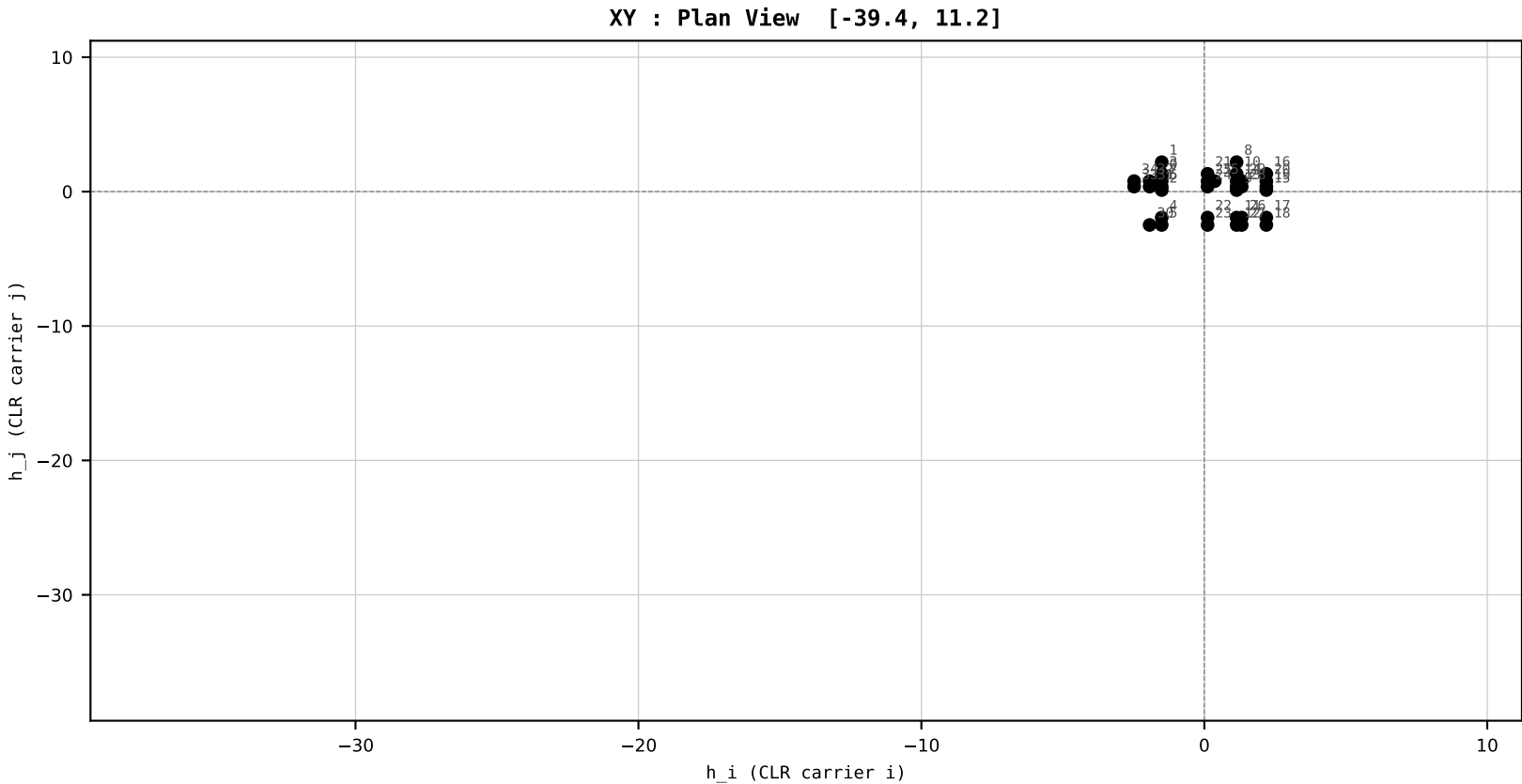
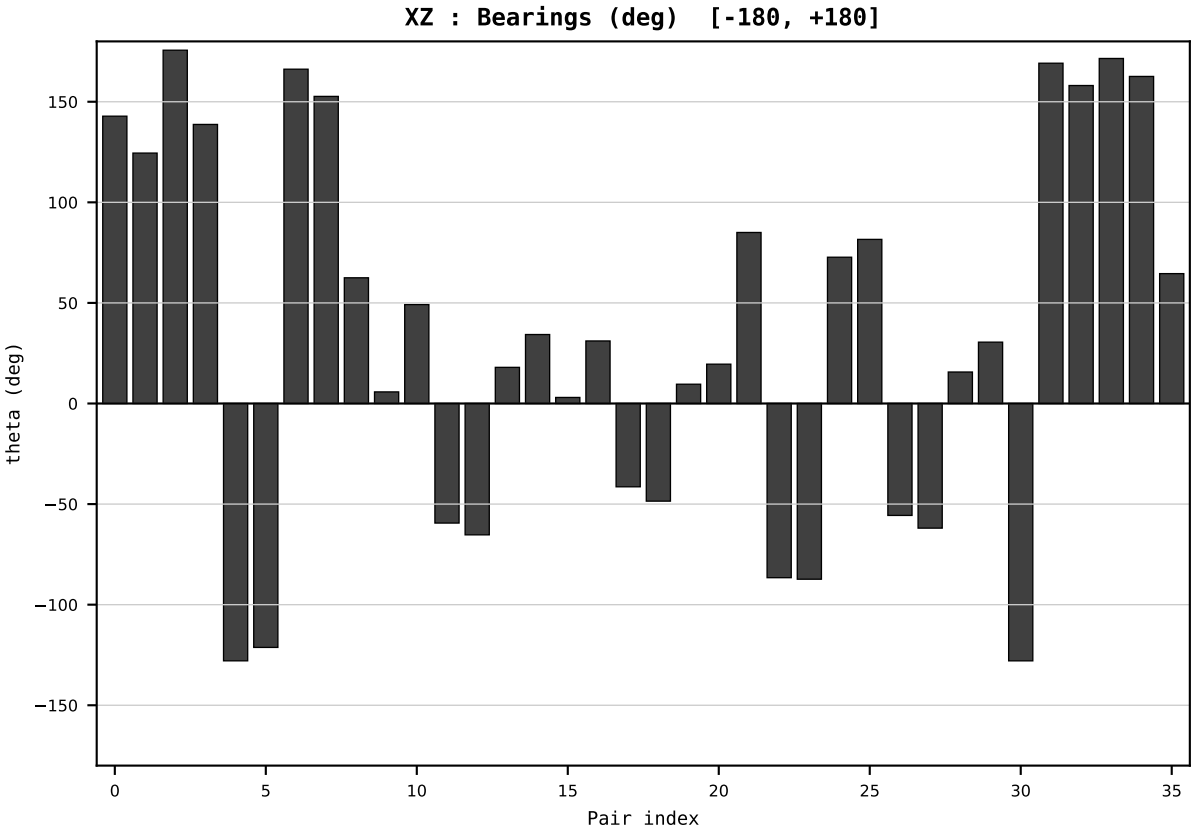
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2024
t : 23 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.256233
Ring = Hs-2
E_metric= 20.7697
kappa_HS= 107.02
omega = 3.1307 deg
d_A = 0.272026
Helm = Solar
Helm d = +0.222573
DR = 4.673 HLR
DR ratio= 107.0x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



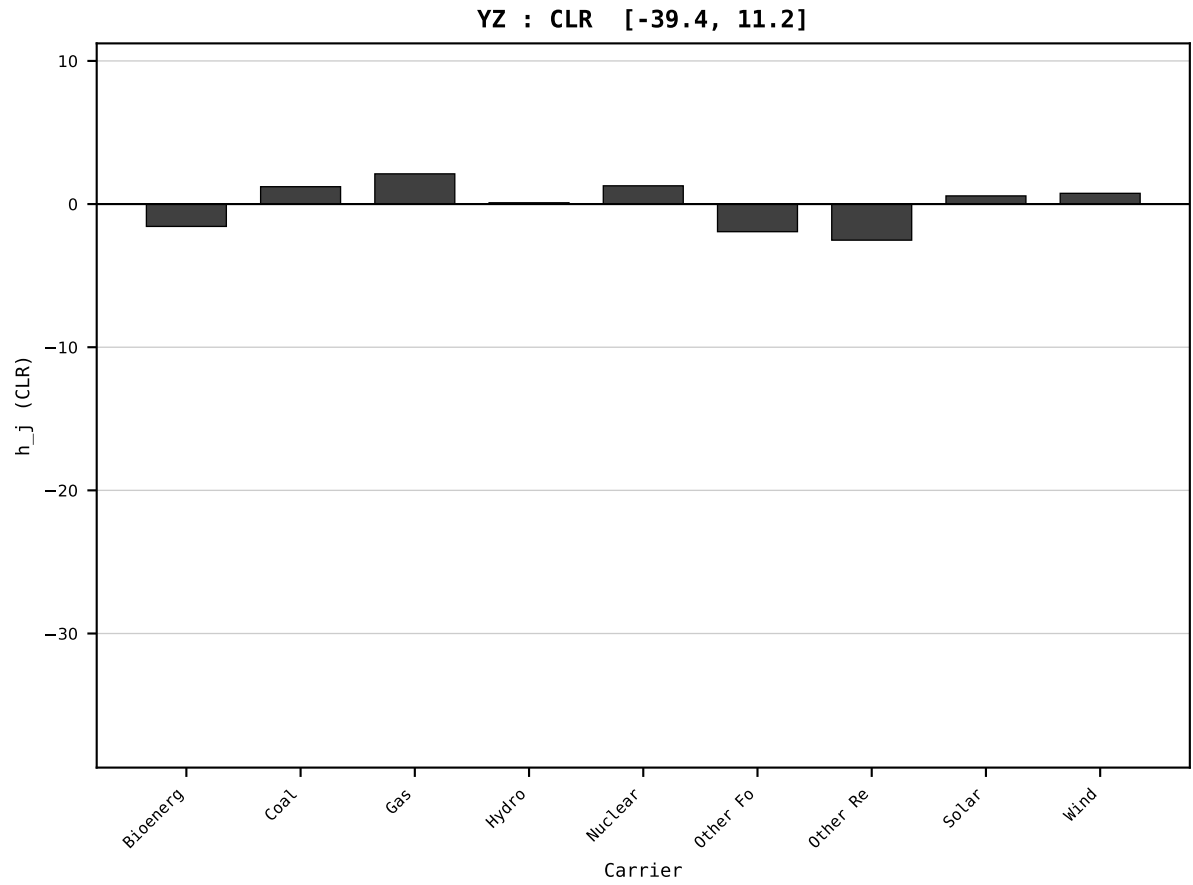
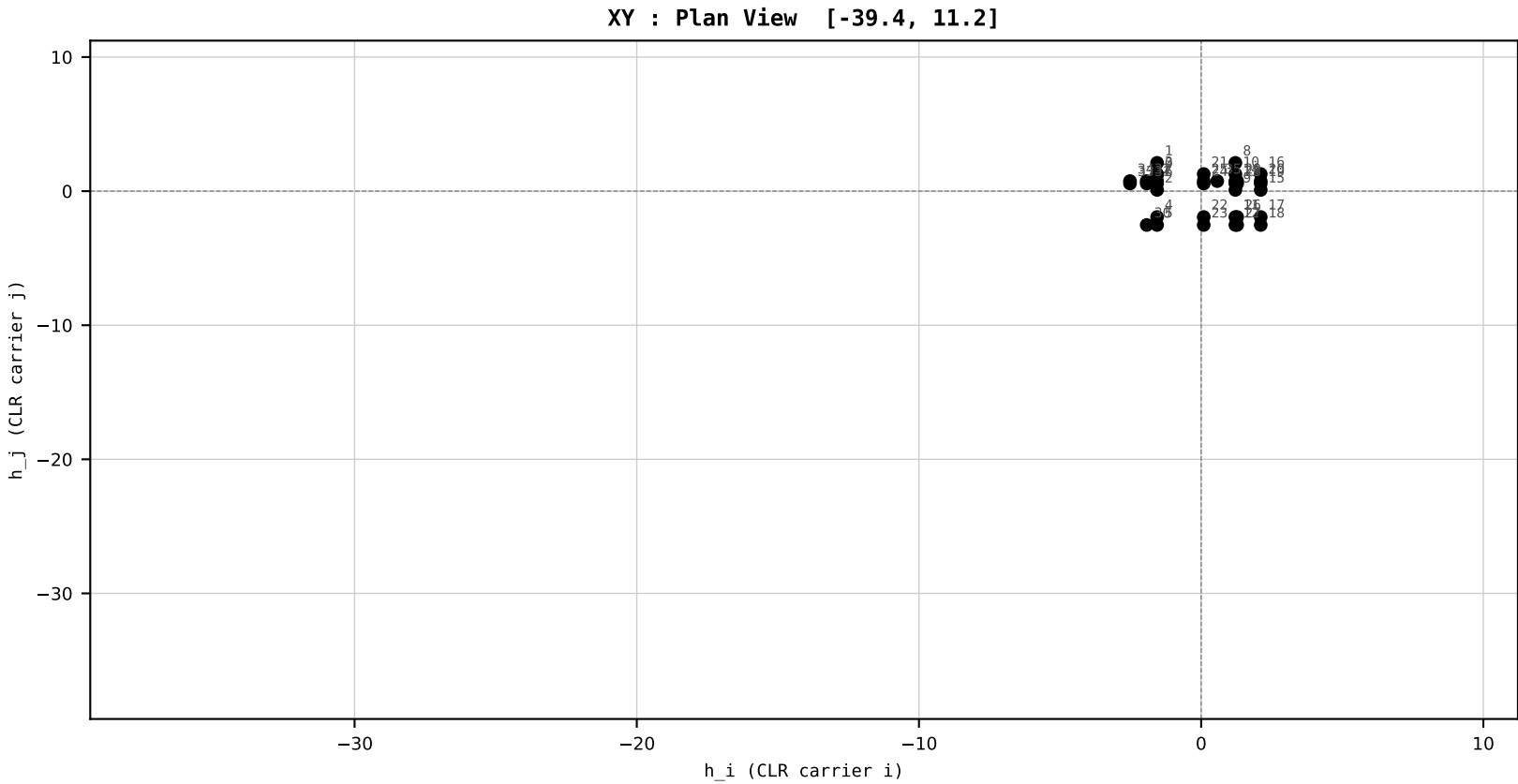
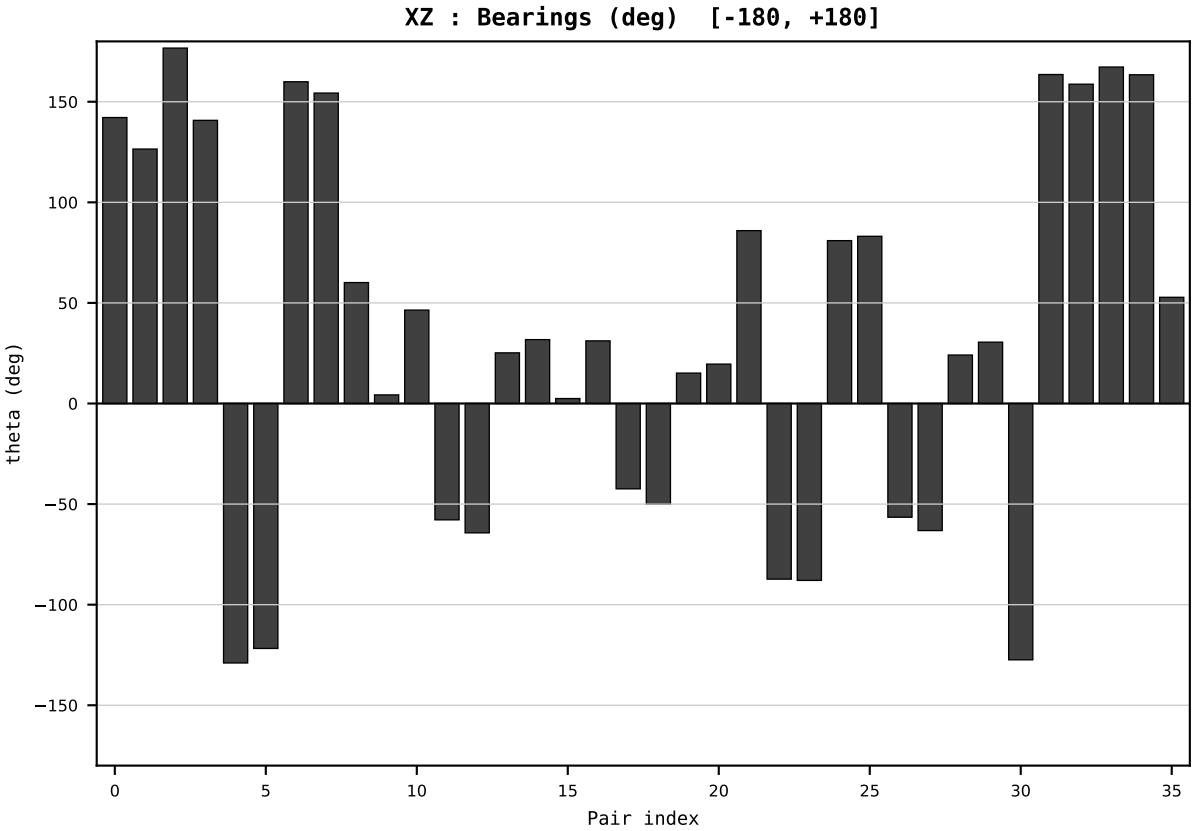
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

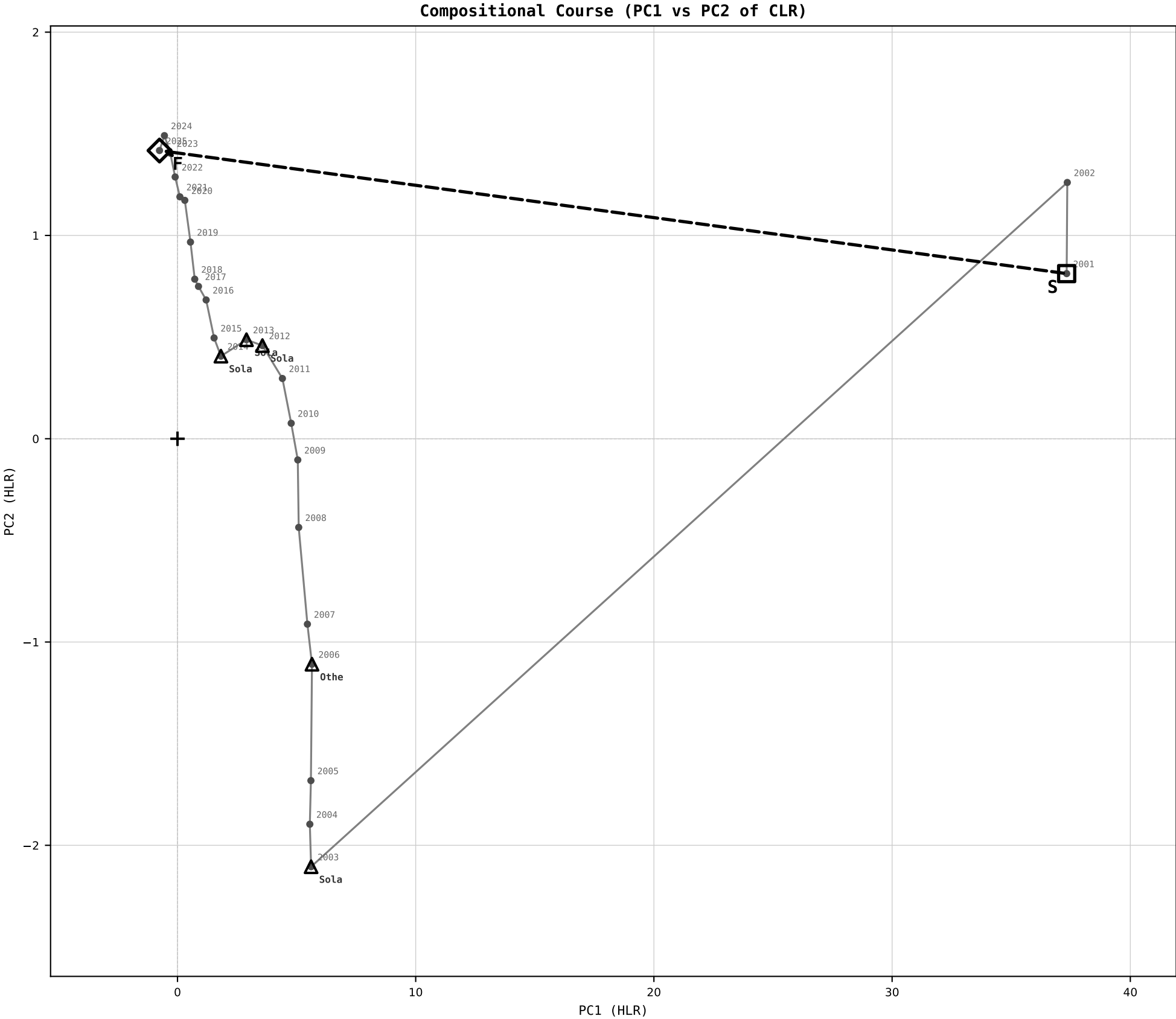
Dataset : ember_USA_United_States_generation_TWh.csv
Year : 2025
t : 24 / 24
D : 9 carriers
Pairs : 36 channels

Hs = 0.240009
Ring = Hs-2
E_metric= 20.9704
kappa_HS= 102.78
omega = 3.0581 deg
d_A = 0.244792
Helm = Solar
Helm d = +0.200013
DR = 4.633 HLR
DR ratio= 102.8x

XZ [-180,180]
CLR[-39.4,11.2]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind





COURSE METRICS

Scale Provenance:
 $Y_j \rightarrow x_j = Y_j / T \rightarrow h_j = \text{CLR}$
 $\rightarrow \text{PC projection}$
Unit: HLR (Higgins Log-Ratio)

Net distance = 38.0885 HLR
Path length = 41.8379 HLR
Directness = 0.9104

DR start = 42.090 HLR
ratio = 1903969992697630976.0x
DR final = 4.633 HLR
ratio = 102.8x

E_metric start = 1411.9935
E_metric final = 20.9704

PC1 loadings:
Solar -0.938
Other Fo +0.159
Coal +0.146

PC2 loadings:
Wind +0.831
Other Fo -0.385
Coal -0.324

Top displacement events:
2003 Solar d=31.925
2014 Solar d=1.080
2012 Solar d=0.882
2006 Other Fo d=0.708
2013 Solar d=0.691

The instrument reads.
The expert decides.
The loop stays open.